

# IT8101 Research Methods and Skills - T2 2026

## Table of Contents

### IT8101 Research Methods and Skills - T2 2026

**Course code:** IT8101 | **Source:** Whitecliffe Canvas (learn.mywhitecliffe.com)

*12 modules · 65 content pages · 2 assignments · 78 files.*

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### Course Introduction

#### Disclaimer and Copyright

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#### Course Outline

**IT8101 Course Overview** Course Description This course will equip you with the knowledge and understanding of using quantitative and qualitative methods in conducting meaningful enquiry and research. Course Level: 8 Course Credits: 15 Course Learning Outcomes On successful completion of this course, you will be able to: LO 1: Develop an appropriate IT related research question and associated methodology, emerging from relevant literature. LO 2: Critically reflect on ethical issues and/or potential barriers to undertaking proposed research in the field of IT. Assessment You can find all assessments for this course within the course modules, or by clicking Assignments on the left panel. This will show all assessed tasks. This course has 2 assessments. The assessment due dates will be noted in your course calendar. An overview of each assessment is provided here: Assessment Name Learning Outcomes Weighting Assessment

1: Literature Review, Research Gap and Questions LO 1 50% Assessment 2: Research Proposal and Reflection LO 2 50% Disclaimer: All assessment grades are provisional and subject to approval by the School of Information Technology (SoIT) Academic Programme Committee. Success Criteria To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%. Extensions and Late Submissions Policy An extension can only be granted under exceptional conditions normally for a maximum of (5 working days) , unless there is evidence to support a longer timeframe. Complete and return the following form to your Course Lecturer: Assessment Extension Request Form.pdf Assessments received one (1) to five (5) working days past the due date, or an approved extended due date, will be marked but will receive a mark capped at the pass mark i.e. 50% for graded assessments. Assessments received more than five (5) working days past the due date, or approved extended due date, will not be marked and will receive a mark of zero or fail. Academic Integrity and Academic Misconduct Policy Whitecliffe is committed to ensuring the highest level of academic integrity and expect staff and students to conduct themselves as responsible and honest members of the academic community. Where there is evidence that breaches of academic integrity may have occurred, the course of Academic Integrity and Academic Misconduct Policy will be followed. This policy considers academic integrity to mean honesty in all matters relating to academic endeavours. Breaches in academic integrity, or academic misconduct, include all forms of cheating incorporates plagiarism (intentional or unintentional), unauthorised collaboration, impersonation, falsification and collusion. Learning Activities Learning activities are provided to develop and consolidate your learning and understanding of the concepts. In preparation for the assessment tasks, make sure you complete each learning activity. You are provided with some reading material, videos, and learning tasks for self-study to support your learning during the course. You are also encouraged to engage with other reliable resources online. The Whitecliffe Library is a helpful place to begin.

#### Links & References

- [Link] [Assessment Extension Request Form.pdf](#)
- [Link] [Academic Integrity and Academic Misconduct Policy](#)
- [Link] [Whitecliffe Library](#)

#### Online Learning Resources

IT8101 Online Learning Resources O'Reilly Learning O'Reilly learning provides individuals, teams, and businesses with expert-created and curated information covering all the areas that will shape our future—including artificial intelligence, operations, data, UX design, finance, leadership, and more. To access O'Reilly Learning, you will need to login using@mywhitecliffe.com email address here: Login to O'Reilly

#### Links & References

- [Link] [Login to O'Reilly](#)

## Ngā Tikanga Paihere

IT8101 Ngā Tikanga Paihere Working in IT requires you to consider the needs and values of the people around you in terms of the systems they interact with. In this course we ask you to integrate Ngā Tikanga Paihere, tikanga Māori, and Te Ao Māori (the worldview principles of the people of New Zealand). Ngā Tikanga Paihere is a framework, originally designed by Stats NZ, to guide ethical and culturally appropriate data use. It consists of 10 tikanga [Māori world view principles], highlighted below, which can be applied to the IT sector in general.

Throughout this course and your postgraduate studies, you will see the Ngā Tikanga Paihere framework referenced. Each module will highlight a specific tikanga [principle] relevant to the content covered within the module. A reflection question will also be included at the end of the module to allow you to think holistically about how that principle applies. For more information on the framework and to help you understand the Māori world view better, have a read through the 18-page Ngā Tikanga Paihere Guidelines. Source : [Nga-Tikanga-Paihere-Guidelines-December-2020.pdf \(data.govt.nz\)](#)

### Links & References

- [Link] [tikanga](#)
- [Link] [Ngā Tikanga Paihere](#)

## Turnitin Canvas Plagiarism Framework and Draft Coach

IT8101 What is the Turnitin Canvas Plagiarism Framework Turnitin Canvas Plagiarism Framework The Turnitin Canvas Plagiarism Framework is a way to integrate Turnitin's plagiarism checking service directly into your Canvas assignments. Here are the key features:

**Similarity Reports** This is the core function, allowing lecturers to see how much a student's work matches existing sources. It highlights potential plagiarism and provides an overall similarity score. Lecturers can choose when to make the report visible to students.

**Simplified Workflow** Turnitin works within the Canvas assignment activity. Lecturers can set it up for assignments and view reports during the marking process. The similarity score label provides a link to the similarity report. Students, if allowed, have access to the report from the submission details.

**Basic Spelling and Grammar Suggestions** Turnitin offers basic suggestions for improving a student's writing mechanics.

**Focus on Originality** The framework prioritizes checking for plagiarism without offering Turnitin's full suite of grading functionalities. This keeps the focus on originality.

**What are the file requirements?** Turnitin accepts files that are less than 100MB and less than 800 pages. The allowed filetypes for upload are: Microsoft Word (DOC and DOCX) Portable Document Format (PDF) Plain text (TXT) Rich Text Format (RTF) OpenOffice (ODT) Powerpoint (PPTX) HTML Adobe PostScript

**Frequently Asked Questions**

**What is Turnitin Canvas Plagiarism Framework?** This framework integrates Turnitin's plagiarism checking service directly within your Canvas assignments. It analyses your submitted work against a vast database of sources to identify potential matches.

**How does it work?** Turnitin has been enabled for an assignment. You submit your work through Canvas as usual. Turnitin generates a Similarity Report highlighting potential matches in your work. Your instructor reviews the report alongside your assignment and provides feedback as part of the assessment.

**What is Turnitin Draft Coach?** Turnitin Draft Coach provides you with the ability to use Turnitin's tools while you draft your assignment. You will be able to run a total of three Similarity Checks

**Links to an external site.** . Draft Coach is available as an add-on in Microsoft

Word online, accessible through the Student Hub . You can only upload documents with filetypes doc , docx or odt . How to use Turnitin Draft Coach? Login to the Student Hub to an external site. Click on All Apps. In the Apps dashboard, click on Word. Upload your draft assignment and open in Word, or paste your draft text into a blank document. In the ribbon, click on Turnitin and next on Draft Coach. This will open the Turnitin Draft Coach sidebar. You must agree with the Turnitin Terms and Condition when you do a similarity check for the first time. Click on Go to Similarity to run the check. Click on View Full Report to review and refine results. Consult your lecturer for details on the threshold of an acceptable similarity score. How many times can I do Similarity Check for an assignment? You will be able to run a total of three Similarity Checks while drafting your assignment. We suggest using these strategically to get as much out of the reports as possible. Where can I learn more about Turnitin? Turnitin has a Student hub available, a page with links to the most important guides about Turnitin.

### Links & References

- [Link] [Similarity Checks](#)
- [Link] [Student Hub](#)
- [Link] [Student hub](#)

## Module 1: Research Fundamentals

### Introduction to Module 1

Module 1 Research Fundamentals: Overview Icon Progress Bar (browser only) Kia ora koutou - Hello everyone Welcome to Module One. This module is an essential introduction to the concept of research and will prepare you for the learning ahead. You will encounter what is meant by Research and learn to distinguish different research types and models. You will discover the typical steps involved in an academic research project, become familiar with common terminologies, encounter popular tools and resources for managing your research. Objectives At the completion of this module, you will be able to: Distinguish between various research types and research models. Identify the characteristics of research. Describe the Scientific Method Be familiar with popular research tools

### What is Research?

Module 1 What is Research? Focus This page introduces the concept of research, and by the end of it, you will be able to: distinguish between various research types and research models identify the characteristics of the research describe the iterative and cyclical nature of research, as well as the steps involved in an academic research project use common research terminologies There are readings and short videos to support your learning. What is Research? Various authors have proposed various definitions of research. Some are general, while others are field-specific: "Research is a process of systematic inquiry that entails collection of data; documentation of critical information; and analysis and interpretation of that data/information, in accordance with suitable methodologies set by specific professional fields and academic disciplines." or "Research is defined as the creation of new knowledge and/or the use of existing knowledge in a new and creative way so as to generate new concepts, methodologies and understandings. This could include synthesis and analysis of previous research to the extent that it leads to new and creative outcomes." If we think of the everyday

use of the term, research is finding out things we do not know. People often use a systematic approach when they collect and interpret information to solve the small problems of daily living. In this course, we will concentrate on formal research, which is research in which we intend to improve our understanding of a phenomenon and expect to communicate our findings to the larger scientific community. Research Overview Watch the YouTube video for a good overview of Research. Research entails more than just answering 'what' questions and gathering data; it also necessitates analysis. In this course, we will look for explanations, relationships, comparisons, predictions, generalisations, and theories. It answers the "why" questions. In Computer Science and related domains, research entails a systematic process of data collection and organisation, information analysis using a suitable methodology, and result interpretation. It is carried out to assess the validity of a hypothesis, expand previous work, and add to existing knowledge. The video below provides a good summary of the information covered on this page and introduces the Scientific Method which we will be talking about later in this module: Different Types of Research Research can be classified in many different ways on the basis of the methodology of research. Some of the common research types are as follows: Pure/Basic research Applied research Quantitative research Qualitative research Pure research Pure research is also called basic or fundamental research. It is exploratory research done within a pre-described boundary. The mindset and willpower of the researcher is the only way to advance such types of research. Research work is driven by the motivation and commitment of the researcher. The work progresses by finding variables and their relationship between each other. Pure research always serves as the foundation for applied research. The primary concern of such type of research will be the designing of internal logic and architecture, to build upon limited knowledge. Research concerning some natural phenomenon or relating to pure mathematics or work that tries to generalize some facts are examples of fundamental research. Applied Research Applied research fundamentals that we have come across in pure research are applied to produce some end products. In short, it is an application of theory into practical solutions of problems. Applied research is basically a group task, trying to create or solve a particular problem. The research becomes truly acceptable when third parties or sponsors use our product with full satisfaction. For example, studying the radio communication channel is a form of pure research, but creating a product that uses radio communication for any particular task comes under applied research. In western countries, about 80% of research carried out is applied research. Basic and applied research have the same goal: to add to or develop a body of knowledge. Basic research is not generally related to a specific problem, and its findings are not intended to be immediately applied. It may be concerned with theoretical issues or fundamental questions concerning the nature of things. Applied research, on the other hand, directly addresses a problem or has immediate practical implications. There are also differences in how the findings are applied. In general, basic research contributes to theory, whereas applied research findings are used directly or immediately. Applied research is designed to address a specific problem, and its findings are typically applied immediately by decision makers based on feasibility and sustainability. Figure 2.2: Basic and Applied research Quantitative Research Quantitative research is that which can be represented or described according to some numerical system. They are normally associated with large-scale analysis works. These are basically chosen to compare and fine tune the products by considering the amount of output that research process delivers. Statistical tests are done to measure validity and reliability. Experimental and descriptive research is the major classification of quantitative research. Descriptive research provides an accurate profile of the group. It describes the

process, mechanism or relationship. It gives information and stimulates new explorations. It does not involve any in-depth study, but it reveals just the concepts of theories. For example, while studying the economic activities of a village, one will look for day-to-day activities of the villagers, analysis it and generate a report. There are no deep evaluations, just what he/she perceives and reflects in his/her report. Experimental research measures variations in varied conditions. For example, if we are trying to find the effect of a particular fertilizer, we take a plot and divide it into two halves. In the first half, we keep the fertilizer and the other half will be kept free. During the time of harvest, we can directly see the effect of the fertilizer. Thus, we have a control group and an experimental group. Before and after the true impact study, the research work will be supervised very closely. Qualitative Research Qualitative research takes an inductive approach. This type of research is commonly found in social sciences where researchers intend to study social and cultural phenomena. Qualitative research does not have a well-placed definition, but it typically takes a “world-view” angle, which has all assumptions but meanings change from situation to situation. Unlike quantitative research, no numerical measures are incorporated in qualitative research, instead it uses an in-depth analysis approach by taking case studies or events to study situations. It is not involved in investigating and developing hypotheses. A common perception of qualitative research is the emphasis on discovery rather than proof. Types of Research Watch this video for an overview of basic and applied research as well as research types. These topics will be explored further in other modules of the course. Basic vs Applied Research Watch this video for an explanation of how your research focus can change throughout the research process and how this will affect the decisions you make throughout: Quantitative vs Qualitative Research When you are conducting your research, you will often need to decide between using Qualitative or Quantitative research methods. In Module 4 we go into these types more deeply. However, it is good to have an understanding of these terms before we look at Research Strategies on the next page. As you watch this video start thinking about what type of research you prefer and why: Research Process Research follows a process and this process is typically cyclical. Take a look at the diagram below before proceeding to the next module page. Sources: <https://holisticinfosec.io/post/detection-development-research-cycle-nist-csf/> Figure 2.1: The Research Cycle (Leedy & Ormrod, 2014, p. 2-6) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=mV0bUQpz468>
- <https://www.youtube.com/watch?v=JEZjwIDNEHY>
- <https://www.youtube.com/watch?v=pg8deU3XTls>
- <https://www.youtube.com/watch?v=a-XtVF7Bofg>

#### Links & References

- [Link] <https://youtu.be/JEZjwIDNEHY>
- [Link] <https://holisticinfosec.io/post/detection-development-research-cycle-nist-csf/>

## Characteristics of Research

Module 1 Characteristics of Research Icon Progress Bar (browser only) Focus This page introduces the characteristics, approaches and terminologies of research. By the end of it, you will be able to: Identify research characteristics Understand research terminologies There are readings and short videos to support your learning. Research Characteristics Good research follows a systematic approach to solving a problem in a particular area of study. The characteristics of research vary from study to study due to the subject matter and methodology. However, every research project should have the following characteristics: Empirical - based on observations and experimentation on theories. Systematic - follows an orderly and sequential procedure. Controlled - all variables except those that are tested/experimented upon are kept constant. Employs hypothesis - guides the investigation process. Analytical - there is a critical analysis of all data used so that there is no error in their interpretation. Objective, unbiased, & logical - all findings are logically based on empirical data. Replicable - It should be able to be repeated with similar results. Employs a relevant and consistent methodology - applies systematic procedures or techniques to ensure that the study achieves valid, reliable results. Ethical - follows agreed ethical standards. source:

[https://en.wikiversity.org/wiki/Characteristics\\_of\\_research](https://en.wikiversity.org/wiki/Characteristics_of_research) Characteristics of Research Watch the "Research Simplified" video and note any different terminology used to convey the characteristics of good research: Research Terminologies As we can see from the definitions above, there are different ways of describing research. However, there are a lot of common terminologies used by researchers that it is helpful to learn. Below is a link to a glossary of key research terminology. Some of the readings we use in this course will have a glossary that you can access as well. Referring to glossaries will assist you in developing your research literacy. Here are some of the commonly used research terminologies used in computer science related domains: Research Terminology Glossary Accuracy a term used in survey research to refer to the match between the target population and the sample. Bias a loss of balance and accuracy in the use of research methods. It can appear in research via the sampling frame, random sampling, or non-response. It can also occur at other stages in research, such as while interviewing, in the design of questions, or in the way data are analysed and presented. Bias means that the research findings will not be representative of, or generalisable to, a wider population. Case Study the collection and presentation of detailed information about a particular participant or small group, frequently including data derived from the subjects themselves. Causal Hypothesis a statement hypothesising that the independent variable affects the dependent variable in some way. Claim a statement, similar to a hypothesis, which is made in response to the research question and that is affirmed with evidence based on research. Control Group the group in an experimental design that receives either no treatment or a different treatment from the experimental group. This group can thus be compared to the experimental group. Controlled Experiment an experimental design with two or more randomly selected groups [an experimental group and control group] in which the researcher controls or introduces the independent variable and measures the dependent variable at least two times [pre- and post-test measurements]. Correlation a common statistical analysis, usually abbreviated as  $r$ , that measures the degree of relationship between pairs of interval variables in a sample. The range of correlation is from -1.00 to zero to +1.00. Also, a non-cause and effect relationship between two variables. Data factual information [as measurements or statistics] used as a basis for reasoning, discussion, or calculation. Dependent Variable a variable that varies due, at least in part, to the impact of the independent variable. In other words, its value

"depends" on the value of the independent variable. For example, in the variables "gender" and "academic major," academic major is the dependent variable, meaning that your major cannot determine whether you are male or female, but your gender might indirectly lead you to favour one major over another. Empirical Research the process of developing systematised knowledge gained from observations that are formulated to support insights and generalisations about the phenomena being researched. Framework the structure and support that may be used as both the launching point and the on-going guidelines for investigating a research problem. Hypothesis a tentative explanation based on theory to predict a causal relationship between variables. Independent Variable the conditions of an experiment that are systematically manipulated by the researcher. A variable that is not impacted by the dependent variable, and that itself impacts the dependent variable. In the earlier example of "gender" and "academic major," (see Dependent Variable) gender is the independent variable. Methodology a theory or analysis of how research does and should proceed. Methods systematic approaches to the conduct of an operation or process. It includes steps of procedure, application of techniques, systems of reasoning or analysis, and the modes of inquiry employed by a discipline. Null Hypothesis the proposition, to be tested statistically, that the experimental intervention has "no effect," meaning that the treatment and control groups will not differ as a result of the intervention. Investigators usually hope that the data will demonstrate some effect from the intervention, thus allowing the investigator to reject the null hypothesis. Population the target group under investigation. The population is the entire set under consideration. Samples are drawn from populations. Sample the population researched in a particular study. Usually, attempts are made to select a "sample population" that is considered representative of groups of people to whom results will be generalised or transferred. In studies that use inferential statistics to analyse results or which are designed to be generalisable, sample size is critical, generally the larger the number in the sample, the higher the likelihood of a representative distribution of the population. Standard Deviation a measure of variation that indicates the typical distance between the scores of a distribution and the mean; it is determined by taking the square root of the average of the squared deviations in a given distribution. It can be used to indicate the proportion of data within certain ranges of scale values when the distribution conforms closely to the normal curve. Statistical Analysis application of statistical processes and theory to the compilation, presentation, discussion, and interpretation of numerical data. Statistical Bias characteristics of an experimental or sampling design, or the mathematical treatment of data, that systematically affects the results of a study so as to produce incorrect, unjustified, or inappropriate inferences or conclusions. Unit of Analysis the basic observable entity or phenomenon being analysed by a study and for which data are collected in the form of variables. Validity the degree to which a study accurately reflects or assesses the specific concept that the researcher is attempting to measure. A method can be reliable, consistently measuring the same thing, but not valid. Variable any characteristic or trait that can vary from one person to another [race, gender, academic major] or for one person over time [age, political beliefs]. Further Reading on Research Terminologies Glossary of Research Terms YouTube Video Watch this video "Complex Research Terminology Simplified" to get an understanding of key research terms: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel



- <https://www.youtube.com/watch?v=9wSnVjWROw0>
- <https://www.youtube.com/watch?v=8xvpXBVCo0c>

## Links & References

- [Link] [Glossary of Research Terms](#)

## The Scientific Method

Module 1 The Scientific Method Icon Progress Bar (browser only) Focus This page introduces the Scientific Method as an integral approach to conducting research. You will learn about the commonly used steps in this method and view a real-world example. There are readings and short videos to support your learning. Suggestion It is recommended that you read the below articles to expand your knowledge of scientific research methods and scientific report writing. The following concise guide from Purdue University is helpful in understanding the process of the scientific method. This guide is helpful in how to write a conclusion for a scientific report. We will go over scientific report writing in greater detail in later modules, but it is critical that you have a basic understanding of it before you begin identifying a research problem from the literature review. The Scientific Method The scientific method is the process of experimentation that is used to explore observations and answer questions. Through the process of the scientific method, researchers investigate a problem and then share their findings with a wider community of other researchers. It is a step-by-step approach to conducting a study. As the video below shows, we all use the scientific method when solving everyday issues. It is a proven approach to problem-solving that forms the basis for the research process. Steps in Scientific Methods Most of us are very familiar with the steps in the Scientific Method from studying science at school. Below is a simple diagram with associated descriptors for each step: Step 1: The Question The first step of the scientific method is the "Question." This step may also be referred to as the "Problem." Your question should be worded so that it can be answered through experimentation. Keep your question concise and clear so that everyone knows what you are trying to solve. The question should have a purpose and answer the following questions: Why do you want to know? How does this matter? Who wants to know? This first step should also have a goal: What purpose will the answer to this question serve? Step 2: Research Now you have your problem formulated into a question the next step is to undertake some Research. You will need to find out more about your subject. It is time to gather some background information before you move on to your next step which is developing the hypothesis. It is time to go out and find what other research or experimentation is out there that addresses this question and then consider how this information will influence your experiment. Where do you find this background information? The first step is usually by looking at the published literature that is available on the topic of interest. That is undertaking a literature review. Step 3: State your Hypothesis Now it is time to state your hypothesis. The hypothesis is an educated guess as to what will happen during your experiment. The hypothesis is often written using the words "IF" and "THEN." For example, "If I do not study, then I will fail the test." The "if" and "then" statements reflect your independent and dependent variables. The hypothesis should relate back to your original question and must be testable. Have a go at writing hypotheses with this hypothesis writing worksheet. Step 4: The Experiment Now it is time to test your hypothesis. This is done through experimentation. This starts by explaining your procedure or process. Remember to keep your experiment fair and unbiased and watch those variables (change only one thing at a time). After you have your process figured out,

make a list of materials you will need to conduct the experiment. When you are ready, you can conduct your experiment. The experiment needs to be repeatable. Repeat your experiment several times to validate your findings. Make sure you keep careful notes and data regarding the process and outcomes. Read this article to review the key points in designing experiments using the scientific method. Step 5: Analysing Data Now it is time to analyse your data. In doing this you are looking to find out: Is the data reasonable? Is your data organised so others will be able to understand it? Does the data support or disprove your hypothesis? Do you need to run the experiment again? We will look further into data analysis and statistics in another module. However, it is important to be aware of key data analytical concepts. Read and learn about these in this article on Uncertainty, Error and Confidence. Step 6: The Conclusion Finally, you have reached your conclusion. In the conclusion, you summarise and explain what happened in your experiment. Your conclusion should answer the question posed in Step 1. Your conclusion should be based solely on your results. In writing your conclusion think about the following questions: Was your hypothesis correct? If your hypothesis was not correct, what can you conclude from that? Do you need to run your experiment again changing a variable? Is your data clearly defined so everyone can understand the results and follow your reasoning? Remember, even a failed experiment can yield a valuable lesson. Example of the Scientific Method The diagram below is a very neat example of the scientific method in action. The researcher begins with her observation about plant growth, forms her question, states her hypothesis, gathers the results, and makes her conclusion - see below: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=SMGRe824kak>

#### Links & References

- [File] [concise guide from Purdue University](#)
- [File] [This guide](#)
- [File] [this hypothesis writing worksheet.](#)
- [Link] [designing experiments using the scientific method.](#)
- [Link] [this article on Uncertainty, Error and Confidence.](#)

### Research Tools

Module 1 Research Tools Icon Progress Bar (browser only) Focus Researchers, like other professionals, have their own set of tools to carry out their work. These tools can vary considerably depending on the discipline, or field of interest, the researcher is researching. A microbiologist needs a microscope and culture media; an attorney needs a library of legal decisions and statute law. But in general, most researchers will need to access the following tools to conduct good research. The below presentation order is random. We will also read further about these tools in later modules. This page introduces the essential research tools necessary for conducting effective research, helping you to gain a better understanding of the resources and techniques for use in your own research projects. There are readings and short videos to support your learning. Critical Mind Critical thinking is an important soft skill to have when conducting research, for several reasons: Evidence-based decision making: Critical

thinking enables researchers to make informed decisions based on evidence and logic, rather than relying on intuition or assumptions. This is particularly important in research, where the goal is to generate new knowledge and advance understanding. Avoiding bias: Critical thinking helps researchers to avoid bias and maintain objectivity. By evaluating evidence objectively and considering multiple perspectives, researchers can minimize the influence of their own beliefs and biases on their work. Improving the quality of research: Critical thinking helps to improve the quality of research by allowing researchers to evaluate their methods and results. Researchers can identify strengths and weaknesses in their work, and make necessary changes to improve their findings. Solving problems: Critical thinking is also essential for solving problems in research. Researchers can use critical thinking to identify and evaluate potential solutions, and make informed decisions about the best course of action. Evaluating the work of others: Critical thinking is also important for evaluating the work of others. By critically analyzing the research of others, researchers can determine its validity and reliability, and build upon the work of others to advance their own understanding.

### Computer and Other Hardware/Tools

Computers and other hardware play an essential role in conducting research for several reasons:

- Data storage and processing:** Computers provide a convenient way to store and process large amounts of data, which is critical in many areas of research. Researchers can use specialized software to analyze data, such as statistical software, simulations, or visualization tools.
- Communication and collaboration:** Computers and the internet allow researchers to communicate and collaborate with other researchers, regardless of their location. This makes it easier to share data, collaborate on projects, and get feedback on results.
- Experimentation and simulation:** In some fields of research, computers and hardware are used to perform experiments or simulations that would not be possible to do manually. For example, in neuroscience, researchers use brain-computer interfaces to study the brain, and in physics, high-performance computing is used to simulate complex systems.

### Internet and Libraries

The internet provides researchers with easy access to a vast amount of information and resources that can help with their work. This includes academic papers, databases, simulations, and other tools. To begin the research process, we need access to published research articles, such as conferences and journal papers, to conduct a literature review. Tip: Whitecliffe College Library is here to help! A literature review is important for understanding the current state of knowledge in a particular field and identifying gaps that need to be filled. Researchers use databases such as PubMed, Scopus, and Web of Science to search for relevant papers. There are numerous reputable publishers, including IEEE, ACM, Taylor and Francis, Springer, and others. Access to conference and journal papers is essential for conducting research for several reasons:

- Keeping up to date:** Conference and journal papers are the primary means by which researchers disseminate their findings and share their work with others in their field. By reading these papers, researchers can keep up to date with the latest developments in their field and learn about new ideas and methods.
- Building knowledge:** Reading conference and journal papers helps researchers build their knowledge and understanding of their field. It provides them with the latest information and helps them to broaden their perspective and develop new ideas.
- Validating results:** Access to conference and journal papers allows researchers to validate their results and ensure that they are based on the latest knowledge. By comparing their results with those of other researchers, they can identify any discrepancies and make sure their findings are robust and reliable.
- Improving methods:** Conference and journal papers often describe new methods and techniques that researchers can use in their work. By reading these papers, researchers can learn about new approaches and tools that can

help them improve their methods and advance their research. Citing sources: Conference and journal papers are an important source of information for researchers and are often cited in their own work. This helps to establish a clear chain of evidence and supports the validity of their findings. Statistics and Data Analysis Tools Statistical software such as SPSS , R , and SAS are essential tools for analyzing data and testing hypotheses. Researchers use these tools to perform statistical tests, generate graphs and charts, and interpret the results of their data analysis. Learning statistics is important in research for several reasons: Data analysis: Statistics is the study of data, and researchers need to be able to analyze data to test hypotheses and draw conclusions. Statistics provides tools and techniques for summarizing, describing, and analyzing data, such as descriptive statistics, inferential statistics, and regression analysis. Understanding variability: Statistics helps researchers understand the variability of their data and the uncertainty in their results. By using statistical methods, they can quantify the uncertainty in their estimates and make inferences about populations based on their sample data. Making informed decisions: Statistics provides a framework for making informed decisions based on data. Researchers can use statistical tests to determine if their results are significant and make decisions about the validity of their findings. Designing experiments: Statistics plays a crucial role in the design of experiments. Researchers use statistical methods to determine the appropriate sample size, randomization procedures, and other factors that influence the validity and reliability of their results. Communication of results: Statistics is also important for communicating research results effectively. Researchers need to be able to present their findings in a way that is easy to understand and interpret, and statistics provides methods for visualizing data and presenting results in a clear and concise manner. Academic Writing Your writing skills are an important part of your researcher toolkit. There are many helpful tools out there, including software that organizes your writing like Scrivener , to programmes that assist you with your grammar and writing style, like Trinka or Grammarly . Take a look at this blog-post from Research.com that lists some of the most useful tools for academic writing. Academic writing is important in conducting research for several reasons: Communication of results: Academic writing is the primary means of communicating research results to others. Researchers use academic writing to share their findings with the broader research community, including their peers, the general public, and policymakers. Establishing credibility: Academic writing is an important tool for establishing credibility and demonstrating the value of research. By writing clear, well-organized, and well-supported papers, researchers can demonstrate their expertise and the quality of their work to others. Preserving knowledge: Academic writing helps to preserve knowledge and ensure that research results are accessible to future generations. By publishing their work in peer-reviewed journals, researchers can contribute to the advancement of their field and ensure that their work has a lasting impact. Collaboration: Academic writing is also important for collaboration and interdisciplinary research. By writing papers and sharing their results, researchers can collaborate with others and build upon their work to make new discoveries. Improving thinking: The process of writing academic papers can also help researchers to clarify their thinking and refine their ideas. By organizing their thoughts and writing down their results, researchers can identify gaps in their knowledge and refine their research strategies. Research Management Software Research management helps researchers allocate their resources effectively and efficiently. By planning and managing their research project, researchers can ensure that they use their time, money, and other resources wisely and avoid unnecessary waste. Better collaboration: Research management can help improve collaboration between

researchers. By establishing clear roles and responsibilities, researchers can work together more effectively and achieve their objectives more efficiently. Follow the link in the image below for a list of some of the most effective research project management tools available: Reference management software, such as EndNote, Mendeley, and Zotero, is used to organize, store, and manage references and bibliographies. These tools help researchers keep track of the literature they have reviewed and ensure that they are able to access and cite the work of others as needed. EndNote The software will allow you to: Export citations from databases, Library Search and Google Scholar into EndNote Attach PDFs of articles to a record in EndNote-store (manually or using find full-text feature) Arrange citations in groups Use the EndNote toolbar in Word to insert in-text citations and create a list of references automatically Create and sync citations between EndNote desktop and EndNote Web Share EndNote libraries with others To learn more, see the EndNote website . Mendeley This software is another reference management tool, as well as an academic social network, that can help you organize your research, collaborate with others online, and discover the latest research. With Mendeley you can: Automatically generate bibliographies Collaborate easily with other researchers online Easily import papers from other research software Find relevant papers based on what you are reading Access your papers from anywhere online To learn more, see the Mendeley website . Zotero Zotero is a free and open-source reference management software to manage bibliographic data and related research materials, such as PDF files. To learn more, see the Zotero website . Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- [https://www.youtube.com/watch?v=IRCHdhdS\\_aU](https://www.youtube.com/watch?v=IRCHdhdS_aU)

#### Links & References

- [Link] [Tip: Whitecliffe College Library is here to help!](#)
- [Link] [SPSS](#)
- [Link] [R](#)
- [Link] [SAS](#)
- [Link] [Scrivener](#)
- [Link] [Trinka](#)
- [Link] [Grammarly](#)
- [Link] [blog-post from Research.com](#)
- [Link] <https://www.linkedin.com/advice/1/what-tools-software-can-you-use-manage-your>
- [Link] [EndNote website](#)
- [Link] [Mendeley website](#)
- [Link] [Zotero website](#)

#### Statistics in Research

Module 1 Statistics in Research Focus This page introduces using statistics in research. We will cover: The role of statistical methods in research Two major areas of statistics known as

descriptive statistics and inferential statistics

### Choosing the right statistical technique

The principle of Pono in statistical research

There are readings and short videos to support your learning.

#### Further Reading

Read the Chapter 4 Basic Statistical Measures in Research Methodology. Read The importance-statistics for more details on how statistics is useful in research.

#### The Role of Statistical Methods in Research

Statistics plays a crucial role in research, serving as a valuable tool for study design, data analysis, and drawing meaningful conclusions. Research studies often generate a substantial amount of raw data, which needs to be organized and condensed for ease of interpretation and further analysis.

#### Why should I use statistics?

Statistics provides the means to develop indices or measures that succinctly capture the essence of the data. These summaries allow researchers to gain insights, identify patterns, and make informed interpretations of the collected information. By employing statistical techniques, you can explore relationships, uncover trends, and test hypotheses with rigour and objectivity. Statistical analysis aids in quantifying the significance of findings, determining the reliability of results, and making informed decisions based on evidence.

#### What can Statistics do?

Informing methods on data collection: Statistics provide guidance on selecting appropriate methods and techniques for collecting data. Supporting or negating hypotheses: Statistics are used to analyze data and test hypotheses. Identifying uncertainty, errors, and outliers: Statistics help researchers identify and address sources of uncertainty, errors, and outliers in the data. Aiding interpretation, summarization, and communication of datasets: Statistics enable researchers to interpret and make sense of complex datasets. Allowing multivariate statistics and modeling: Statistics provide researchers with tools to analyze and model relationships among multiple variables.

#### Inferential and Descriptive Statistics

##### Descriptive statistics

This branch deals with summarising and describing data sets without making inferences or drawing conclusions about a larger population.

##### Inferential statistics

This branch involves making predictions, inferences, or generalisations about a population based on a sample of data. For example, it is impractical to measure the height of every adult woman, but you can measure the heights of a random sample and use that information to make generalizations about the heights of all women. For example, a confidence interval provides a range that the population mean height is likely to fall in. source: <https://statisticsbyjim.com/glossary/inferential-statistics/>

#### Why choose Descriptive Statistics?

Descriptive statistics offer several advantages, including the ability to succinctly summarize data, create visual representations for better understanding, facilitate comparisons between datasets, detect outliers or errors, maintain simplicity in calculations, and serve as an initial step in data exploration, providing valuable insights for subsequent analyses.

#### Why choose Inferential Statistics?

Inferential statistics are valuable for drawing broader conclusions from limited data samples. They allow researchers to make predictions, test hypotheses, and generalize findings to larger populations, based on the analysis of a representative subset. This enables data-driven decision-making, hypothesis testing, and the assessment of relationships between variables. Inferential statistics provide a powerful means to extract meaningful insights and make informed judgments beyond the scope of descriptive statistics. A study using descriptive statistics is simpler to perform. However, if you need evidence that an effect or relationship between variables exists in an entire population rather than only your sample, you need to use inferential statistics. See the video below for a great example: Nga Tikanga Paihere: Pono Pono : to be true, valid, honest, genuine, sincere

Statistical analyses may impact individuals, communities, and cultural contexts. Applying the Pono principle in the use of statistics in research means reflecting on how you can ensure that you employ statistical methods with integrity , fairness , and cultural sensitivity .

#### Cultural Values

When conducting

research within a specific cultural community, researchers can reflect on whether their chosen statistical methods align with the cultural values and norms of that community. Community The Pono principle encourages researchers to engage in meaningful collaboration and reciprocity with the community or stakeholders affected by the research. Integrity By embracing the Pono principle, researchers can ensure that statistical analyses are conducted with cultural sensitivity and integrity. Activity: Your Colleague Wants to Know Scenario: You have taken a break from your job in an IT software development company to pursue your postgraduate studies. One day, you run into a former colleague who is very interested to learn about what you are up to. Although she is a senior manager in the Marketing Department, she did not get the opportunity to do postgraduate study and she is really interested in hearing about your research projects. She asks you all sorts of questions like the ones listed below. Write down your answers in your own spoken voice - they need to be authentic to you. Try to give real-life examples to demonstrate your meaning. Share your answers with your lecturer or class peers for feedback. What exactly is research? How do you know if something is good research? What are some of its characteristics? What is some research terminology you have learned? How does applied research differ from basic research? How does research apply to everyday life? What topics in technology research are you interested in? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=edEXEyvG4Wk>

#### Links & References

- [Link] [Chapter 4 Basic Statistical Measures](#)
- [Link] [The importance-statistics](#)
- [Link] [confidence interval](#)
- [Link] <https://statisticsbyjim.com/glossary/inferential-statistics/>

## Module 2: Research Design

### Introduction to Module 2

Module 2 Research Design: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Two. This module introduces you to the concept of research design. You will learn what it means to design research, and the philosophies and approaches that can be applied to a research design, as well some different strategies involved. Designing your research before you get started is an essential step in ensuring that your time and energy is pointed in the right direction, allowing you to answer your fundamental research questions. Objectives At the completion of this module, you will be able to: Explain the research onion model and how the steps guide your research decision-making process Use research terminology in everyday conversations Start the high-level thinking necessary to complete your research proposal

## What is Research Design?

Module 2 What is Research Design Icon Progress Bar (browser only) Focus This page introduces you to research design as a concept. There are readings and short videos to support your learning. Research Design Research design is the plan or framework that outlines the methods and procedures that will be used to conduct a research study. It is the blueprint that guides the entire research project, from defining the research question and objectives to collecting and analyzing data and reporting the findings. Research design is an important part of the research process as it determines the feasibility and credibility of the study, and it affects the quality and validity of the results. A well-designed research study can provide reliable and accurate results that contribute to the advancement of knowledge in a particular field. Some of the key elements of a research design include the type of research method (e.g., qualitative, quantitative, mixed methods), the sample size and selection process, the data collection methods, the measurement and data analysis techniques, and the ethical considerations. Research design is often confused with the choice of research method - the decision to use either qualitative or quantitative methods. Choosing which methods to use is an important part of the research design process, but it is not the only one. Consider research design to have two levels. The first level describes the research's logic, framework, or structure. It is concerned with the overall structure of the research. The second level is concerned with the mechanics of the research, specifically the type of data used. So, the second level is concerned with decisions about how to gather evidence. This is represented in figure 3.1 below. Figure 3.1: Research Design Levels (Chandra, V., Harindran, A. Research Methodology (2017)) Research Process Onion We will use Saunders et al 2019 Research Onion to guide you through Level 1. Then we will move into Level 2, the detail, in the following modules. The research design will depend on various factors. The Research Process Onion, developed by Saunders et al. in 2007, illustrates the numerous decisions that a researcher must make when starting research. Some may begin by focusing on the type of data they plan to collect and the method they will use. However, Saunders et al. argue that researchers should start at the outer layers of the onion and work inward, to create a well-considered and cohesive research approach. The outer layer of the diagram in Figure 3.2 raises the question of which research philosophy to adopt. From there, you work inward, determining the subject of your research approach, which will stem from your philosophy. Next, you consider your research strategy, or general plan for answering your research question, which will determine your choice of research method. Then, as suggested by Saunders et al. (2003), you must consider the time horizon, deciding whether to take a "snapshot" at a specific time (cross-sectional) or to document a "journey" over time (longitudinal). These decisions will be made independently of the chosen research strategy. Figure 3.2 The Research Process 'Onion'. source: Saunders webinar (2020) There are numerous videos that explain the Research Onion. The video linked below provides a clear and thorough explanation. I highly recommend watching it at least once, as it will give you a comprehensive understanding of how all the layers are interconnected. Research Onion Model Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=-dXnfKgWK0g>

## Research Approach

Module 2 Research Approach Icon Progress Bar (browser only) Focus This page introduces the different types of research approach commonly applied so that you can choose the correct kind for your research design: Inductive approach Deductive approach Abductive approach Deciding on your research approach and strategy There are readings and short videos to support your learning. Further Reading Tip Sheet: Deductive, Inductive and Abductive Reasoning What is a Research Approach? A research approach is an overall strategy that a researcher adopts to carry out a study. Inductive and deductive are two broad research approaches that guide the process of formulating research questions, collecting and analyzing data, and making conclusions. Inductive research approach: This approach involves collecting data first and then developing theories or hypotheses based on the data. The process begins with observations and ends with the formation of general principles or theories that explain the data. Example: Imagine that a researcher wants to understand the factors that influence user satisfaction with a new mobile app. They conduct a survey of users to gather data on their experiences with the app, including their levels of satisfaction, the features they use most frequently, and any issues they have encountered. Based on the data collected from the survey, the researcher identifies patterns and trends in user behavior and satisfaction. They then develop a set of hypotheses about the factors that contribute to user satisfaction, such as the app's design, user experience, and technical performance. Finally, the researcher uses additional methods, such as user testing and observational studies, to validate their hypotheses and gain a deeper understanding of the factors that influence user satisfaction. The process is iterative, with the researcher refining and adjusting their theories as they gather more data and insights. This approach is inductive because it starts with observations and data and proceeds to the formulation of hypotheses and theories. The goal is to gain a deep understanding of the factors that influence user satisfaction and to generate new insights into how to improve mobile app design and development. Deductive research approach: This approach involves formulating theories or hypotheses first and then collecting data to test them. The process begins with general principles or theories and ends with specific conclusions that confirm or disconfirm the original hypotheses. Example: Imagine that a researcher has a hypothesis that a certain type of machine learning algorithm, decision trees, can be used to improve the accuracy of predictions made by predictive models in a certain domain (e.g. stock prices). The researcher begins by reviewing existing literature on decision trees and predictive modeling to gain a deeper understanding of the relevant concepts and methods. They then develop a set of hypotheses about the relationship between decision trees and predictive accuracy. Next, the researcher designs a controlled experiment to test their hypotheses. They use a dataset of historical stock prices and apply both decision tree algorithms and other types of predictive models to make predictions. They then compare the accuracy of the predictions made by each model and use statistical methods to test their hypotheses about the relationship between decision trees and predictive accuracy. This approach is deductive because it begins with a set of hypotheses and uses data to confirm or disconfirm those hypotheses. The goal is to test a specific theory or idea and to generate new knowledge that supports or refutes the original hypothesis. By using a systematic, controlled approach, the researcher can ensure that their findings are reliable and rigorous, and that they can make confident conclusions about the relationship between decision trees and predictive accuracy in the stock price domain. Both inductive and deductive approaches have their own strengths and weaknesses. Inductive research is often considered more flexible and open-ended, while deductive research is considered more structured and

focused. In practice, many research projects use a combination of both approaches. Source: Adapted from Burney (2008:6) Understanding one's assumptions and philosophical position is an important step in choosing the appropriate research approach. According to Abbott and McKinney (2012), the Wheel of Science has two starting points: deductive and inductive. These two approaches are often used as a starting point for the research process and inform the choices that researchers make about their methods and techniques. By reflecting on their own assumptions and philosophical position, researchers can make informed decisions about the approach that best aligns with their research questions and goals. This helps to ensure that the research is rigorous, relevant, and meaningful, and that the findings can be used to advance knowledge and understanding in the field. In addition to deductive and inductive reasoning, there is a third type called abductive reasoning, also known as "taking your best shot." This approach starts with an incomplete set of observations and finds the most likely explanation based on the available information. We use abductive reasoning frequently in daily decision-making, where we make the best decision based on the information available, even if it is not complete. Source: <https://5mingrowth.com/3-learning-art-design-and-reasonings-28911c97ee72> Example of an Abductive Approach: Imagine that a computer scientist is trying to diagnose an issue with a complex software system that is causing the system to crash frequently. They have a general understanding of how the system works, but do not have enough information to determine the root cause of the problem. The computer scientist begins by gathering data on the system's behavior and performance, including log files, error messages, and other relevant information. They use this information to identify patterns and trends in the system's behavior. Next, the computer scientist generates several possible explanations for the problem based on their observations and the information they have gathered. They then use additional tools and techniques, such as simulation and debugging, to test each explanation and see if it fits the available data. Based on the results of their tests, the computer scientist selects the explanation that is most likely to be the root cause of the problem. They then use this explanation to guide their further investigation and to develop a solution to fix the issue. This approach is abductive because it starts with an incomplete set of observations and proceeds to the likeliest possible explanation based on the available information. The goal is to find the best explanation for a problem or issue, given the available data and constraints. By using a systematic, iterative process, a computer scientist can generate new insights and solutions to complex problems and ensure that their findings are reliable and meaningful. Deduction, Induction, Abduction Watch the following two videos to gain more insights into deductive, inductive and abductive approaches to research: Choosing your Approach It is most likely that you are only going to choose between an Inductive or a Deductive approach. Please watch the video below, which contrasts both approaches and considers how you should make your decision. Think about which is more likely to fit your research topic. Research Strategy Once you have decided on your approach, you then need to look at which strategies best fit with the structure. The figure below shows some strategies that fit with different approach choices. We will return to discuss strategies more in a later module page. Figure 3.4 Research Approaches and Associated Strategies. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=-nn3XMoPC7s>

- [https://www.youtube.com/watch?v=ZSj\\_Y6O9m8w](https://www.youtube.com/watch?v=ZSj_Y6O9m8w)
- [https://www.youtube.com/watch?v=\\_T2BmiHEVOI](https://www.youtube.com/watch?v=_T2BmiHEVOI)

### Links & References

- [Link] [Tip Sheet: Deductive, Inductive and Abductive Reasoning](#)
- [Link] <https://www.butte.edu/departments/cas/tipsheets/thinking/reasoning.html>

## Research Methodologies

Module 2 Research Methodologies Icon Progress Bar (browser only) Focus This page introduces you to the concept of research methodologies. You will learn what a research methodology is, how it differs from "methods", and consider how one fits in with the different purposes research can take. There are readings and short videos to support your learning. Further Reading Chapter three: Framing Research What is a Research Methodology? Let's begin by asking the fundamental question of what exactly a research methodology is: Research methodology is the systematic, theoretical analysis of the methods applied to a field of study. It comprises the theoretical analysis of the body of methods and principles associated with a branch of knowledge. This can include the design of the study, the sample selection process, data collection methods, data analysis techniques, and the interpretation of results. The methodology should be chosen based on the research question, the type of data being collected, and the goals of the study. In computer science, research methodology refers to the systematic and theoretical analysis of the methods used in conducting research in the field of computing. The methodology used in computer science research can vary depending on the specific area of study and the research question being addressed. It is important to understand the difference between the terms 'method' (tools) and 'methodology' (a form of inquiry). In practical research, we acknowledge that the distinction between method and methodology can become blurred. Although we present method as a means of doing research and methodology as a way of thinking about research design, some argue that much of research is about method (Wolcott, 1990). Nevertheless, being able to differentiate between the two will enhance your research skills. Framing Research Read chapter three, "Framing Research", from Research Methods and Applications for Student Affairs . This will give you an understanding of Methodology and research methods and types. It will also further explore how your research methodology, or perspective, connects to the research decisions. What is Research Methodology? Now watch the video below to learn why it is important for you to know the difference between Research Methodology and Research Methods. Types of Research Research can be classified in terms of its purpose. All research can be broadly classified into three groups: exploratory research, descriptive research and explanatory or causal research. Just as you may employ different strategies in your research, you may also have more than one purpose. Your research methodology will be influenced by the purpose of your research. Nature of Research Watch the video below and start thinking about the nature of your research topic. Types of Research To summarise your understanding of this topic, read section 2.2.51 Types of Research for more about the three groupings of research identified above: Exploratory, Descriptive and Explanatory or Causal Research. Which of these three kinds do you think will suit your project best? Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=vifGBH-0Wns>
- <https://www.youtube.com/watch?v=FlBFdEgrTBM>

## Links & References

- [Link] [Chapter three: Framing Research](#)
- [Link] <https://www.oreilly.com/library/view/research-methods-and/9781119299707/c03.xhtml#c03anchor-1>
- [Link] [2.2.51 Types of Research](#)

## Research Types

Module 2 Research Types Icon Progress Bar (browser only) Focus This page introduces several further classifications research can take, as determined by the research methodology chosen. There are readings and short videos to support your learning. The Different Types of Research Research can be classified in many different ways on the basis of the methodology of the research. Some of the most common research types are as follows: Pure/Basic research Applied research Quantitative research Qualitative research Research Types To begin, please read Chapter 1.3 Links to an external site. in the chapter "Research Methodology" for definitions of each type of research. The tables below summarise these types. Research Types While basic research explores the "why" and "how" of phenomena, applied research focuses on the "how to" and "what for," aiming to create practical innovations and improvements. Review the table below and then ask yourself the following question: Pure/Basic Research Basic research, also known as fundamental or pure research, is a scientific inquiry driven by curiosity and a quest for knowledge about the underlying principles of the natural world. It seeks to expand our understanding of fundamental concepts and theories without any immediate practical application in mind. Applied Research In contrast, applied research is focused on solving specific real-world problems or addressing practical challenges . It takes the knowledge gained from basic research and applies it to develop practical solutions, products, or technologies that can have direct and tangible benefits in areas such as technology, medicine, engineering, and industry. When you start working through the research process you will probably move between basic and applied research. Research Focus Changing Watch this video for an explanation of how your research focus can change throughout the research process and how this will affect the decisions you make throughout: Qualitative vs Quantitative Research Qualitative and quantitative research are also two distinct and contrasting approaches, often used in social science research. View the table below to find out more: Qualitative Research Qualitative research involves gathering non-numerical data, such as words, images, or observations, to understand the underlying meaning and context of a phenomenon. It often uses methods like interviews, focus groups, or content analysis to explore subjective experiences, perspectives, and social dynamics. Quantitative Research In contrast, quantitative research collects numerical data through structured surveys, experiments, or statistical analysis to identify patterns, relationships, and trends. It aims to measure and quantify variables, making it suitable for hypothesis testing and statistical generalization. While qualitative research delves into the depth and richness of human experiences, quantitative research provides statistical rigour and quantifiable insights into the studied phenomena. Researchers often choose between these methods based on their research objectives and the nature of the research question. These two approaches are not mutually exclusive, however,

and a blend of both can make for a robust study. Quantitative vs Qualitative Research As you watch this video start thinking about what type of research you prefer and why: When you are conducting your research you will often need to decide between using Qualitative or Quantitative research methods. In Module 4 we go into these types more deeply. However, it is good to have an understanding of these terms before we look at Research Strategies on the next page. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=pg8deU3XTls>
- <https://www.youtube.com/watch?v=a-XtVF7Bofg>

### Links & References

- [Link] [Chapter 1.3](#)

## Research Strategies

Module 2 Research Strategies Focus This page introduces the idea of a research strategy and how it defines the actions you will take during your research. There are readings and short videos to support your learning. Research Strategy Your research strategy will be a general plan about how you will go about answering the research question/s you have set. First, we need to clarify what we mean by strategy and how it is different from the methods you will use to collect your data. The strategy is concerned with the overall approach you adopt whereas the methods are more concerned with the tools you will use to collect the data. Common strategies are: Experiments Surveys Case studies Grounded theories Ethnography Narrative Action research Cross-sectional and longitudinal studies These research strategies can be grouped into the categories of Qualitative or Quantitative research. Which of the above strategies do you think are Qualitative strategies? Write out your answers, then check them below: Answers Case studies: Qualitative, as they involve in-depth exploration of a single subject, organisation, or phenomenon, often using various qualitative data sources. Grounded theories: Qualitative, as they aim to develop theories from qualitative data by systematically analysing and interpreting patterns. Ethnography: Qualitative, as it involves immersion in a cultural or social context, relying on participant observation and rich, descriptive data. Narrative: Qualitative, as it focuses on the storytelling aspect of personal experiences or events. Action research: Can be both qualitative or mixed methods, depending on the approach taken. It often involves collecting both quantitative and qualitative data to inform action and drive change. Which of the above strategies do you think are Quantitative strategies? Write out your answers, then check them below: Answers Experiments: Typically quantitative as they involve manipulating variables and measuring the effects using numerical data. Surveys: Quantitative, as they gather structured, numerical data from a large number of participants. Cross-sectional and longitudinal studies: Quantitative methods that collect numerical data to analyse trends and relationships over time. Which Research Strategy? Go through this series of informative videos on research types/strategies. While watching, start to think about making a decision on your research strategy for your research proposal. The first video below describes what you need to think about when choosing research strategies to use: The following videos will introduce you to specific strategies. It is important for you to consider all potential strategies which you could

use to conduct your research, so view each one below. Experiment Strategy Survey Strategy Case Study Strategy Narrative Strategy Grounded Theory Strategy Action Research Strategy Now that you have heard a little about these types of research strategies, have a think about which strategies appeal to you and how they could be useful to your research project. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=kqkVStFLzL4>
- <https://www.youtube.com/watch?v=piuiBvUTXLM>
- [https://www.youtube.com/watch?v=ru\\_GHuJdInU](https://www.youtube.com/watch?v=ru_GHuJdInU)
- <https://www.youtube.com/watch?v=ectSlote8uA>
- <https://www.youtube.com/watch?v=uWeCzxZr7II>
- <https://www.youtube.com/watch?v=NTgiRQqa3m0>
- <https://www.youtube.com/watch?v=Ov3F3pdhNkk>

## Lab 1: Research Design

IT8101 Lab 1: Research Design Overview In this lab, you will explore how philosophy drives research and is essential to formulating a research design. First you must do the reading linked below, and then follow the tasks outlined in the instructions. Source Text Biddix et al., Research Methods and Applications for Student Affairs , (2018). Chapter 3: Framing Research Instructions In this chapter, the authors give examples of how a particular philosophy drives research. They explain the structure Research Perspective/Philosophy>Type>Design>Method and how these four components work together to form the methodology of a research project. They gave the three examples below - let's take another look: Note: You may have realised that the words used to describe each step in Biddix et al are different to those used by Saunders et al. This is often the case in Research. In example 1, we have added in Saunders et al. 2019 terminology. Example 1 Broido, Brown, Styles, and Bronkema (2015) explored changes in the experiences of women who worked at one university for 25–40 years. Data sources included interviews with 28 classified, administrative, and faculty women and documents related to aspects of their gendered experiences such as employee handbooks and policies. Findings focused on how gender manifested in the institution and how participants responded. Perspective (Philosophy) > Constructivism (Feminist Epistemology) Type (Methodological choice) > Qualitative Design (Strategies) > Case Study Methods (Techniques and procedures) > Interviews and Document Analysis Example 2 Mayhew, Seifert, Pascarella, Nelson Laird, and Blach (2012) sought to understand how institutional type affected moral reason and development of first-year students. Data sources were responses to a survey given to first-year college students ( n = 3,081) at 19 four-year colleges and universities from across the United States in the fall and spring of their first year. Findings suggested moral development was associated (but not conditional) with specific institution types and that researchers should explore other institutional variables when examining moral reasoning. Perspective > Positivism Type > Quantitative Design > Quasi-experimental (Pre/post-design) Methods > Difference (ANOVA) and prediction (multilevel modelling) Example 3 Barnhardt (2015) studied the relationship between student capacity for democratic participation and the organizational context of collective action. Data sources included national data from a random sample of 149

postsecondary institutions followed by a content analysis of 683 newspaper articles. Findings suggested that distinctive institutional characteristics, including required diversity courses and enrolling large numbers of students in area studies degree programs, exercised a supportive role in fostering student collective action. Perspective > Pragmatism Type > Mixed Methods Design > Embedded Sequential Methods > Prediction and Document Analysis Instructions 1. Briefly describe a research scenario of your choice. 2. Choose two different philosophies/perspectives 3. For each one, go through the steps of the Research Onion model identifying which approaches and strategies would be linked to your chosen philosophy and how they could relate to your research topic. 4. Complete a summary of the 4-part structure for each of your philosophies/perspectives that matches the examples given above [using the components 1. Perspective, 2. Type, 3. Design, 4. Methods] 5. Write 2-3 paragraphs comparing the affect the different choices at each stage will have on your research outcomes. Select which one you feel would be best suited for your research agenda. 6. Send your work for this lab to your lecturer to review and provide formative feedback. Suggested Support We highly recommend watching the webinar by Dr Mark Saunders on research philosophies, approaches and strategies. It is long (2:34:52 mins), so watching it is entirely optional. It is very useful, however, for thinking about the philosophies that drive research. Dr Saunders Webinar

#### Videos

- <https://www.youtube.com/watch?v=b3kKFNOwOY>

#### Links & References

- [Link] [Chapter 3: Framing Research](#)

## Module 3: Your Research

### Introduction to Module 3

Module 3 Your Research: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Three. In this module you will focus on arriving at your independent research topic. You will learn the best ways of narrowing down a broad topic to a defined research question. You will consider how to plan a research design that is best suited for your research agenda and will think carefully about what sort of data you will use and how it will be collected. Objectives At the completion of this module, you will be able to: Understand how to narrow down a research topic into a defined question Design your research according to an identified design methodology and philosophy Identify types of research data Choose appropriate forms of data collection for your research design

### Finding your Topic

Module 2 Finding Your Topic Focus This page introduces the process behind finding your research topic: from the general topic of interest up to the definition of a final research question. There are readings and short videos to support your learning. Further Reading From Topic to Research Question Getting your topic sorted Research Problem Before you can define the research question, you need to identify the research problem. A research problem refers to the specific issue or topic a researcher wants to investigate and study. It is the question or issue that the researcher wants to explore, analyse, and ultimately answer through their research.

The research problem should be a clear and concise statement outlining the gap or knowledge deficit in the current research and identifying the specific research questions to address. A well-defined research problem is crucial for producing meaningful and valuable research outcomes.

**Research Question** The research question guides the study and is specific, focused, and answerable. The research question is typically derived from the research problem and helps narrow the study. It provides a clear and concise direction for the research and helps ensure the study is relevant and feasible. The research question should be specific and answerable and avoid being too broad or too narrow. It should be designed to provide insight into the research problem and address the research gap identified in the literature review: The importance of a good research question cannot be overstated in the research process. A well-crafted research question is the cornerstone of a successful research endeavor, guiding researchers toward valuable insights and impactful outcomes. It plays an important role in refining the scope and aligning research objectives. Clarity, feasibility, relevance, and significance are all features that distinguish a strong research question.

**The Path to a Research Question** Narrowing down a broad research topic to a specific question is an essential step in the research process. Here are some useful steps for how to go about it:

1. Identify your general area of interest Begin by brainstorming the topic you want to investigate. Consider your interests and any questions you may have.
2. Conduct preliminary research Conduct a preliminary search of the existing literature on your topic to get a sense of what is out there and to identify gaps in the literature.
3. Refine your research question After you have identified your general area of interest, narrow it down to a specific research question. Your research question should be specific, clear, and able to be answered.
4. Consider your research goals Consider what you want to accomplish with your research. Is your goal to close a gap in the literature, test a hypothesis, or investigate a specific phenomenon?
5. Consult with experts Consult with experts in the field to get feedback on your research question and ensure that it is feasible and relevant.
6. Finalise your research question After taking into account all the above factors, finalise your research question. Make it specific enough to guide your research while also being broad enough to allow for interesting discoveries.

**Example: Climate Change Research** Take a look at how the steps listed above have been applied by a researcher wanting to conduct research into climate change.

1. Identify your general area of interest Climate change and its impact on the environment and society.
2. Conduct preliminary research Read academic papers, news articles, and reports on climate change to gain a general understanding of the topic and identify gaps in the literature.
3. Refine your research question How has climate change affected agriculture in New Zealand?
4. Consider your research goals The goal is to understand the specific impacts of climate change on agriculture in New Zealand and identify potential adaptation strategies.
5. Consult with experts Consult with agricultural scientists and climate change experts to ensure that the research question is relevant and feasible.
6. Finalise your research question How has the changing climate affected milk yields and the overall productivity of the dairy industry in New Zealand, and what are some effective adaptation strategies for farmers?

By following this process, we have gone from a broad topic (climate change) to a specific research question that is focused on a particular aspect of the topic (climate change's impact on milk production in New Zealand). This specific research question allows us to conduct a more targeted and meaningful study that can contribute to the existing literature on the topic.

**Your Topic** Now it is time to start putting your knowledge into action. You may have already decided on your Topic. If not, brainstorm some general subjects of interest before selecting the one that interests you the most. Now it is time to narrow your

chosen topic down into something you can use to form your research question. As well as the general steps outlined above, there are several frameworks you could choose to adopt to help you on this process to narrow your research interests. Follow the link below to a helpful resource that guides you through several of these frameworks, using IT subjects as examples: Narrowing Down Frameworks Just for Fun! Once you have your question, and before you embark any further into the design of your research, it is good to remember some research essentials and hear some top tips. Watch this helpful video, comparing the process of research to cooking, to learn how to set up good research habits. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=r9fkSzdLf7M>

### Links & References

- [Link] [From Topic to Research Question](#)
- [Link] [Getting your topic sorted](#)
- [Link] <https://www.baeldung.com/cs/design-research-problem>

### Activity: Narrowing down your Topic

**Activity Narrowing down your Topic** There is no universal set of criteria for a good research question. Different disciplines have different priorities and requirements. A good research question for a history paper will differ from a good research question for a biology paper. In general, however, a good research question should be: Clear and focused. In other words, the question should clearly state what the writer needs to do. Not too broad and not too narrow. The question should have an appropriate scope. If the question is too broad it will not be possible to answer it thoroughly within the word limit. If it is too narrow you will not have enough to write about and you will struggle to develop a strong argument (see the activity below for examples). Not too easy to answer. For example, the question should require more than a simple yes or no answer. Not too difficult to answer. You must be able to answer the question thoroughly within the given timeframe and word limit. Researchable. You must have access to a suitable amount of quality research materials, such as academic books and refereed journal articles. Analytical rather than descriptive. In other words, your research question should allow you to produce an analysis of an issue or problem rather than a simple description of it source: <https://www.monash.edu/rlo/research-writing-assignments/understanding-the-assignment/developing-research-questions> **Activity Task** It is likely that when you find a topic you will have a million ideas about what you would like to research. So, it is useful to have a process to use to narrow down your topic. Work through the following steps, devised by the education team at Monash University in Melbourne Australia, to help you refine your research topic. There is an example of the exercise in the table below. Create three columns on a piece of paper, in a word document, or in a spreadsheet. Select a broad topic or choose a topic that you are interested in. In the first column write down the items contained in the first column below. Add any other items that may be relevant to your topic. In the second column write down potential sub-topics and other limitations. If you get stuck use the examples in the second column below to guide you. In the third column write the potential value of what you have

written in the second column. What would that sub-topic or other limitation allow you to argue or demonstrate? Circle or highlight the items in column two that have the strongest potential value. If you get stuck use this example below to guide you. Send a copy of your table to your lecturer to provide feedback.

### Links & References

- [Link] <https://www.monash.edu/rlo/research-writing-assignments/understanding-the-assignment/developing-research-questions>

## Deciding Your Research Design

Module 3 Deciding Your Research Design Icon Progress Bar (browser only) Focus This page asks you to consider applying what you have learned about research design to the design of your own research. There are readings and short videos to support your learning. Recap on Research Design Let's summarise what we have learned so far about research design, before applying it to your own work: Research design is a plan to answer your research question. A research method is a strategy used to implement that plan. Research design and methods are different but closely related because good research design ensures that the data you obtain will help you answer your research question more effectively. The design and methods you will use depend on your research goal. It depends on what subjects (and who) you want to study. Let's say you are interested in studying what makes people happy, or why some students are more conscious about recycling on campus. To answer these questions, you need to make a decision about how to collect your data. The most frequently used methods include: Observation / Participant Observation Surveys Interviews Focus Groups Experiments Secondary Data Analysis / Archival Study Mixed Methods (combination of some of the above) One particular method could be better suited to your research goal than others, because the data you collect from different methods will be different in quality and quantity. For instance, surveys are usually designed to produce relatively short answers, rather than the extensive responses expected in qualitative interviews. What other factors should I consider when choosing one method over another? Time for data collection and analysis is something you want to consider. An observation or interview method, the so-called qualitative approach, helps you collect richer information, but it takes time. Using a survey helps you collect more data quickly, yet it may lack details. So, you will need to consider the time you have for research and the balance between strengths and weaknesses associated with each method (e.g., qualitative vs. quantitative). source <https://guides.lib.vt.edu/researchmethods/design-method> Research Design Before you continue any further, we recommend watching the following video to check you are clear on the differences between research design, research methodology and research methods. Example of Research Design for a Computer Science Project Here's an example of a 3-step process for choosing a research design in the computer science domain. Analyse it carefully, and think about how you can apply a similar strategic step-by-step process to the design of your own research. Remember, the chosen research design should enable the researcher to obtain reliable and valid data that can answer the research question and support the conclusions drawn from the study. Research Question: Does the use of a personalised recommendation algorithm improve the user engagement on an e-commerce website? 1. Research Design Options Experimental Design: A/B testing where half of the users receive personalised recommendations while the other half does not. Quasi-Experimental Design: Observing user engagement on an e-commerce website before and after the implementation of

personalised recommendation algorithm. Survey Design: Conducting a survey to gather data on user engagement and satisfaction with personalised recommendations. 2. Considerations: Experimental Design: Requires a large user base, careful selection of the control and experimental groups, and avoiding biases in the selection process. Quasi-Experimental Design: Lack of control over external factors that may affect user engagement and satisfaction with the website. Survey Design: Risk of response bias and the need for a large sample size to ensure statistical significance. 3. Chosen Research Design: Experimental Design with A/B testing where half of the users receive personalised recommendations while the other half does not. By choosing the experimental design with A/B testing, we can compare the engagement and satisfaction levels of users who received personalised recommendations to those who did not. This design allows for control over external factors that may affect user engagement and satisfaction and provides us with quantitative data to support our findings. However, it is crucial to ensure careful selection of the control and experimental groups and avoid biases in the selection process to ensure the validity of our results. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=eJK1L6E1uFU>

#### Links & References

- [Link] <https://guides.lib.vt.edu/researchmethods/design-method>

### Data in Research

Module 3 Data in Research Icon Progress Bar (browser only) Focus This page introduces the importance of data in research, allowing you to contemplate the type of data you will use for your research, and how you can ethically collect and use the data. There are readings and short videos to support your learning. Further Reading Read Chapter 3 Data Collection Data and its Importance in Research Data are units of information that are collected through some kind of observation. If we want to get more specific, in research data are a set of values of qualitative or quantitative variables about one or more objects or people. We usually see data represented in some kind of a visual like a graph, table or chart. As we have previously seen, it can be qualitative or quantitative, and it can be gathered using a variety of methods, including surveys, interviews, experiments, observations, and existing databases. Data's importance in research cannot be overstated. It provides the evidence required to support or refute a hypothesis, and thus serves as the foundation of scientific enquiry. Researchers can use data to gain insights into phenomena, identify patterns, and test theories. It serves as the foundation for making informed decisions and recommendations in a wide range of fields. Furthermore, reliable and valid data is critical for establishing trust in research findings and ensuring their acceptance by the scientific community and the general public. As a result, rigors and transparent data collection and analysis are critical to producing high-quality research that contributes to our understanding of the world around us. Data in Computer Science In computer science research, data can be broadly classified into two types: quantitative and qualitative data. Quantitative Data: This type of data is numerical in nature and can be analyzed using statistical methods. It includes data such as measurements, counts, and ratings. Examples of quantitative data in

computer science research include the number of users visiting a website, the time taken to execute a program, or the number of errors encountered during testing. This type of data is useful for testing hypotheses and making predictions. Qualitative Data: This type of data is non-numerical in nature and is often gathered through methods such as interviews, focus groups, or observations. It includes data such as opinions, perceptions, and attitudes. Examples of qualitative data in computer science research include user feedback on a website or app, software developer's perceptions of a programming language, or the factors influencing the adoption of a new technology. This type of data is useful for gaining a deeper understanding of the context and motivations underlying the research question. Working with Data To derive meaningful insights that can inform the research conclusions, the collected data must be organised, analysed, and interpreted. Descriptive statistics, inferential statistics, and qualitative analysis are examples of data analysis techniques. Data must be collected, analysed, and reported transparently to ensure the validity and reliability of research findings. It's crucial to be mindful of the privacy laws and regulations governing the collection, utilization, and reporting of data. As researchers contemplate collecting, processing, or using data, they must assess whether their actions are ethical. However, answering the question of ethicality as a simple "yes" or "no" isn't always sufficient, as there can be a range of complexities and nuances that require careful consideration. Data Analysis According to LeCompte and Schensul, research data analysis is a process used by researchers for reducing data to a story and interpreting it to derive insights. The data analysis process helps in reducing a large chunk of data into smaller fragments, which makes sense. Three essential things take place during the data analysis process — the first data organization. Summarization and categorization together contribute to becoming the second known method used for data reduction. It helps in finding patterns and themes in the data for easy identification and linking. Third and the last way is data analysis – researchers do it in both top-down or bottom-up fashion. Source:

<https://www.questionpro.com/blog/data-analysis-in-research/> Ethics in Data Science When we are collecting, using or reporting on data we need to be very aware of privacy act, regulations and laws that govern the use of that data. When considering the capture, processing, or use of data, researchers need ask themselves “Is this ethical?” The implication in the framing of the question is that the answer is either “yes” or “no.” In many cases, that is fine, but in many more cases, the answer is not nearly as cut and dried as the question implies. In the book 97 things about Ethics everyone in Data Science should know, Bill Franks has collected a cross-section of short articles concerning ethics in Data Science. He suggests that in the trend where analytical processes are being fully automated Humans are increasingly suspicious about what is happening to those processes. The other trend he notes is the rise of artificial intelligence (AI) and its increasing sophistication and the impact this has on our lives. Franks states: you can't wait for someone else to take the lead when it comes to ensuring that your analytics and data science activities are ethical. You must step up and help take the lead yourself! Be intentional about considering ethics every day, and you will sleep well at night while protecting the integrity and reputation of both yourself and your organization. Source: [https://learning.oreilly.com/library/view/97-things-about/9781492072652/](https://learning.oreilly.com/library/view/97-things-about/9781492072652/preface01.html#how_the_book_is_organized)

Activity: Ethical Data Preparation: 1. From Bill Franks's collection of articles, read Ethics are "Fuzzy". Then, choose another couple of articles from the book that relate to your area of research. 2. Consider the tikanga or principle Tapu, which stands for sensitivity and risk. It's about balancing benefits and risks. Tapu can be described as follows: Source: Nga-Tikanga-Paihere-Guidelines-December-2020 Task 1. Create a

mindmap that brainstorms the ways you can ensure that the type of data, its collection, its analysis, its storage and its use are ethical. 2. Your mindmap must address the following ethical data questions that relate to Tapu : sensitivity and risk: How will you address the privacy issues around data collection and data reporting when conducting your research? What processes will you need to put in place to raise awareness of who the researcher is? How will you inform the public what the research is about and what the information collected will be used for? How will you address any concerns related to your research? To answer this, identify at least three concerns that might arise from your research and then provide information about the process and actions you would put in place to address each of them. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [Chapter 3 Data Collection](#)
- [Link] [https://www.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter003.xhtml#ch3\\_1](https://www.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter003.xhtml#ch3_1)
- [Link] <https://www.questionpro.com/blog/data-analysis-in-research/>
- [Link] [privacy act](#)
- [Link] [97 things about Ethics everyone in Data Science](#)
- [Link] [Ethics are "Fuzzy"](#)
- [File] [Nga-Tikanga-Paihere-Guidelines-December-2020](#)

### Lab 2: the Role of the Researcher

IT8101 Lab 2: the Role of the Researcher Summary As a researcher you will be involved in working with different communities. This Lab is about understanding the groups you may be working with to conduct your research and the appropriate research protocols for working with that community. We will be looking at qualitative research methods as they usually require more in-depth relationships than quantitative methods. However, in practice it would be wise and professional to engage with your research audience in the same way no matter what the research methods. Completing this Lab is expected to take you a couple of hours. You may do it alone or work with one or two other students to complete the tasks. Reading Read these four articles about conducting qualitative research with specific communities in mind. NGO Qualitative Research Methodology Download NGO Qualitative Research Methodology Qualitative Research: Getting Started Download Qualitative Research: Getting Started Kaupapa Māori research excerpt - Leonie Pihama Links to an external site. Old ways of having new conversations - basing qualitative research within Tikanga Māori protocols to an external site. As a researcher it is important to be guided by good principles. In the qualitative research, the researcher will be engaging closing with a section of the community. 1. Note down the advice given about engaging with research participants in each article. Then notice the similarities and differences in between the notes for each article. 2. The last two readings are about conducting research with Māori. What are some of the ways the researchers whakapapa, develop relationships with the Māori research participants? 3. In the article Qualitative Research: Getting Started, the section 'the role of the researcher', 4 questions are identified that are important for a researcher to answer. These are: Why am I interested in this topic? What do

I think the answer is (to my research question)? What am I going to get out of it (doing the research)? What do others in my professional community thinking of this work and of me? Instructions Take the topic that you would like to research and the hypothesis or research question/s you are looking to answer. Then, think about an audience you would like to work with. Write the answers to the questions in a way that you can share with your research participants. Send your answers to your lecturer to review and provide feedback.

### Links & References

- [File] [NGO Qualitative Research Methodology](#)
- [File] [Download NGO Qualitative Research Methodology](#)
- [File] [Qualitative Research: Getting Started](#)
- [File] [Download Qualitative Research: Getting Started](#)
- [Link] [Kaupapa Māori research excerpt - Leonie Pihama](#)
- [Link] [Old ways of having new conversations - basing qualitative research within Tikanga Māori protocol](#)

## Module 4: Literature Review

### Introduction to Module 4

Module 4 Literature Review: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Four. In this module you will learn everything you need to know about preparing a literature review. This will be important for Assessment 1, in particular. A literature review is an important part of any research project, as it sets your research in context and identifies how it fits with the research that has been done before. In this module, you will learn some best practices for reading and analysing existing literature, structuring your findings, and preparing the review. Objectives At the completion of this module, you will be able to: Understand why a literature review is important Know how to structure an literature review Use critical thinking to analyse what you are reading Be able to identify what makes a good literature review

### Literature Review: Introduction

Module 4 Literature Review: Introduction Focus This page introduces you to the literature review and explores why they are important. There are readings and short videos to support your learning. Further Reading Read the section Why write a literature review? Read 2.24 Objectives of the Literature Review to understand the four important objectives you need to accomplish in preparing your literature review. What is a Literature Review? Most of you will be familiar with literature reviews and will have completed them for other courses already in your studies. They are especially important for larger, independent research projects, and so understanding their role and function is critical. A literature review is an important part of any research project, as it sets your research in context and identifies how it fits with the research that has been done before. You may be asked to write a literature review as part of a dissertation, thesis, or longer project, or as a separate assignment to develop the research, synthesis and analytical skills involved. A literature review involves two parts: A Process that involves you in sourcing, reading, organising, and analysing the literature, and A Final Product that communicates your understanding and your interpretation of that literature. Continue by

watching the video below for a helpful summary of the Literature Review: As we go through this topic, we will follow more of the Scribbr series of videos on Literature Reviews. Make sure you take the time to watch them and make notes. They are practical with step-by-step instructions to help you get your literature review started and completed. Why write a Literature Review? New discoveries do not materialise out of nowhere; they build upon the findings of previous experiments and investigations. A literature review shows how the investigation you are conducting fits with what has gone before and puts it into context. A literature review demonstrates to your reader that you are able to: Understand and critically analyse the background research Select and source the information that is necessary to develop a context for your research It also: Shows how your investigation relates to previous research Reveals the contribution that your investigation makes to this field. Does it fill a gap? or Build on existing research? Provides evidence that may help explain your findings later Reasons to write literature review To provide context : A literature review sets the stage for a research project by highlighting the existing knowledge in the field. It provides background information and helps the reader understand the significance of the research question and objectives. To identify gaps in knowledge: By reviewing the literature, researchers can identify areas where further research is needed. This can help to generate research questions and hypotheses that will advance the field. To evaluate previous research: A literature review allows researchers to critically evaluate the quality and validity of previous studies. This can help to identify any limitations or weaknesses in the existing literature and suggest ways to overcome them. To synthesise existing knowledge: A literature review helps to synthesise the key findings of previous studies and identify patterns and trends in the research. This can help to establish a theoretical framework for the research project. To establish credibility: A literature review demonstrates that the researcher has a thorough understanding of the existing literature and has considered it carefully when designing the research project. This can enhance the credibility of the study and its findings. If you are doing a thesis, dissertation, or a long report, it is likely that you will need to include a literature review. If you are doing a lab write-up or a shorter report, some background reading may be required to give context to your work, but this is usually included as an analysis in the introduction and discussion sections. A Literature Review will help your Research! Conducting a literature review adds significant value to your research. It does the following important jobs: Provides context: It establishes the current state of knowledge and sets the stage for the research project by highlighting gaps, inconsistencies, and contradictions in the literature. Identifies research questions: A literature review can help identify research questions that have not yet been answered or need further exploration. It can also help refine research questions by clarifying the current state of knowledge in the field. Enhances research quality: A literature review ensures that the research project is well-informed and builds on prior research. It can help to avoid redundant or unnecessary research and ensure that the study is relevant and impactful. Identifies research methods: A literature review can also inform the choice of research methods and design. It can help identify appropriate data collection and analysis methods and establish a theoretical framework for the research. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=KkAnKGuX7fs>

## Links & References

- [Link] [Why write a literature review?](#)
- [Link] [2.24 Objectives of the Literature Review](#)

### Getting Started with the Literature Review

Module 4 Getting Started with Literature Review Focus This page provides information on how to start a literature review. The topics covered include: Getting started Finding sources Making decisions The Tika Principle There are readings and short videos to support your learning. Further Study Scribbr - Literature Review How to get started on a Literature Review Take a look at the diagram below, which shows a cyclical process for producing a literature review, made of 7 steps. In this page, we will be focusing on the first 4 steps. Figure 5.1: Seven Steps to Producing a Literature Review. How do get started with a Literature Review? Once you have identified your topic and some research questions, you need to find out what has already been discovered and researched. Start by identifying what you will need to know to inform your research: What research has already been done on this topic? What are the sub-areas of the topic you need to explore? What other research (perhaps not directly on the topic) might be relevant to your investigation? How do these sub-topics and other research overlap with your investigation? Note down all your initial thoughts on the topic. You can use a spidergram or list to help you identify the areas you want to investigate further. It is important to do this before you start reading so that you do not waste time on unfocused and irrelevant reading. A literature review takes time so you need to plan out how you will get to the finished review within the necessary timeframe. Think about whether this subject spans multiple disciplines. If so, consider finding sources outside from your discipline to broach the subject, if relevant. Searching for Sources It is easy to think that the best way to search for texts is to use the Internet - to 'Google it'. There are useful online tools that you may use, like Google Scholar. However, for most literature reviews you will need to focus on academically authoritative texts like academic books, journals, research reports, government publications. Searching Google will give you thousands of hits, few of them authoritative, and you will waste time sorting through them. A better idea is to use databases to search for journal articles: They normally have the most up-to-date research and you will be expected to refer to them in your literature review. Another good way to find sources is to look at the reference lists in articles or books already identified as relevant to your topic. You will be expected to prioritise recent research, but it is also important to acknowledge the standard texts in your field. An easy way to identify these is to check reference lists to see which texts are frequently cited. There are several popular sources for finding literature reviews in the computer science domain, including: IEEE Xplore Digital Library ACM Digital Library ScienceDirect SpringerLink arXiv DBLP (Digital Bibliography & Library Project) CiteSeerX These sources provide access to a wide range of academic journals, conference proceedings, and other publications in computer science and related fields, making them valuable resources for conducting a literature review in the computer science domain. What should you do once you find a paper? The flowcharts below are a good resource for understanding how to prioritise, organise and collate your information. Your decision-making process for choosing which texts to include in your literature review should be similar to that shown below: Source: [aviz.fr/literature](http://aviz.fr/literature) As the flowcharts demonstrate, not all texts you find will be useful, relevant, trustworthy, interesting or influential. You will have to make judgements about which articles, research papers, and book

chapters will be most helpful to your research question and can be considered scholarly. Tika: Accurate, Truthful, Fair Sources Using the Tika principle as a reflective tool The Tika principle, which means Value for All, emphasizes accuracy, truthfulness, and fairness. It can be helpful in conducting a literature review. Adhering to this principle can ensure that the information you gather from the literature is accurate and reliable, and that the review is conducted in a fair and unbiased manner. In a literature review, it is essential to search for and analyze relevant sources of information to develop a comprehensive understanding of the research topic. By applying the Tika principle, you can ensure that the sources used are credible and trustworthy, and that any claims made are supported by evidence. This can enhance the reliability and validity of the literature review and its findings. Applying the Tika principle, which emphasises correctness, can help to avoid any potential biases or inaccuracies that may arise during the literature review process. It can ensure that all relevant viewpoints and perspectives are considered and that the review is conducted in an objective and impartial manner. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [Scribbr - Literature Review](#)
- [Link] <https://www.scribbr.com/methodology/literature-review/>
- [Link] [IEEE Xplore Digital Library](#)
- [Link] [ACM Digital Library](#)
- [Link] [ScienceDirect](#)
- [Link] [SpringerLink](#)
- [Link] [arXiv](#)
- [Link] [DBLP \(Digital Bibliography & Library Project\)](#)
- [Link] [CiteSeerX](#)
- [Link] [Using the Tika principle as a reflective tool](#)

### Undertaking your Literature Review

Module 4 Undertaking your Literature Review Icon Progress Bar (browser only) Focus This page introduces some tips for undertaking a literature review, including how to structure and organise your reading. There are readings and short videos to support your learning. Further Reading A guide to reading literature Undertaking your Literature Review When you read and analyse a lot of sources, it is important to have a good system for taking notes and keeping track of what you read. This will make it easier to find what you need later, especially in a larger project or dissertation conducted over a longer period of time. A systematic process will also make it easier to start grouping the literature into themes or key concepts, and help you start to see where authors reach consensus or disagree with each other. In this section we will learn how to structure your reading and start to turn your notes into a coherently planned literature review. Structuring your reading If you have thought about the areas you need to research and have conducted some searches for literature, you should be ready to set down some draft topic headings to structure your literature review. Select one of your headings and choose a few key texts to read first - three is ideal to start with. Remember that you may eventually be writing about the same text under different headings, so bear that in mind when you are reading and

making notes. When you have finished reading your chosen texts, write a draft section summarising and commenting on what you have read, taking special care to show how it is relevant to your research. Then look to see what you need to discuss further, and do more reading to enable you to plug the gaps. Is reading not your strong point? Reading critically is a skill. If you feel daunted, there are helpful resources out there that provide some great tips for reading mindfully and productively. The University of Reading Library, for example, has compiled very useful guides and videos that help you to build the critical analysis skills that are key to doing a good literature review: [Managing Academic Reading](#) When to stop reading? You should be guided by how long your literature review needs to be - it is no good reading hundreds of texts if you only have 1,000 words to fill. Try to set limits on how long you will spend reading. Then plan backwards from your deadline and decide when you need to move on to other parts of your investigation, e.g., gathering the data. You need to show you have read the major and important texts in your topic, and that you have also explored the most up-to-date research. If you have demonstrated both of these, you are on the right lines. If you keep coming across very similar viewpoints and your reading is no longer providing new information, this is a sign you have reached saturation point and should probably stop. Be guided by your research questions. When reading, ask yourself, "How does this relate to my investigation?" If you are going off into unrelated areas, stop reading and refocus on your topic.

**Organising Literature Review with a Spidergram** A spidergram, also known as a mind map or concept map, is a visual tool that can be used to organise a literature review. Here are the steps to organise a literature review using a spidergram: Choose a central topic or research question for your literature review. Write the central topic or research question in the centre of the spidergram. Identify key themes, concepts, or subtopics related to the central topic and write them as branches radiating out from the centre of the spidergram. Identify relevant literature for each of the themes, concepts, or subtopics and link them to the appropriate branch on the spidergram. Continue to add additional themes, concepts, or subtopics as needed and link relevant literature to each branch. Using a spidergram to organise a literature review can help to visually display the connections between different themes, concepts, or subtopics, and how they relate to the central topic or research question. It can also help to identify any gaps or overlaps in the literature and guide the development of the literature review. Once the spidergram is complete, it can serve as a useful reference tool throughout the literature review process. [Figure 5.2 Spidergram diagram](#)

Another thing you can do is to group what you have read into different topics or themes. These can provide useful headings when you come to write up your literature review. Use different coloured highlighters to identify which topic or heading each article fits into. [Identify Themes and Gaps in Literature Review](#) Watch this video to hear about some practical steps you can take to identify themes and gaps in the literature: Be selective - you do not have to include everything you have read in your literature review. Only include research which is relevant and which helps you understand more about your own investigation. What you leave out will not be wasted as it helped you refine your understanding of the wider issues and identify what was relevant to your own investigation. You do not have to refer to everything in the same depth in your literature review. You are usually expected to prioritise recent research. Some scientific research that was crucial in the past is now out of date. For instance, there may be a few older studies that were important in starting research in the field, but their methods have been surpassed by more accurate methods. You only need to demonstrate your awareness of these older, dated studies in a few sentences, then move on to discussing in greater depth the up-to-date methods and why they are more accurate. Outline

Your Literature Review's Structure Watch this video for guidance on how to develop an outline for the structure of your literature review. By following the steps outlined, you can ensure that your literature review is comprehensive, well-organised, and provides a solid foundation for your research project. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=hMGdwIhiwzU>
- <https://www.youtube.com/watch?v=pOK1-H8Vafs>

#### Links & References

- [Link] [A guide to reading literature](#)

### Writing your Literature Review

Module 4 Writing Your Literature Review Icon Progress Bar (browser only) Focus This page provides information and tutorials on how to write a literature review. In this section we will cover: Your first literature review writing Analysing literature review Summarising literature review There are readings and short videos to support your learning. Further Reading Writing your Literature Review from start to finish Writing your Literature Review well Scribbr - literature-review Structuring your literature review Writing your Literature Review A key feature of any literature review is how you choose to group the literature into sub-sections or themes to enable comparison. This shows how you are conceptualising the topic. The structure you create helps you (and your readers) navigate and understand the literature. In a longer project, it is normal to refer back to the concepts in the literature review to help analyse your own results and provide potential reasons to explain what you have found. So it is important to set up these concepts clearly, and to explore and evaluate them in the literature review. Like an essay, a literature review has an introduction, main body, and conclusion: Introduction: This explains the broad context of your research area and the main topics you are investigating. It briefly highlights the relevant issues or debates that have characterised your field of research. It should also include some signposting for the reader, explaining the organisation / sequence of topics covered, and the scope of your survey. Main body: An analysis of the literature according to a number of themes or topics that overlap with your research. It may have headings. You can write your literature review one section at a time, but make sure you read through them all to check they link together and tell a coherent "story". This should show how your research builds on what has been done before. Based on previous research, you provide justifications for what you are doing, why you are doing it, and how you are going to do it. Conclusion: This should summarise the current state of the research in your field as analysed in the main body. It should identify any gaps or problems with the existing research, and explain how your investigation is going to address these gaps or build on the existing research. 4 TIPS on Writing a Literature Review Watch this helpful video, which covers the elements needed in the written literature review. It offers useful tips for this crucial writing stage: Analysing the literature critically Your review's Main Body paragraphs need to do some serious work analysing the existing body of knowledge related to your research questions. Simply summarising the literature is not enough; you also need to analyse it in order to evaluate the

strength of the arguments presented in previous research. This analytical work will help you to draw connections between your own research and the existing literature. To develop a strong literature review, it is important to first group and categorise the literature you have read in order to identify the key arguments presented. However, this is just the first step. You also need to critically evaluate each source and offer your own judgment on the validity of the arguments presented. This analysis will help you to draw your own conclusions about how your research relates to previous findings. By conducting a thorough and critical analysis of the existing literature, you will be better equipped to explain how your research fits into the larger context of the field. For example, you may be able to explain why your results are in agreement or disagreement with previous research, and what implications this has for future work in the field.

**Critical or Descriptive?** You can ensure you are analysing critically by testing out your own views against those you are reading about: What do you think about the topic? Then as you read each new study, does the evidence presented confirm your view, or does it provide a counter-argument that causes you to question your view? Also think about the methods used to gather the evidence - are they reliable or do they have gaps or weaknesses? When writing up your literature review, use each of your headings or themes to compare and contrast the differing views put forward in the relevant studies and explain how they relate to your investigation. Your literature review needs to tell an interesting "story" which leads up to how and why you are doing your investigation. If you are writing a story which reads like one thing after another, this is likely to be descriptive. However, if your story is comparing, contrasting and evaluating the previous literature, you are on the right track, creating a critical and analytic review. See the example below:

**Descriptive** Summarises what other people have found without saying what these findings mean for your investigation. Usually a chronological list of who discovered what, and when. Analytical Synthesises the work and succinctly passes judgement on the relative merits of research conducted in your field. Reveals limitations or recognises the possibility of taking research further, allowing you to formulate and justify your aims for your investigation. For example: "Green (1975) discovered ...." "In 1978, Black conducted experiments and discovered that ...." "Later Brown (1980) illustrated this in ...." For example: There seems to be general agreement on x, (see White 1987, Brown 1980, Black 1978, Green 1975). However, Green (1975) sees x as a consequence of y, while Black(1978) puts x and y as .... While Green's work has some limitations in that it ....., its main value lies in ...." Be a Critical Student! Below you will find a helpful video that encourages you to learn to be a critical student. You can apply these tips to your literature review:

**How to Write a Literature Review**

**Define your research question or problem statement:** The first step in writing a literature review is to define the research question or problem statement you want to address. This will help you determine what literature you need to review. For example, let's say your research question is: "What are the latest trends in natural language processing (NLP) techniques for sentiment analysis?"

**Search for relevant literature:** Once you have your research question or problem statement, you can begin searching for relevant literature. Use academic search engines such as Google Scholar or IEEE Xplore to find articles, conference papers, and other sources related to your topic. For our example, you might search for "NLP sentiment analysis techniques" or "latest developments in NLP sentiment analysis".

**Read and evaluate the literature:** Once you have found relevant literature, read each source carefully and evaluate its relevance and quality. Ask yourself questions such as: Does this source directly relate to my research question? Is the information accurate and up-to-date? What are the strengths and weaknesses of the methodology used in this research? For our example, you might evaluate

each source based on how recent it is, how relevant it is to NLP sentiment analysis, and how rigorously the methodology was applied. Organize and synthesize the literature: Once you have read and evaluated the literature, organize it into categories or themes. For example, you might categorize sources based on the NLP techniques used, or based on the types of data analyzed. Synthesize the information from each source and identify key trends or patterns in the literature. For our example, you might synthesize the literature based on the different NLP techniques used for sentiment analysis, such as rule-based, machine learning, or deep learning approaches. Write the literature review: Finally, use your organization and synthesis of the literature to write your literature review. Start with an introduction that provides background information on your research question or problem statement, then describe the categories or themes you identified in the literature. Within each category or theme, summarize the information from each source and identify key trends or patterns. Conclude your literature review by summarizing the major findings and identifying any gaps or areas for future research. For our example, you might write a literature review that summarizes the latest trends in NLP techniques for sentiment analysis, such as the increasing use of deep learning approaches and the need for better training data. Example Introduction Here's a brief example of an Introduction in a literature review on the latest trends in NLP techniques for sentiment analysis. It outlines the background to the topic and summarises key trends and patterns: Natural language processing (NLP) techniques have been widely used for sentiment analysis, the process of identifying and classifying the emotional tone of a piece of text. In recent years, there have been several new developments in NLP techniques for sentiment analysis, including the use of deep learning approaches and the need for better training data. This literature review aims to summarize the latest trends in NLP techniques for sentiment analysis. NLP techniques for sentiment analysis: Rule-based approaches: Rule-based approaches use manually defined rules to classify the sentiment of a piece of text. While these approaches are relatively simple, they are limited by the accuracy of the rules and the difficulty of defining rules for complex emotions. Machine learning approaches: Machine learning approaches use statistical models to learn the sentiment of a piece of text from training data. These approaches can be more accurate than rule-based approaches, but require large amounts of annotated training data. Deep learning approaches: Deep learning approaches use neural networks to learn the sentiment of a piece of text. These approaches can be more accurate than machine learning approaches, but require even larger amounts of training data. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=kW6Uzn-8uMI>
- <https://www.youtube.com/watch?v=YVLjziA5U2o>

#### Links & References

- [Link] [Writing your Literature Review from start to finish](#)
- [Link] [Writing your Literature Review well](#)
- [Link] [Scribbr - literature-review](#)
- [Link] [Structuring your literature review](#)

## Module 5: Developing Research Questions

### Introduction to Module 5

Module 5 Developing Research Questions: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Five. In this module, you will learn how to apply the findings of your literature review to the development of precise research questions. You will explore the role of the research question, as well as different types of research questions and the characteristics of really good ones. Objectives At the completion of this module, you will be able to: Understand the role of research questions Identify what makes a good research question Identify different types of research questions Apply strategies for developing research questions

### What is a Research Question?

Module 5 What is a Research Question? Icon Progress Bar (browser only) Focus This page introduces the research question in more detail. You will explore what these questions are, what they need, and how to begin developing them. There are readings and short videos to support your learning. Further Reading Developing Your Research Question – Ohio State University Libraries FINER Research Framework: Crafting Strong Research Questions - Elsevier Author Services "Improving your Research Question" by the University of Maryland Global Campus "Developing Research Questions by Monash University Writing Research Questions What is a Research Question? A research question is a succinct and clear enquiry that a researcher seeks to answer through their research. It serves as the foundation of any research study. A well-crafted research question should be relevant, feasible, and answerable through research methods, while also being narrow enough to enable in-depth exploration of the topic, but broad enough to be of interest to the research community. The development of research questions is typically driven by gaps or uncertainties in the existing literature, observations or phenomena that require further investigation, or problems or issues that demand resolution. The research question should be targeted towards a specific aspect of the topic and be formulated in a way that facilitates the collection and analysis of data that will aid in answering the question. Illustrative examples of research questions may include enquiries into the impact of social media on mental health, or the identification of factors contributing to employee job satisfaction in the healthcare industry, for example. A research question is a question central to your research. It should be: Clear: it provides enough specifics that one's audience can easily understand its purpose without needing additional explanation. Focused: it is narrow enough that it can be answered thoroughly in the space the writing task allows. Concise: it is expressed in the fewest possible words. Complex: it is not answerable with a simple "yes" or "no," but rather requires synthesis and analysis of ideas and sources prior to composition of an answer. Arguable: its potential answers are open to debate rather than accepted facts. Example What is the impact of using artificial intelligence (AI) techniques on enhancing the accuracy of medical image classification? This research question addresses the use of AI techniques in medical image classification, a topic of significant interest in the field of computer science and healthcare. The question is specific in focusing on the impact of AI on the accuracy of medical image classification, and it is answerable through research methods such as empirical evaluation and comparison of AI-based and non-AI-based approaches. This research question could potentially guide a research study that seeks to develop and evaluate

AI-based algorithms for medical image classification with the aim of improving diagnostic accuracy and patient outcomes. Why is a research question essential to the research process? Research questions help writers focus their research by providing a path through the research and writing process. The specificity of a well-developed research question helps writers avoid the “all-about” paper and work toward supporting a specific, arguable thesis. Developing Research Questions Review the following steps to developing a research question: Choose an interesting general topic. Most professional researchers focus on topics they are genuinely interested in studying. Writers should choose a broad topic about which they genuinely would like to know more. An example of a general topic might be “The use of gadgets in our society” or “Future of artificial intelligence.” Do some preliminary research on your general topic. Do a few quick searches in current periodicals and journals on your topic to see what’s already been done and to help you narrow your focus. What issues are scholars and researchers discussing, when it comes to your topic? What questions occur to you as you read these articles? Consider your audience. For most college papers, your audience will be academic, but always keep your audience in mind when narrowing your topic and developing your question. Would that particular audience be interested in the question you are developing? Start asking questions. Taking into consideration all of the above, start asking yourself open-ended “how” and “why” questions about your general topic. For example, “Why do people use gadgets?” or “How is artificial intelligence affecting our lives?” Evaluate your question. After you’ve put a question or even a couple of questions down on paper, evaluate these questions to determine whether they would be effective research questions or whether they need more revising and refining. Is your research question clear? With so much research available on any given topic, research questions must be as clear as possible in order to be effective in helping the writer direct his or her research. Is your research question focused? Research questions must be specific enough to be well covered in the space available. Is your research question complex? Research questions should not be answerable with a simple “yes” or “no” or by easily found facts. They should, instead, require both research and analysis on the part of the writer. They often begin with “How” or “Why.” Begin your research. After you’ve come up with a question, think about the possible paths your research could take. What sources should you consult as you seek answers to your question? What research process will ensure that you find a variety of perspectives and responses to your question? How to Develop a STRONG Research Question We recommend watching the video below for a summary of all the ingredients necessary for a strong question: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=71-GucBaM8U>

#### Links & References

- [Link] [Developing Your Research Question – Ohio State University Libraries](#)
- [Link] [FINER Research Framework: Crafting Strong Research Questions - Elsevier Author Services](#)
- [Link] ["Improving your Research Question" by the University of Maryland Global Campus](#)
- [Link] ["Developing Research Questions by Monash University](#)

- [\[Link\] Writing Research Questions](#)

## Types of Research Questions

Module 5 Types of Research Questions Icon Progress Bar (browser only) Focus This page introduces the types of research questions that align with either quantitative or qualitative research. You will explore some relevant examples that will be helpful for your own research development. There are readings and short videos to support your learning. Further Reading Research Questions - Types Types of Research Questions In Module 2, we explored the Research Onion and its various layers. This likely heightened your awareness of the numerous decisions required when planning research, decisions that can also impact the types of research questions you generate. The following section, excerpted from an article by Imed Bouchrika in Guide2Research (May 2021), outlines the various categories of questions that research can ask. The table below the list illustrates and contrasts the differences between qualitative and quantitative questions. Keep in mind that you may have more than one research question and they may fit different categories. Different Types Research questions can be classified into different categories, depending on the type of research to be done. Knowing what type of research one wants to do—quantitative, qualitative, or mixed-methods studies—can help in determining the best type of research question to use. Doody and Bailey (2016) suggest a number of common types of research questions, as outlined below. Quantitative Research Questions Quantitative research questions are precise. These questions typically include the population to be studied, dependent and independent variables, and the research design to be used. They are usually framed and finalized at the start of the study (Berger, 2015). Quantitative research questions also establish a link between the research question and the research design. Moreover, these questions are not answerable with “yes” or “no” responses. As a result, quantitative research questions don’t use words such as “is,” “are,” “do,” or “does.” Quantitative research questions usually seek to understand particular social, familial, or educational experiences or processes that occur in a particular context and/or location (Marshall & Rossman, 2011). They can be further categorized into three types: descriptive, comparative, and relationship. Descriptive research questions aim to measure the responses of a study’s population to one or more variables or describe variables that the research will measure. These questions typically begin with “what.” Comparative research questions aim to discover the differences between two or more groups for an outcome variable. These questions can be causal, as well. For instance, the researcher may compare a group where a certain variable is involved and another group where that variable is not present. Relationship research questions seek to explore and define trends and interactions between two or more variables. These questions often include both dependent and independent variables and use words such as “association” or “trends.” Ritchie et al. (2014) and Marshall and Rossman (2011) have also further categorized qualitative research questions into a number of types, as listed below: Contextual research questions seek to describe the nature of what already exists. Descriptive research questions attempt to describe a phenomenon. Emancipatory research questions aim to produce knowledge that allows for engagement in social action, especially for the benefit of disadvantaged people. Evaluative research questions assess the effectiveness of existing methods or paradigms. Explanatory research questions seek to expound on a phenomenon or examine reasons for and associations between what exists. Exploratory research questions investigate little-known areas of a particular topic. Generative research questions aim to provide new ideas for the development of theories and actions. Ideological research questions

are used in research that aims to advance specific ideologies of a position. Qualitative Research Questions Qualitative research questions are open-ended questions that seek to explore and understand a particular phenomenon from the perspective of the participants or the researcher. According to Creswell (2014), qualitative research questions typically aim to explore, describe, or understand a particular phenomenon in detail, and may include questions related to the experiences, perceptions, beliefs, attitudes, or behaviors of the participants. Qualitative research questions are often used in studies that seek to generate new ideas, theories, or hypotheses about a particular phenomenon, or to gain a deep understanding of a particular social or cultural context. They are often open-ended and flexible, allowing the researcher to adapt and modify the research questions as the study progresses. There are different types of qualitative research questions that can be used in qualitative studies, depending on the research goals and objectives. According to Creswell (2014), the types of qualitative research questions include: Phenomenological questions: These questions aim to explore and describe the lived experiences of the participants related to a particular phenomenon. Ethnographic questions: These questions seek to explore and describe the cultural practices and beliefs of a particular group or community. Grounded theory questions: These questions aim to develop new theories or concepts based on the data that emerges from the study. Case study questions: These questions seek to explore a particular case or situation in detail, often with the goal of understanding a particular problem or issue. Narrative questions: These questions aim to explore and describe the personal stories and experiences of the participants. Participatory questions: These questions seek to involve the participants in the research process, often with the goal of empowering them and promoting social change. These different types of qualitative research questions can be used in combination or separately in qualitative research studies, depending on the research goals and objectives. Example The following table illustrates the differences between quantitative and qualitative research questions.

| Topic                   | Quantitative Research Questions                                                                            | Qualitative Research Questions                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Mental health diagnoses | How does race affect rates of mental health diagnosis among teens in foster care?                          | What is the experience of African-American teens seeking help for mental health concerns?               |
| Career choice           | What is the relationship between household income and career choice among American university students?    | How do American university students from low-income households experience making a career choice?       |
| Social media use        | What proportion of Australians aged 55 to 75 use popular social media sites?                               | How do Australians aged 55 to 75 use social media?                                                      |
| Early-stage dementia    | What are the differences in the perceptions towards people with early-dementia before and after diagnosis? | How do people with early-stage dementia experience other people's reactions to their changed condition? |

Source: Guide2Research by Imed Bouchrika, May 2021 <https://www.guide2research.com/research/how-to-write-a-research-question> Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Links & References

- [Link] [Research Questions - Types](#)
- [Link] <https://research.com/research/how-to-write-a-research-question>

## Writing good research questions

Module 5 Writing Good Research Questions Icon Progress Bar (browser only) Focus This page spends more time exploring what makes a really effective research question compared to a poor one. There are readings and short videos to support your learning. Further Reading Clarifying your research questions Research question-examples Key Success Strategies for Writing Good Research Questions

Deciding on a research topic can be a difficult task. You will need subject expertise. You will need to think and read critically within your area of interest. Here are some suggestions:

- Begin thinking early: As you discuss ideas with academics, bear in mind you will have to select a supervisor for your research. Start discussing your research project with potential faculty members.
- Do preliminary research on your topic: Generate keywords and use the library resources to gain knowledge on a subject. Read recent theses available online on your topic and see how they conducted the research. Successful completion often depends on whether the topic continues to hold your interest.
- Start asking questions: Ask open-ended “how” and “why” questions about your topic.
- Review your research question: It is likely that some of your initial ideas will be challenged. Adapt your work and be flexible.
- Consider your timeline: Be sure your question is specific and manageable in the time you have to complete the research.
- Set achievable goals: Familiarise yourself with recent developments in your field. This will ensure your idea is achievable.
- Originality: Refer to the databases and reference sources in your related subject guide to ensure your proposed research topic has not already been presented at Masters or Doctoral degree level.

Example Here is an example of a well-written question in the computer science domain:

Research question: How does the use of machine learning algorithms affect the accuracy of fraud detection in financial transactions?

This research question focuses on the impact of machine learning algorithms on the accuracy of fraud detection in financial transactions. It is specific and measurable, as it focuses on a particular technology (machine learning algorithms) and a particular application (fraud detection in financial transactions). It is also relevant, as fraud detection is a critical issue in the financial industry, and machine learning algorithms are becoming increasingly popular in this context. To further develop this research question, the researcher may need to identify the key variables or concepts involved, such as the types of machine learning algorithms used, the data sets used for training and testing, and the metrics used to evaluate the accuracy of fraud detection. Additionally, the researcher may need to consider the appropriate research methodology, such as experimental design or case study analysis, to address this research question.

Tips for Writing Research Questions Learn five top tips from the author of Writing Research Questions, Patrick White: Good Research Questions Are...

There is no universal set of criteria for a good research question. Different disciplines have different priorities and requirements. A good research question for a history paper will differ from a good research question for a biology paper. In general, however, a good research question should be:

- Clear and focused. The question should clearly state what the writer needs to do. Not too broad and not too narrow. The question should have an appropriate scope. If the question is too broad it will not be possible to answer it thoroughly within the word limit. If it is too narrow you will not have enough to write about and you will struggle to develop a strong argument.
- Not too easy to answer. The question should require more than a simple yes or no answer.
- Not too difficult to answer. You must be able to answer the question thoroughly within the given timeframe and word limit.
- Researchable. You must have access to a suitable amount of quality research materials, such as academic books and refereed journal articles.
- Analytical rather than descriptive. Your research question should allow you to produce an analysis of an issue or

problem rather than a simple description of it. source: <https://www.monash.edu/rlo/research-writing-assignments/understanding-the-assignment/developing-research-questions> Sample Research Questions source: <https://nesslabs.com/good-questions> If you ask good questions, you are much more likely to get good answers. On this page we look at some research questions that are not so good and then rewrite them to make them clearer. The reasoning for the changes is set out clearly for you to replicate when clarifying your questions. The following are a few examples of some sample research questions: Unclear: How should social networking sites address the harm they cause? Clear: What action should social networking sites like Twitter and Facebook take to protect users' personal information and privacy? The unclear version of this question does not specify which social networking sites or suggest what kind of harm the sites might be causing. It also assumes that this "harm" is proven and/or accepted. The clearer version specifies sites (Twitter and Facebook), the type of potential harm (privacy issues), and who may be experiencing that harm (users). A strong research question should never leave room for ambiguity or interpretation. Unfocused: What is the effect on the environment from global warming? Focused: What is the most significant effect of glacial melting on the lives of penguins in Antarctica? The unfocused research question is so broad that it could not be adequately answered in a book-length piece, let alone a standard college-level paper. The focused version narrows down to a specific effect of global warming (glacial melting), a specific place (Antarctica), and a specific animal that is affected (penguins). It also requires the writer to take a stance on which effect has the greatest impact on the affected animal. When in doubt, make a research question as narrow and focused as possible. Too simple: How are doctors addressing diabetes in New Zealand? Appropriately Complex: What main environmental, behavioural, and genetic factors predict whether New Zealanders will develop diabetes, and how can these commonalities be used to aid the medical community in prevention of the disease? The simple version of this question can be looked up online and answered in a few factual sentences; it leaves no room for analysis. The more complex version is written in two parts; it is thought-provoking and requires both significant investigation and evaluation from the writer. As a general rule of thumb, if a quick Google search can answer a research question, it is likely not very effective. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=2lqiGXIO70I>

#### Links & References

- [File] [Clarifying your research questions](#)
- [Link] [Research question-examples](#)

## Module 6: The Research Method

### Introduction to Module 6

Module 6 The Research Method: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Six. This module explores different research methods in more depth. Research methods, as we learned earlier, refer to the procedures, techniques, and

strategies utilized to collect and analyse data in order to answer research questions or test hypotheses. It involves the systematic and structured approach to gathering and interpreting information for the purpose of generating new knowledge or verifying existing knowledge. Different research methods are utilized depending on the nature of the research questions and the type of data that is being collected, such as quantitative methods, qualitative methods, or mixed methods. In this module, you will take a closer look at three different research methodologies used in Computer Science - Qualitative Research, Quantitative Research, and Mixed Model Research - and will study the research strategies, tools and techniques that align with each methodology. Objectives At the completion of this module, you will be able to:  
Understand characteristics associated with diverse research methods used in Computer Science  
Identify various tools and techniques used for different research methods  
Make appropriate choices for research methods that align with personal research goals

## Qualitative Research

Module 6 Qualitative Research Icon Progress Bar (browser only) Focus This page introduces qualitative research as it is performed in Computer Science / Information Technology. We will cover several topics: What is Qualitative Research? Why Qualitative research? Advantages and Disadvantages Qualitative Methods There are readings and short videos to support your learning. Further Reading Chapter 4 from Qualitative Research: An Introduction to Methods and Designs Qualitative Research Methods Overview Qualitative Research Methods: A Data Collector's Field Guide Research Methods in Computer Science Qualitative, quantitative, and mixed research methods are all methods commonly used in computer science to investigate and understand different aspects of computing and technology. Have a read through the descriptions of these methods below, noting how they can be put to use in computer science/ IT research: Qualitative Research Qualitative research methods involve exploring and understanding the underlying reasons, opinions, and motivations of users, stakeholders, and developers. Qualitative research often involves collecting data through interviews, focus groups, observation, or case studies. It is useful for understanding the context, attitudes, and behaviors of users, and it can be used to inform the design and development of computer systems. Quantitative Research Quantitative research methods, on the other hand, involve the collection and analysis of numerical data. Quantitative research often involves surveys, experiments, or other types of statistical analysis. It is useful for measuring and analyzing user behavior, system performance, or other objective metrics. Quantitative research can provide empirical evidence to support design decisions and to evaluate the effectiveness of computer systems. Mixed Research Mixed research methods combine both qualitative and quantitative research techniques. This approach involves collecting and analyzing both numerical and non-numerical data. Mixed methods research is useful when a more comprehensive understanding of a research problem is required. It can help to triangulate data from different sources to gain a more complete picture of a particular phenomenon. As you previously learned, the selection of an appropriate research methodology is critical to the successful execution of your study, as it determines the approach for information acquisition, research strategies, and methods employed. This is why it is important to look in further depth at the qualities and characteristics of the three research methodologies highlighted above. Qualitative Research in Computer Science As you learned before, qualitative research is an exploratory and subjective form of inquiry that seeks to understand and interpret social phenomena from the perspective of the participants. It involves the collection and analysis of non-numerical data , such as

words, images, and observations, to generate descriptive and contextual insights. The research design is typically flexible and iterative, with the goal of generating rich and detailed data that captures the complexity and diversity of the phenomenon under investigation. Qualitative research methods include techniques such as ethnography, interviews, focus groups, and content analysis, among others. The findings of qualitative research are based on the researcher's interpretations of the data, which are influenced by their own subjectivity, theoretical assumptions, and contextual factors. Qualitative research is used in a wide range of academic fields, including sociology, psychology, anthropology, education, and health sciences, among others. Denzin and Lincoln provide a definition of qualitative research in the Sage Handbook of Qualitative Research (2011) : "Qualitative research involves the studied use and collection of a variety of empirical materials—case study, personal experience, introspection, life story, interview, artefacts, and cultural texts and productions, along with observational, historical, interactional, and visual texts—that describe routine and problematic moments and meanings in individuals' lives." Qualitative Research in Computer Science Qualitative research in computer science involves the collection and analysis of non-numerical data to gain an in-depth understanding of a phenomenon or explore research questions from the perspectives of the participants. This type of research is often used in computer science to investigate the user experience, social and cultural aspects of technology, or the development and adoption of new technologies. Example An example of qualitative research in computer science is a study that explores the challenges and opportunities for incorporating artificial intelligence (AI) into educational technologies. The researchers could conduct in-depth interviews with teachers, students, and educational technology developers to understand their perspectives on the use of AI in education, the potential benefits and risks of AI, and the ethical and social implications of AI in educational settings. The researchers could also analyze online discussion forums or social media platforms to gain insights into the public discourse around AI in education. The findings from this study could provide valuable insights for the development of educational technologies that incorporate AI in a responsible and effective manner.

### Why Choose Qualitative Research?

Qualitative research methods are often a good approach when researching the following:

- New product idea generation and development
- Investigating current or potential product/service/brand positioning and marketing strategy
- Strengths and weaknesses of products/brands
- Understanding dynamics of purchase decision dynamics
- Studying reactions to advertising and public relations campaigns, other marketing communications, graphic identity/branding, package design, etc.
- Exploring market segments, such as demographic and customer groups
- Studying emotions and attitudes on societal and public affairs issues
- Assessing the usability of websites or other interactive products or services
- Understanding perceptions of a company, brand, category and product
- Determining consumer language as a preliminary step to develop a quantitative survey

### Why Go Qualitative?

### Challenges with Qualitative Research

Some of the challenges with Qualitative Research include:

- It can be more time consuming
- You often end up with masses of data to transcribe
- It is often more difficult to code data
- It is not always directly applicable to widely dispersed social settings
- Due to the specificity of the participants, there may be limited applicability to other situations
- It usually gives only nominal level data, so it is difficult to quantify
- It can be difficult to control for researcher bias

### Qualitative Research Methods

Qualitative Research uses a variety of methods. These are six main types of Qualitative research methods:

#### One-to-one interviews

One-to-one interviews are a qualitative research method that involves a structured conversation between a researcher and a participant, where the participant is asked open-ended questions about their experiences,

opinions, beliefs, or attitudes towards a particular topic. In the computer science domain, one-to-one interviews can be a valuable method for gathering data from users, developers, or other stakeholders about their experiences with technology, software, or systems. An example of a one-to-one interview in computer science might be a study on the usability of a new software application. The researcher could conduct one-to-one interviews with individual users of the application, asking them about their experiences using the software, what features they found helpful or challenging, and how they would suggest improving the application. The researcher could use this data to identify common themes or issues with the software, and to develop recommendations for improving the usability of the application. Another example of a one-to-one interview in computer science might be a study on the experiences of software developers working on a particular project. The researcher could conduct one-to-one interviews with individual developers, asking them about their experiences with the project, what challenges they encountered, and how they overcame those challenges. The researcher could use this data to identify common issues faced by developers on the project, and to develop recommendations for improving the development process.

**Focus Groups** A focus group is a qualitative research method that involves a group of participants who are brought together to discuss a specific topic or issue. In computer science, focus groups can be used to gather feedback on new software, website designs, or technology products. The goal of a focus group is to explore participants' perceptions, attitudes, beliefs, and behaviors related to the topic of interest. The focus group is typically moderated by a researcher who guides the discussion, asks open-ended questions, and encourages participants to share their opinions and experiences. The discussion is often recorded or transcribed for later analysis. Data analysis involves examining the themes and patterns that emerge from the discussion. For example, a computer science researcher might conduct a focus group with users of a new mobile app to gather feedback on the user interface, features, and overall usability. The researcher might ask questions such as: What do you think of the app's design? Are there any features that you find particularly useful or confusing? How easy was it to navigate through the app? Would you recommend this app to others? The participants might also be asked to provide suggestions for improvements or new features. By analyzing the data collected from the focus group, the researcher can gain insights into users' experiences and perspectives, identify strengths and weaknesses of the app, and inform future design and development decisions.

**Ethnographic Research** Ethnographic research is a qualitative research method that aims to understand the culture and behavior of people within a particular context. This approach involves observing and interacting with people in their natural settings and involves collecting data through interviews, surveys, field notes, and artifacts. In computer science, ethnographic research is often used to understand how people use technology in their daily lives, the social and cultural factors that influence technology adoption and usage, and how technology can be designed to better meet the needs of users. An example of ethnographic research in computer science is a study of how people use social media platforms like Facebook or Twitter. The research team may conduct field observations by observing people using these platforms in different settings, such as at home, work, or public places. They may also conduct interviews with users to understand their motivations for using social media, their social networks, and how they interact with others online. The research team may also collect data on the types of content that people share on social media and the reactions they receive from others. By analyzing the data collected from these different sources, the research team can gain insights into how social media is used in different contexts, the role it plays in people's lives, and the social and cultural factors that

influence its use. This information can then be used to design better social media platforms that meet the needs of users and support their social and cultural practices.

### Case Studies

The case study is a qualitative research method that involves in-depth investigation and analysis of a single entity or a phenomenon within its real-life context. It is a useful research method in computer science, where researchers can examine a particular technology, system, or process to gain insights into its behavior, performance, and impact. Case studies are usually conducted through observation, interviews, and analysis of documents, artifacts, and other relevant data sources. Researchers use various analytical techniques to analyze the data and identify patterns, themes, and relationships.

Example Here is an example of a case study in computer science: Example: A case study of the implementation of Agile methodology in software development. In this case study, the researcher would investigate the implementation of Agile methodology in software development in a particular organization or project. The researcher would gather data by conducting interviews with software developers, project managers, and other stakeholders involved in the project. The researcher would also analyze project documentation and artifacts, such as user stories, sprint backlogs, and product demos. The researcher would use various analytical techniques, such as content analysis and thematic analysis, to analyze the data and identify patterns and themes related to the implementation of Agile methodology. The researcher would examine how the implementation of Agile methodology affected the software development process, the quality of the software, and the satisfaction of stakeholders. The results of the case study could provide insights into the benefits and challenges of implementing Agile methodology in software development and inform future research and practice in the field.

### Record Keeping

Record keeping is a qualitative research method in computer science that involves collecting and analyzing data through the examination of documents, artifacts, and other sources of information. This method is particularly useful for exploring how technology is used in different contexts and how people interact with technology in their daily lives. One example of record keeping in computer science research is a study that examines the use of mobile devices in the classroom. Researchers may collect data through the examination of classroom materials, such as lesson plans and student work, as well as through observations of how teachers and students use mobile devices in the classroom. They may also collect data from student surveys or interviews to gain a better understanding of how students perceive the use of mobile devices in their learning. By analyzing this data, researchers can gain insights into how mobile devices are used in the classroom, how they affect teaching and learning, and how they can be effectively integrated into educational settings. For example, they may find that the use of mobile devices in the classroom is associated with increased engagement and participation among students, but that it also poses challenges related to distraction and misuse.

### Observation

Observation is a qualitative research method used in computer science to collect data by observing and recording behaviors and activities in natural settings. The purpose of observation is to gain insight into how individuals or groups use technology and interact with computer systems in real-world situations. An appropriate example of using observation in computer science is in the evaluation of a user interface design. Observing users as they interact with the system can help identify issues with the design, such as confusing or unintuitive features, that may not be apparent from user feedback or surveys alone. For instance, imagine a researcher is evaluating a new mobile application that helps users manage their finances. To understand how users interact with the app, the researcher may conduct observation studies where users are observed as they complete tasks, such as adding a new expense or setting up a budget. The

researcher can observe how easily users navigate the app and identify any issues that arise during the process. During the observation, the researcher may take notes or record video of the user's interactions with the application. The researcher may also ask the user to think aloud while using the app to provide additional insights into their thought process and decision-making. After the observation study is complete, the researcher can analyze the data to identify common patterns, areas of difficulty, and potential improvements for the app's user interface design. This information can then be used to refine the design and improve the overall user experience. Qualitative Research Methods Watch the video below, which covers the qualitative research process, including research methods, to undertake Qualitative Research. (14:29mins): Although there are many different types of research methods used in qualitative research, they all generally share the characteristics shown in the figure below: source: <https://phantran.net/qualitative-research-definition-types-methods-and-examples/characteristics-of-qualitative-research-methods-min/> Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=dC1JLc42Yxw>
- [https://www.youtube.com/watch?v=\\_uapR0qiN6s](https://www.youtube.com/watch?v=_uapR0qiN6s)

### Links & References

- [Link] [Chapter 4 from Qualitative Research: An Introduction to Methods and Designs](#)
- [Link] [Qualitative Research Methods Overview](#)
- [Link] [Qualitative Research Methods: A Data Collector's Field Guide](#)
- [Link] [Sage Handbook of Qualitative Research \(2011\)](#)

## Quantitative Research

Module 6 Quantitative Research Icon Progress Bar (browser only) Focus This page introduces the subject of quantitative research in computer science. We will cover several topics: What is Quantitative Research? Why Quantitative Research? Advantages and Disadvantages Quantitative Methods There are readings and short videos to support your learning. Further Reading Read Chapter 3 in Understanding and Applying Research Design for an overview of Quantitative Research Design. Read this short overview on scribbr for Quantitative Data Analysis techniques. Read this article for a description of the types of Quantitative research. Overview As you learned previously, quantitative research is the process of collecting and analysing numerical data. It relies primarily on describing or measuring phenomena in quality. It can be used to find patterns and averages, make predictions, test causal relationships, and generalize results to wider populations. It is all about understanding numerical information to make decisions about a process. Quantitative methods use statistics to describe large populations with survey data or use theoretical generalisations gained from experiments. Quantitative research is one of the systematic research techniques. Quantitative researchers use different tools to collect numeric data in terms of numbers and statistics. This data is represented in forms such as charts, figures and tables. Types of Quantitative Research in Computer Science There are five main types of quantitative research, which are used in computer science studies: Descriptive Research This type of research is used to describe a

phenomenon or population. Descriptive research is often used to gather information about a large population, and it may involve collecting data through surveys or other forms of data collection. Descriptive research aims to identify patterns, trends, and relationships among variables. For example, a researcher might conduct a descriptive research study on the types of operating systems used in a specific industry. The researcher might gather data on the number of companies using each type of operating system, the features and functionalities of each system, and the benefits and drawbacks of each system. To conduct this study, the researcher could use a survey to collect data from companies in the industry. The survey could include questions about the company's size, the type of operating system used, the reasons for choosing the system, and the perceived benefits and drawbacks of the system. The researcher could then analyze the data to identify patterns and trends in the types of operating systems used and the reasons for their use. The results of the study could be used to provide insights into the strengths and weaknesses of different operating systems in the industry. This information could be used by companies to make more informed decisions about which operating system to use, or by developers to create more effective operating systems.

**Correlational Research** This type of research is used to examine the relationship between two or more variables. Correlational research aims to identify whether there is a relationship between variables, and if so, what type of relationship exists (positive, negative, or neutral). Correlational research often involves collecting data through surveys or questionnaires. An appropriate example of using correlational research in computer science is examining the relationship between website load time and user satisfaction. In this case, website load time is the independent variable, and user satisfaction is the dependent variable. To conduct this study, a researcher may collect data from a sample of website users, including information about website load time and user satisfaction ratings. The researcher can then use statistical analysis to examine the relationship between website load time and user satisfaction. If the analysis shows that website load time is negatively correlated with user satisfaction, it suggests that users are less satisfied with the website as load time increases. This information can be used to improve website performance by optimizing load time and improving user satisfaction. It is important to note that correlational research does not imply causation. In the example above, a negative correlation between website load time and user satisfaction does not necessarily mean that slower load times cause lower satisfaction. Other variables, such as website design or content, may also be contributing factors.

**Causal-Comparative Research** This type of research is used to compare two or more groups or conditions to determine whether there is a cause-and-effect relationship between variables. Causal-comparative research is often used in educational research to evaluate the effectiveness of interventions or programs. An appropriate example of using causal-comparative research in computer science is a study that compares the effectiveness of two different programming languages in producing high-quality code. The researcher may select two groups of software developers: one group who have experience using a certain programming language (Group A) and another group who have experience using a different programming language (Group B). The dependent variable in this study would be the quality of code produced by each group, as measured by the number of errors, readability, and maintainability. The researcher cannot randomly assign participants to each group, as they would in experimental research, but instead selects participants who have naturally been exposed to the programming language they use. The researcher collects data on the quality of code produced by each group and analyzes the data to determine whether there is a significant difference in the quality of code produced by each group. If the data analysis shows that Group

A produces higher quality code than Group B, the researcher may conclude that the programming language used by Group A is more effective than the programming language used by Group B. However, the researcher cannot conclude with certainty that the programming language is the only factor that affected the quality of code produced, as there may be other variables, such as the experience level of each group or the type of project being worked on, that could have influenced the results.

**Experimental Research** This type of research is used to test a hypothesis or theory by manipulating one or more variables and measuring the effect on another variable. Experimental research is often conducted in a laboratory setting, and it involves randomly assigning participants to different conditions or groups. In the computer science domain, experimental research can be used to test hypotheses related to software or hardware systems, user interfaces, or other technological innovations. An appropriate example of experimental research in computer science is a study that investigates the effect of different website layouts on user engagement. In this experiment, the researcher may manipulate the website layout by changing the placement of images, text, or other elements on the page. The dependent variable in this study could be user engagement, which may be measured by the time spent on the site, the number of pages visited, or other engagement metrics. The experiment would be conducted by randomly assigning participants to different groups or conditions, such as a control group and one or more experimental groups. The control group would be shown the website in its current layout, while the experimental groups would be shown the website with different layouts. The researcher would then collect data on user engagement for each group and compare the results. If the experimental groups showed a significant difference in user engagement compared to the control group, the researcher could conclude that the website layout had a causal effect on user engagement.

**Survey Research** This type of research is used to collect data from a large population through standardized questionnaires or surveys. Survey research aims to gather information about attitudes, beliefs, opinions, and behaviors of the population being studied. Survey research can be conducted in various ways, such as in-person, phone, mail, or online surveys. Survey research can also be used for descriptive, correlational, or causal-comparative research. An appropriate example of using survey research in computer science is in the evaluation of a software product or service. Survey research can be used to gather feedback from users on their experience with the product, such as ease of use, satisfaction with features, and overall usability. For instance, imagine a company that develops a new cloud-based collaboration tool. To evaluate the product, the company may conduct a survey of users to gather feedback on the tool's usability, features, and overall experience. The survey may include questions such as: How easy was it to navigate the tool? How satisfied were you with the collaboration features? How likely are you to recommend this tool to a colleague? The survey may also include open-ended questions to gather qualitative feedback on specific features or areas for improvement.

**Types of Quantitative Research** To consolidate what you have just learned, watch this video to learn about the types of Quantitative research design: Advantages of Quantitative Research for Computer Scientists

**Objectivity** : Quantitative research is based on numerical data, which makes it more objective and less prone to researcher bias. This objectivity helps to ensure the validity and reliability of the research findings.

**Generalisability** : Quantitative research often involves larger sample sizes, which can increase the generalisability of the research findings. This means that the results can be applied to a larger population beyond the sample.

**Replicability** : Quantitative research methods often use standardized procedures and instruments, making it easier to replicate the study and test the validity and reliability of the findings.

**Precision** : Quantitative

research involves measuring data using precise instruments and methods, which allows for greater precision and accuracy in the data analysis. Disadvantages of Quantitative Research for Computer Scientists Limited understanding : Quantitative research focuses on numerical data, which can limit the researcher's understanding of the context and complexity of the data. This may lead to a limited understanding of the research problem. Inability to capture nuances : Quantitative research is less effective at capturing the nuances and complexity of human experiences, emotions, and behaviors. This can be particularly problematic in areas of computer science that involve human-computer interaction or user experience. Lack of depth : Quantitative research tends to focus on identifying patterns and correlations in the data, but it may not provide a deep understanding of the underlying causes of those patterns. Researcher bias : Although quantitative research is based on numerical data, researcher bias can still impact the study design, data collection, and analysis. Researchers must be aware of their own biases and take steps to minimize their impact. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=vfi3Nrnpo-U>

#### Links & References

- [Link] [Chapter 3](#)
- [Link] [scribbr](#)
- [Link] [types of Quantitative research.](#)

### Choosing between Qualitative, Quantitative, and Mixed methods

Module 6 Choosing between Qualitative, Quantitative, and Mixed Methods Icon Progress Bar (browser only) Focus This page introduces the decision-making process for selecting which methodology will work best for your research project. We will focus on: How to select research methods? Qualitative vs Quantitative? Mix Research Methods Activities There are readings and short videos to support your learning. Further Reading Quantitative and Qualitative research methods Chapter 3 Observational and empirical generalisation Read strengths and weaknesses of the two approaches in Quantitative and Qualitative research methods Simultaneous Triangulation: Mixing User Research and Data Science Methods 6 types of mixed method design Research Methods in Human-Computer Interaction, 2nd Edition, eds Jonathan Lazar et. al. Research Methods for Cyber Security, Thomas Edgar, David Manz. How do I choose my Methodology? Choosing between qualitative and quantitative research can be a challenging task as it depends on several factors such as the research question, the nature of the phenomenon being studied, and the goals of the study. Here are some considerations that can assist you in deciding which approach to use: Qualitative research may be more appropriate if: The research question is exploratory or seeks to understand a complex phenomenon in-depth. The research is focused on the subjective experiences, perspectives, or attitudes of participants. The sample size is small, and the goal is to gain a deep understanding of the experiences of a few participants. The research involves the collection of non-numerical data such as text, images, or audio. Quantitative research may be more appropriate if: The research question is hypothesis-driven and seeks to test specific relationships between variables. The research is

focused on objective, measurable data that can be analysed statistically. The sample size is large and representative, and the goal is to generalise findings to a larger population. The research involves the collection of numerical data through surveys, experiments, or other quantitative methods. Research Examples Here are some examples of how qualitative and quantitative research methods can be used in computer science research: Qualitative research examples: A study that uses semi-structured interviews to explore the experiences of software developers in open-source projects, aiming to understand the factors that influence their participation and motivation. An ethnographic study of the ways in which people use social media, aiming to understand how social media is changing our communication patterns and social interactions. A case study of a company's agile software development process, aiming to understand how the company has adapted and customised agile methodologies to fit its specific needs. Quantitative research examples: An experiment that tests the effect of different user interface designs on users' task completion times and error rates. A survey that investigates the factors that influence users' adoption of new technologies, aiming to identify the most important drivers and barriers to adoption. A data mining analysis that identifies patterns and trends in large datasets, such as user behaviour data or logs data from an online service. Qualitative and Quantitative Research Watch this video and consider the methods for your research topic: Mixed-Methods Research In some cases, a mixed-methods approach that combines both qualitative and quantitative methods may be appropriate. This can provide a more comprehensive understanding of the phenomenon being studied and can help address the limitations of each approach. Mixed methods research involves the use of both qualitative and quantitative data for better predictions. It can involve both interviews and surveys approaches to make meaningful decisions. In mixed-methods design, the research combines qualitative and quantitative methods and uses primary and secondary data. There are two main advantages of the same study. Firstly, different methods can be used for different purposes in a study. Secondly, it enables a researcher to use triangulation. This refers to the use of different data collection methods within one study to ensure that data is telling you what you think it is.

Figure: Different research methodologies source:

[https://learning.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter001.xhtml#ch1\\_3](https://learning.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter001.xhtml#ch1_3) Examples of using a mixed-method approach in Computer Science research: A study that aims to evaluate the effectiveness of a new educational programming tool. The study could use a mixed-methods approach to gather both quantitative data on student learning outcomes (e.g., test scores) and qualitative data on student perceptions and experiences using the tool (e.g., through surveys or interviews). By combining both types of data, the study could provide a more comprehensive understanding of the tool's effectiveness. A study that investigates the impact of social media on mental health. This study could use a mixed-methods approach to collect quantitative data on social media usage patterns and mental health outcomes (e.g., through surveys or self-report measures) and qualitative data on individuals' experiences and perceptions of social media and its effects on mental health (e.g., through interviews or focus groups). By combining both types of data, the study could gain a more nuanced understanding of the complex relationship between social media and mental health. A study that seeks to improve the accessibility of a website for users with disabilities. This study could use a mixed-methods approach to gather both quantitative data on website accessibility (e.g., through automated accessibility testing) and qualitative data on users' experiences and perceptions of the website's accessibility (e.g., through user testing or surveys). By combining both types of data, the study could identify specific accessibility issues and

develop more effective solutions based on user feedback. Mixed Methods Research For more consideration of this approach, watch this introduction to mixed methods research (6:35 min): Activities: Deciding on your Research Method(s) Here are some activities you can do to help you choose your research design: Read academic papers: Read academic papers on topics related to your research question and observe how other researchers have approached similar questions. Look for papers that use both qualitative and quantitative methods and try to identify the strengths and limitations of each approach. Conduct a literature review: Conduct a literature review on your research topic to identify what has already been studied, the methods used, and the gaps in the literature. This will help you determine what kind of data you need to collect, and what research methods will be most appropriate. Create a research question: Develop a clear research question that outlines what you want to know or understand. This will help you determine what type of research method is most appropriate for your study. Consider your resources: Consider the resources available to you, including time, budget, and access to participants. Depending on your resources, some research methods may be more feasible than others. Seek feedback: Seek feedback from other researchers or mentors in your field. They can provide valuable insights and help you identify potential issues or limitations with your research design. Pilot test your research design: Conduct a pilot test of your research design to test the feasibility and effectiveness of your chosen research method. This will help you identify any issues or areas for improvement before conducting your full study. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- [https://www.youtube.com/watch?v=69qP8kFJp\\_k](https://www.youtube.com/watch?v=69qP8kFJp_k)
- <https://www.youtube.com/watch?v=yc--5XwQ5b0>

### Links & References

- [Link] [Quantitative and Qualitative research methods](#)
- [Link] [Chapter 3 Observational and empirical generalisation](#)
- [Link] [Simultaneous Triangulation: Mixing User Research and Data Science Methods](#)
- [Link] [6 types of mixed method design](#)
- [Link] [Research Methods in Human-Computer Interaction, 2nd Edition, eds Jonathan Lazar et. al.](#)
- [Link] [Research Methods for Cyber Security, Thomas Edgar, David Manz.](#)
- [Link] [https://learning.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter001.xhtml#ch1\\_3](https://learning.oreilly.com/library/view/research-methodology/9789353067090/xhtml/chapter001.xhtml#ch1_3)

## Module 7: Design of Experiments (DOE)

### Introduction to Module 7

Module 7 Design of Experiments: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Seven. In this module, you will study the design of experiments, also known as DOE. You will learn about its advantages and will explore the steps in DOE processes. You will discover when and why it is a useful approach in computer science

research. Objectives At the completion of this module, you will be able to: Understand DOE and why it is used in research Perform the steps of the DOE design process Understand the different types of DOE and when to use them Practise using DOE processes

## What is DOE?

Module 1 What is DOE? Icon Progress Bar (browser only) Focus This page introduces Design of Experiments. You will learn fundamentally what is meant by this term, DOE, and why we would choose to use this approach in our research. There are readings and short videos to support your learning. Reminder: This topic is critical to prepare for and complete Assessment 2. Further Reading A Quick History of the Design of Experiments Chapter 1 - Introduction in Fundamentals of Statistical Experimental Design and Analysis, R. Easterling 2017 for an overview of DOE. Design of Experiments (DOE) provides a brief but comprehensive overview of DOE. Steps for planning, conducting and analysing an experiment Chapter 2 - Fundamentals of Experimental Design in Fundamentals of Statistical Experimental Design and Analysis, R. Easterling 2017. Introduction to Design of Experiments (5:01 mins) Design of Experiments Design of Experiments (DoE) is a systematic and efficient approach used to plan, conduct, and analyse experiments in order to gain reliable insights into the factors influencing a specific system or process. By strategically manipulating independent variables and observing the corresponding changes in the dependent variable(s) of interest, DoE enables you to optimise your experiments, reduce variability, and identify the most significant factors. An effectively designed experiment serves as a valuable tool for learning about the world. Typically, an experiment involves varying multiple factors and observing the response values based on combinations of these factor settings. The mathematical challenge lies in selecting the appropriate settings to extract the maximum information from the resulting data. DoE, a branch of applied statistics, encompasses the planning, conducting, analysis, and interpretation of controlled tests to evaluate the factors that impact a parameter or a group of parameters. It provides a powerful framework for data collection and analysis, applicable to various experimental scenarios. One of the key advantages of DoE is its ability to manipulate multiple input factors simultaneously, thereby determining their effects on the desired output or response. By considering multiple inputs concurrently, DoE can unveil important interactions that might be overlooked when studying one factor at a time. This exploration can be conducted using all possible combinations (full factorial) or by examining only a subset of the potential combinations (fractional factorial). What is the Design of Experiments (DoE)? We recommended watching this video for a good introduction to DOE: For more information, we also recommend reading: Chapter 1 - Introduction in Fundamentals of Statistical Experimental Design and Analysis, R. Easterling 2017 for an overview of DOE. Why should we use DOE? A strategically planned and executed experiment may provide a great deal of information about the effect on a response variable due to one or more factors. Many experiments involve holding certain factors constant and altering the levels of another variable. This "one factor at a time" (OFAT) approach to process knowledge is, however, inefficient when compared with changing factor levels simultaneously (Montgomery, 2020). Many of the current statistical approaches to designed experiments originate from the work of R. A. Fisher in the early part of the 20th century. Fisher demonstrated how taking the time to seriously consider the design and execution of an experiment before trying it helped avoid frequently encountered problems in analysis (Fisher, 1935). Key concepts in creating a designed experiment include blocking, randomisation, and replication. Blocking: When randomising a factor is impossible or too

costly, blocking lets you restrict randomisation by carrying out all of the trials with one set of the factor and then all the trials with the other setting (Box, Hunter, & Hunter, 2005).

Randomisation : Refers to the order in which the trials of an experiment are performed. A randomised sequence helps eliminate the effects of unknown or uncontrolled variables (Montgomery, 2020).

Replication : Repetition of a complete experimental treatment, including the setup (Montgomery, 2020). A well-performed experiment may provide answers to questions such as: What are the key factors in a process? At what settings would the process deliver acceptable performance? What are the key, main, and interaction effects in the process? What settings would bring about less variation in the output? A repetitive approach to gaining knowledge is encouraged, typically involving these consecutive steps: A screening design that narrows the field of variables under assessment. A "full factorial" design that studies the response of every combination of factors and factor levels, and an attempt to zone in on a region of values where the process is close to optimisation. A response surface is designed to model the response (Montgomery, 2020).

DOE Process DOE is really a four-step process. This is outlined in Figure 7.1 below.

Figure 7.1: Design of Experiments Process We can break this down further into the following key steps:

- Identify the research question: Clearly define the objective of the experiment and the specific variables that need to be investigated.
- Define the factors and levels: Determine the independent variables (factors) that may influence the dependent variable(s) and define the levels or values at which these factors will be set.
- Select the experimental design: Choose an appropriate experimental design based on the research question and available resources. Common designs include completely randomised designs, randomised block designs, factorial designs, and response surface designs.
- Determine the sample size: Decide on the number of experimental units or observations required to achieve sufficient statistical power and precision.
- Randomise the experiment: Randomly assign the experimental units to different treatment groups or conditions to minimise bias and control for extraneous variables.
- Conduct the experiment: Implement the experimental conditions, collect data, and record any relevant observations.
- Analyse the data: Apply statistical methods to analyse the collected data and draw conclusions about the effects of the factors on the dependent variable(s).
- Draw conclusions and make recommendations: Interpret the results of the analysis, draw meaningful conclusions, and make recommendations based on the findings.

Introduction to Design of Experiments Process Watch this video and learn about the process of setting up a DOE: Introduction to Design of Experiments (5:01 mins)

Example of How to Design Experiments

Research question: What factors affect the performance of a computer network?

Factors and levels: Network bandwidth: Low (10 Mbps), Medium (100 Mbps), High (1 Gbps)

Network latency: Low (1 ms), Medium (10 ms), High (100 ms)

Network topology: Bus, Ring, Star

Experimental design: A full factorial design can be used to investigate the effects of all possible combinations of the factors and levels. This would result in  $3 \text{ (bandwidth)} \times 3 \text{ (latency)} \times 3 \text{ (topology)} = 27$  treatment combinations.

Sample size: Determine the number of repetitions or observations needed for each treatment combination based on statistical considerations and available resources.

Randomise the experiment: Randomly assign the treatment combinations to different experimental runs to ensure unbiased results.

Conduct the experiment: Set up the computer network according to each treatment combination, measure the network performance metrics (e.g., throughput, latency), and record the results.

Analyse the data: Apply statistical analysis techniques, such as analysis of variance (ANOVA), to determine the significance of the factors and their interactions on the network performance.

Draw conclusions and make recommendations: Based on the analysis, identify the most

significant factors affecting the network performance and provide recommendations for optimising network design or configuration. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=Srq9Q-yd1Rk>

## Links & References

- [Link] [A Quick History of the Design of Experiments](#)
- [Link] [Chapter 1 - Introduction](#)
- [Link] [Design of Experiments \(DOE\)](#)
- [Link] [Steps for planning, conducting and analysing an experiment](#)
- [Link] [Chapter 2 - Fundamentals of Experimental Design](#)
- [Link] [Introduction to Design of Experiments](#)

## When to use DOE?

Module 7 When to Use DOE? Focus This page introduces when to use DOE along with some examples. There are readings and short videos to support your learning. Further Reading Read the Integrated circuits (ICs) Case Study in Fundamentals of Statistical Experimental Design and Analysis Easterling 2017. Paper: "Design of Experiments for Performance Evaluation of Algorithms" by R. A. Waltz and J. M. Waltz Waltz, R. A., & Waltz, J. M. (2014). Design of Experiments for Performance Evaluation of Algorithms . In Handbook of Research on Computational Science and Engineering: Theory and Practice (pp. 280-305). IGI Global. Paper: "Applying Design of Experiments to Software Testing" by L. Briand, Y. Labiche, and M. L. Williams Briand, L., Labiche, Y., & Williams, M. L. (2002). Applying Design of Experiments to Software Testing. IEEE Transactions on Software Engineering, 28(12), 1115-1129. Paper: "Design of Experiments for Software Process Improvement" by M. S. P. C. S. Brito and A. F. da Silva Brito, M. S. P. C. S., & da Silva, A. F. (2018). Design of Experiments for Software Process Improvement. International Journal of Software Engineering and Knowledge Engineering, 28(06), 871-895. Paper: "A Design of Experiments Framework for Recommender Systems" by J. M. P. Vieira, A. H. P. Costa, and A. A. Freitas Vieira, J. M. P., Costa, A. H. P., & Freitas, A. A. (2017). A Design of Experiments Framework for Recommender Systems . Expert Systems with Applications, 73, 95-109. When to use DOE? DOE offers a systematic and efficient approach to experimentation, enhancing the quality and efficiency of research outcomes. Statistical design should be employed whenever conducting an experiment, considering the specific objectives and requirements of the study. Various types of designs, including comparative, screening, and optimization experiments, are available to suit different research purposes. Design of Experiments (DOE) is particularly valuable when there are multiple input factors that may impact an output. For instance, when investigating the strength of a glue bond, it becomes important to understand how temperature and pressure influence the outcome (see diagram below). By using DOE, researchers can systematically manipulate these factors to determine their effects and establish robust input-output relationships. Furthermore, DOE serves as a powerful tool for validating suspected input-output relationships and constructing predictive equations that facilitate what-if analysis. This predictive modeling enables researchers to

explore various scenarios and assess the potential impact of changes in input factors on the desired output. By incorporating statistically designed experiments and employing appropriate data analysis techniques, researchers can acquire a deeper understanding of the system under study while minimizing the need for excessive experimentation. Why Use DOE? Watch the two videos below on the DOE process, the value it adds to analysis, and when to use it: Integrated Circuits Case Study DOE takes seemingly impossibly large amounts of data and makes it accessible and usable in a cohesive manner. Read the Integrated circuits (ICs) Case Study in Fundamentals of Statistical Experimental Design and Analysis Easterling 2017. After reading, reflect on the following question: What value did DOE add to the research? Applying DOE in Computer Science Design of Experiments (DoE) can be applied in computer science research in various situations: Performance Optimization: When developing software or algorithms, DoE can help optimize performance by systematically varying input parameters and measuring the corresponding output metrics. It allows researchers to identify the most influential factors and their interactions, leading to improved performance and efficiency. Configuration Tuning: DoE can assist in finding the optimal configuration settings for computer systems, databases, networks, or other components. By systematically exploring different parameter combinations, DoE helps identify the configuration that maximizes performance, minimizes resource usage, or achieves other desired objectives. Software Testing: DoE can aid in efficient and effective software testing by determining the most important factors affecting software behavior and performance. It helps in selecting appropriate test cases, identifying key input variables, and understanding their impact on the output, thus enhancing testing coverage and effectiveness. User Interface Design: DoE can be used to optimize user interface (UI) design in computer applications. By systematically varying UI elements (e.g., layout, color scheme, font size) and measuring user responses, DoE helps identify the design factors that lead to improved user experience, usability, and satisfaction. Network and System Security: DoE can assist in evaluating the effectiveness of security measures and configurations in computer networks and systems. By systematically manipulating security parameters, DoE helps identify vulnerabilities, potential attack vectors, and optimal security configurations. Data Analysis and Machine Learning: In data-driven research, DoE can aid in experimental design and feature selection. It helps determine which variables or features significantly impact the outcome, identify interactions between variables, and optimize the model performance in machine learning tasks. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=9JTVq-UGurU>
- <https://www.youtube.com/watch?v=rejkbil3SsM>

#### Links & References

- [Link] [Integrated circuits \(ICs\) Case Study](#)
- [Link] [Design of Experiments for Performance Evaluation of Algorithms](#)
- [Link] [Applying Design of Experiments to Software Testing.](#)
- [Link] [A Design of Experiments Framework for Recommender Systems](#)

## Key Concepts and Terminology DOE

Module 7 Key Concepts and Terminology DOE Icon Progress Bar (browser only) Focus This page introduces some of the essential concepts and terms commonly used when talking about the design of experiments. There are readings and short videos to support your learning. Key Concepts As with any subject area, there are specific terminology and concepts that you need to use and understand. Below are some keywords that you will need to know. One-Way Experiment: An experiment that focuses on studying the influence of a single factor on a process output. Response (Y, KPOV): The process output or Key Process Output Variable (KPOV) that is dependent on the factors being studied. Factor (X, KPIV): The variable that is being studied and can be controlled or uncontrolled. Also known as the Key Process Input Variable (KPIV). Inference Space: The range of operating conditions or settings of the factors that are under study. Factor Level: The specific setting or value assigned to a factor during the experiment. Treatment Combination (run): A specific set of factor levels that are used together to obtain a response. Replicate: The number of times a treatment combination is repeated in the experiment, usually randomized, to estimate error and account for variability. ANOVA: Analysis of Variance, a statistical method used to analyze the variation in the response variable and determine the significance of the factors studied. Blocking Variable: A variable that the experimenter chooses to control during the experiment but is not the main variable of interest. Interaction: The occurrence when the effects of one treatment vary depending on the levels of another treatment, indicating non-additivity. Collinear: Variables that are linearly related or can be expressed as linear combinations of each other. Confounding: Variables that are not controlled by the experimenter but can have an effect on the output, leading to mixed estimates of effects. Sensitivity: The ability to identify significant treatment differences in the response variable. Additional Jargon: KPOV: Key Process Output Variable KPIV: Key Process Input Variable Pure Trial Error: The inherent variability or error within the experiment. Experimental Error: The overall variability or error observed in the experiment. Correlation: A measure of the linear relationship between two variables. Additive: Referring to the combined effects of factors being simply summed. Statistical Significance: The likelihood that observed differences are not due to random chance. Factorial Design: An experiment design that studies the combined effects of multiple factors. Main Effect: The individual effect of a factor on the response variable. Replication: Repeating the experiment with the same treatment combination to obtain more reliable results. source: <https://www.six-sigma-material.com/Design-of-Experiments.html> Videos Now that you have read through these terms, work through the following two videos on this page. They are designed to further familiarise you with DOE terms and concepts: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=aWhIlCOImXg>
- <https://www.youtube.com/watch?v=pRdG01y-OrI>

## Types of DOE

Module 7 Types of DOE Icon Progress Bar (browser only) Focus This page introduces you to several types of experiment designs commonly used in computer science research. Pay

attention to these, as you may like to propose using some of these types of experiments for your research. You will: Work through the readings. Watch the videos. Observe DOE in Practice.

Further Reading Chapter 5: Completely Randomized Design with Multiple Treatment Factors  
Chapter 6: Randomized Complete Block Design DOE experiments Types of DOE As you have learned, Design of Experiments (DoE) refers to a systematic approach for planning, conducting, and analyzing experiments. It aims to maximize the information obtained from a limited set of experiments while minimizing the effects of confounding variables. There are several types of experimental designs commonly used in computer science and other fields. Here are some of the fundamental types:

**Completely Randomized Design (CRD):** In a CRD, the experimental units are randomly assigned to different treatments or conditions (Smith, 2020). Example: A researcher studying the performance of different sorting algorithms randomly assigns a set of computers to each algorithm to compare their execution times (Johnson et al., 2018).

**Randomized Complete Block Design (RCBD):** RCBD is useful when there are known sources of variation that could affect the response variable but are not of primary interest (Montgomery, 2017). Example: An experimenter comparing the effectiveness of various web browsers divides participants into different age groups (blocks) and then randomly assigns different browsers to each group (Sullivan, 2019).

**Factorial Design:** Factorial designs involve manipulating multiple factors simultaneously to study their main effects and interactions (Kirk, 2013). Example: A researcher investigating the impact of two independent variables, such as screen resolution (low, high) and font size (small, large), on user reading speed and comprehension (Liu et al., 2021).

**Latin Square Design:** Latin square design is useful when there are two sources of variation (rows and columns) that may affect the response (Montgomery, 2017). Example: A software tester evaluating the performance of different algorithms on multiple datasets, ensuring that each algorithm is tested with each dataset exactly once (Das et al., 2019).

**Response Surface Design:** Response surface designs are used when the relationship between the response variable and the factors is nonlinear (Box et al., 2005). Example: A researcher exploring the optimal combination of CPU frequency (GHz) and cache size (MB) to maximize the performance of a computational task (Zhang et al., 2020).

Three types of DOE Watch the following series of videos on the different DOE types. The first video gives an overview, and the following videos give more in-depth training with associated examples: DOE in Computer Science For an overview of other types of DOE watch this series of videos on the Improve phase of Six Sigma. This will give you an understanding of the purpose of each DOE type, commonly used in Computer Science. Improve Phase Overview (2.16mins) Two-level fractional factorial experiments (3.42 mins) Full-factorial experiments (9:19mins) DOE with Curvature (6.27mins) Response surface methods (8.37mins) Then take the Chapter Quiz Full Factorial DOE Watch this video to learn how to design a simple two-factor, two-level experiment and analyse the results of the experiment: Three-factor and two-level experiments Watch this video to learn how to design a three-factor, two-level experiment and analyse the results of the experiment: Coding values In this video learn how to convert uncoded to coded values: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=tZWAYbKYVjM>
- <https://www.youtube.com/watch?v=pTAUa6qXV6E>
- <https://www.youtube.com/watch?v=cIXYKynq1-o>

- <https://www.youtube.com/watch?v=W-H8b86zJcI>

## Links & References

- [Link] [Chapter 5: Completely Randomized Design with Multiple Treatment Factors](#)
- [Link] [Chapter 6: Randomized Complete Block Design](#)
- [Link] [DOE experiments](#)
- [Link] [Improve Phase Overview](#)
- [Link] [Two-level fractional factorial experiments](#)
- [Link] [Full-factorial experiments](#)
- [Link] [DOE with Curvature](#)
- [Link] [Response surface methods](#)
- [Link] [Chapter Quiz](#)

## Lab 3: Design of Experiments (DOE)

IT8101 Lab 3: Design of Experiments (DOE) Summary As this module has examined, the design of experiments (DOE) is a method for determining the link between elements affecting a process and the process' output. To put it another way, it is utilised to discover cause-and-effect linkages. This data is required for managing process inputs and optimising output. DOE is a way for scientists and engineers to examine the relationship between various input variables (also known as factors) and important output variables in a systematic and efficient manner (aka responses). It is a method for gathering data and producing discoveries that is well structured. It is a method for transforming ideas, hypotheses, and/or assumptions into a specific, well-defined set of experiments that can be carried out to validate those ideas, hypotheses, and assumptions, and therefore give us with useful information. Instructions Work through the questions and provide your answer to each one. These questions have been designed for you to demonstrate your understanding of key design of experiment concepts. Support your answers by explaining your reasoning.

1. Describe and explain a research scenario in which you intend to conduct a balanced experiment but end up with an uneven number of individuals in each group.
2. A new feeding supplement has been developed to help farmed fish grow faster. You must determine whether this supplement is more commercially effective in sea trout or rainbow trout. What would you do if you were in this situation?
3. A driver in their twenties is three times more likely than a driver in their sixties to be involved in a traffic collision. One explanation for this could be that as people get older, they become safer drivers. Can you think of any other third-variable effects that could explain this observation differently?
4. According to a major scientific magazine, students who play a musical instrument perform better in a general memory test than those who do not. As a result, the researcher came to the conclusion that playing a musical instrument improves memory. The feedback from the researcher's supervisor is that this conclusion is premature. Why do you think the supervisor decided this conclusion is premature? How do you suggest the researcher goes about gathering further evidence to test this hypothesis?
5. On their main search page, Google decides they want to include a new button. How do you suggest they find out if people like the new button feature? How will they test their hypothesis that the new button is of value?
6. Facebook is experimenting with new user homepage designs. They've come up with 50 different options. They want to put them all to the test and chose the one that best suits their

customers' requirements. What would be the best way to put up this experiment? What measurements would you use? How would you present your findings?

## Module 8: Design Thinking

### Introduction to Module 8

Module 8 Design Thinking: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Eight. This module introduces you to the concept of design thinking. You will explore design thinking as a research tool that is often used in business and technology design and can be very useful for Information Technology projects, depending on their research objectives. Objectives At the completion of this module, you will be able to: Describe what design thinking is and its application to the research process Explain how design thinking developed and came to be common in design process Implement design thinking in a research project

### Introduction to Design Thinking

Module 8 Design Thinking Icon Progress Bar (browser only) Focus This page introduces the topic of design thinking. You will learn about how this problem-solving approach works and why it can be a useful tool. There are readings and short videos to support your learning. Further Reading The introduction to What is Design Thinking? Stanford et al and Chapter 1. Introduction to what Design Thinking is in Understanding Design Thinking, Lean and Agile. Note the overlaps between Design thinking, Agile and Lean and where the design thinking mindset differs from these two approaches. What is Design Thinking? Design thinking is a human-centered approach to problem-solving that has gained significant recognition in various fields. It involves a series of iterative steps and tools aimed at understanding users, defining problems, generating innovative ideas, and prototyping solutions (Brown, 2008). One of the key aspects of design thinking is its emphasis on empathy and gaining deep insights into the needs and experiences of the people for whom the solution is being designed. By immersing themselves in the users' context, designers can uncover latent needs and create solutions that truly address their challenges (Kelley & Kelley, 2013). Design thinking has proven to be effective in tackling complex problems that may lack clear definitions or have multiple stakeholders with diverse perspectives. The process helps in reframing the problem, exploring different possibilities, and identifying novel approaches to finding solutions (Plattner, Meinel, & Leifer, 2011). Moreover, design thinking promotes collaboration and cross-disciplinary teamwork. It encourages the involvement of individuals with diverse backgrounds and expertise, facilitating a holistic approach to problem-solving (Brown, 2008). This collaborative aspect fosters creativity and allows for the integration of multiple perspectives in the ideation and prototyping phases. The design thinking process typically consists of five stages: empathize, define, ideate, prototype, and test. These stages are not strictly linear but rather iterative, allowing for continuous refinement and improvement of ideas and solutions (Brown, 2008). Key Points of Design Thinking Take a look at the video below for a nice summary of the concept of Design Thinking: Design Thinking Case Study Now watch the video below for a case study of how Design thinking can be applied to business problems: References Brown, T. (2008). Design thinking. Harvard Business Review, 86(6), 84-92. Kelley, D., & Kelley, T. (2013). Creative confidence: Unleashing the creative potential within us all. Crown Business. Knemeyer, A. M. (2015). Design thinking in supply chain management. Business Horizons, 58(6), 663-672. Hogue,

V. E. (2018). Design thinking and the innovation process. In The Palgrave Handbook of Creativity at Work (pp. 299-314). Palgrave Macmillan. Plattner, H., Meinel, C., & Leifer, L. (2011). Design thinking: Understand–improve–apply. Springer. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=gnWj97CEjeo>
- <https://www.youtube.com/watch?v=rWyClv8bico>

### Links & References

- [Link] [introduction](#)
- [Link] [Chapter 1. Introduction](#)

## Design Thinking Mindset

Module 8 Design Thinking Mindset Icon Progress Bar (browser only) Focus This page introduces you to the mindset and process applied to design thinking. There are readings and short videos to support your learning. A Design Thinking Mindset Design thinking is a non-linear and iterative way of researching that seeks to understand users and solve user problems. The process is perfect for situations where the overall challenge is not clear or you have problems that are as yet unknown. Design thinking is a human-centric approach, it allows designers and researchers to focus on what is best for the user. Underpinning design thinking process is the concept of iteration and the need for prototyping, as both are used to better support the understanding of the user and further develop the design of a product. Each iteration is used to better inform the design and test and clarify issues that occur as you move through the stages of development. This means that if you get to the evaluation/testing phase of the design and you uncover a significant issue in your design you can go back and reevaluate your original concept to either improve or reshape it to fit this new understanding. Iteration will also happen in the design of the prototype. A prototype is an early working model of a design, which you will use with your stakeholders to share, test and explore your ideas of the design of the product and gain feedback (Dam & Siang, 2018). Prototypes are used early within the design phase and themselves are developed over many different iterations. Figure 8.1: The design thinking process source: <https://uxplanet.org/wondering-if-design-thinking-works-102e7b3a3604> Mindset Before we go to the process of Design Thinking is it important to understand the design thinking mindset, which is the foundation of the process. Read this section : How it works : Mindsets from What is Design Thinking by Stanford et al. Read the Airbnb example and see how the thinking has been applied. Philosophy of Design Thinking Watch this accessible overview of the philosophy of Design thinking and its links to other techniques such as Agile and Lean thinking. This series looks at how to get into doing design thinking. Work through the videos in each section. As you are doing this answer these questions: How does this way of researching and designing differ from the ways you have previously done things? What value would this process add to technology research? Design thinking - Understanding the Process Introduction (2:22mins) Where Design thinking fits in a business (7:11mins) Design thinking is not magic (6mins) The Design Thinking Process (6:30) The Pros and Cons of Design Thinking (6.14 mins) Making Design Thinking Happen (9:33) Need

Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] <https://uxplanet.org/wondering-if-design-thinking-works-102e7b3a3604>
- [Link] : [How it works : Mindsets](#)
- [Link] [Introduction](#)
- [Link] [Where Design thinking fits in a business](#)
- [Link] [The Design Thinking Process](#)
- [Link] [The Pros and Cons of Design Thinking](#)
- [Link] [Making Design Thinking Happen](#)

### The Design Thinking Process

Module 8 The Design Thinking Process Focus This page introduces the process of Design Thinking. You will explore: Phases in design thinking Examples of design thinking The non-linear nature of the thinking process There are readings and short videos to support your learning. Further Reading Design Thinking for Educators" by IDEO and Riverdale Country School Read the section on Process and Tools in What is Design Thinking? Read this article on Why design thinking works and note down the key reasons it is impactful in business and innovation. Design Thinking Steps There are many variants of the design thinking process in use today, ranging from 3 to 7 steps, phases, or modes. The underlying thinking that they embody is largely the same. For our purposes, we will be using the five-phase model proposed by the Hasso-Plattner Institute of Design at Stanford, also known as the d.school. The five phases of Design Thinking, according to d.school, are as follows: Empathise Define Ideate Prototype Test Empathise: In this stage, the focus is on understanding the users and gaining empathy for their experiences, needs, and challenges. Designers aim to gather insights through observations, interviews, and immersion in the users' environment. By empathising with users, designers can uncover meaningful opportunities for innovation (Brown, 2008). Define: The define stage involves synthesising the information collected during the empathize stage to identify the core problems or opportunities. Designers analyse and interpret the data to create a clear and actionable problem statement, which serves as a guiding framework for the subsequent stages (Plattner, Meinel, & Leifer, 2011). Ideate: In the ideation stage, designers generate a wide range of ideas without judgment or evaluation. The goal is to foster creativity and encourage "outside-the-box" thinking. Techniques such as brainstorming, mind mapping, and sketching are commonly employed to stimulate idea generation (Brown, 2008). Prototype: The prototype stage involves transforming selected ideas from the ideation stage into tangible representations. Prototypes can take various forms, including physical models, digital simulations, or even storyboards. Prototyping allows designers to test and refine their ideas, gather feedback, and gain valuable insights for further iterations (Kelley & Kelley, 2013). Test: In the final stage, the prototypes are tested and evaluated with the target users. This stage helps designers gather feedback, validate assumptions, and assess the viability and desirability of their solutions. Testing provides valuable insights that inform future iterations and refinements (Brown, 2008). Example of Design Thinking Empathize: The designers begin by empathizing with the target users, which in this case are teachers and students. They conduct interviews, observations, and surveys to understand the challenges faced by educators and the

needs of students in their learning journey. They might visit classrooms, engage in discussions with teachers and students, and collect data to gain deep insights into the user experience.

**Define:** Based on the information gathered during the empathize stage, the designers analyze the data and identify key pain points and opportunities. They might discover that teachers struggle with managing student engagement and personalized learning, while students express a need for interactive and engaging learning experiences. From this analysis, the designers define the problem statement, such as "How might we create an educational app that promotes personalized and interactive learning?"

**Ideate:** In the ideation stage, the designers generate a wide range of ideas for the educational app. They brainstorm ideas individually or in a collaborative session, encouraging a free flow of creativity. They might use techniques like mind mapping or sketching to visualize and explore different possibilities. The focus is on quantity and variety, aiming to generate innovative concepts that address the defined problem.

**Prototype:** Selected ideas from the ideation stage are transformed into tangible prototypes. The designers create a visual representation or interactive mock-up of the educational app. This prototype might include features like interactive lessons, personalized learning paths, and gamified elements. The prototype is developed with the intention of testing and gathering feedback from users to refine the design.

**Test:** The prototype is tested with a small group of teachers and students. The users interact with the app, providing feedback on its usability, functionality, and effectiveness in addressing their needs. The designers observe and collect feedback, noting areas of improvement and identifying any unforeseen issues. This feedback informs further iterations and refinements of the app design.

**Design Thinking Phases** In the previous page we explored what is the design thinking mindset. Here we are going to work through the Design Thinking Process. Follow this chapter of the LinkedIn course on Design thinking: Empathise Define Ideate Prototype Test Chapter Quiz. [Links to an external site](#)

**A Non-Linear Process of Design Thinking** Design thinking is a problem-solving methodology that provides a solution-based approach by considering the bigger picture and acting informatively. It can be applied to various life situations, not just confined to design-related endeavors. While we previously described the design thinking process as a linear sequence of stages leading to user testing, it is important to note that in practice, it is a more flexible and non-linear approach. Design thinking is particularly effective in tackling complex and ill-defined problems. It involves reframing the problem from a human-centric perspective and developing prototypes to test ideas iteratively. This process emphasizes understanding user requirements and constantly seeking feedback from the user audience. Continuous feedback allows for adjustments, redefining ideas, and brainstorming new concepts based on testing the prototypes. By embracing a continuous feedback loop and iterating on ideas, design thinking fosters innovation and encourages creative problem-solving. It empowers individuals to respond to evolving user needs and adapt their solutions accordingly. Different groups within the design team may conduct more than one stage concurrently, or the designers may collect information and prototype during the entire project so as to enable them to bring their ideas to life and visualise the problem solutions. Also, results from the testing phase may reveal some insights about users, which in turn may lead to another brainstorming session (Ideate) or the development of new prototypes (Prototype).

Author/Copyright holder: Teo Yu Siang and Interaction Design Foundation. Copyright licence: CC BY-NC-SA 3.0 It is important to note that the five stages are not always sequential — they do not have to follow any specific order and they can often occur in parallel and be repeated iteratively. As such, the stages should be understood as different modes that contribute to a project, rather than sequential steps.

However, the amazing thing about the five-stage Design Thinking model is that it systematises and identifies the 5 stages/modes you would expect to carry out in a design project — and in any innovative problem-solving project. Every project will involve activities specific to the product under development, but the central idea behind each stage remains the same (Interaction Design, 2023) References Brown, T. (2008). Design thinking. Harvard Business Review, 86(6), 84-92. Kelley, D., & Kelley, T. (2013). Creative confidence: Unleashing the creative potential within us all. Crown Business. Plattner, H., Meinel, C., & Leifer, L. (2011). Design thinking: Understand–improve–apply. Springer. Interaction Design Org Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [Design Thinking for Educators" by IDEO and Riverdale Country School](#)
- [Link] [Read the section on Process and Tools in What is Design Thinking?](#)
- [Link] [Read this article on Why design thinking works and note down the key reasons it is impactful in business and innovation.](#)
- [Link] [Empathise](#)
- [Link] [Define](#)
- [Link] [Ideate](#)
- [Link] [Prototype](#)
- [Link] [Test](#)
- [Link] [Chapter Quiz.](#)

## Module 9: Research Ethics

### Introduction to Module 9

Module 9 Research Ethics: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Nine. In this module you will dive into the topic of research ethics. We will explore some fundamental ethical principles researchers need to keep in mind, encounter relevant professional codes of practice. As well as examining how to maintain research integrity, you will investigate how to conduct research in a responsive way to Kaupapa Māori - i.e, in "a Māori way". Finally, you will learn strategies for avoiding research misconduct and will gain the skills and knowledge necessary for being able to submit a Research Ethics Application. Objectives At the completion of this module, you will be able to: Understand the basics of Research Ethics and explain why it matters Be familiar with some professional codes of practice for research in Aotearoa New Zealand Apply the principles of Kaupapa Māori Research ethics to your research thinking Apply Research Ethics into your Research Design Construct a Research Ethics Application

### Research Ethics

Module 8 Research Ethics Focus This page introduces you to the topic of research ethics. You will: Study ethical principles of research, Analyse professional codes of conduct relevant to research in Aotearoa New Zealand There are readings and short videos to support your learning. Further Reading Royal Society Te Apārangi's New Zealand Code of Professional

Standards and Ethics Institute of Electrical and Electronics Engineers (IEEE): IEEE Compliance and Ethics. Association for Computing Machinery (ACM): ACM Code of Ethics and Professional Conduct International Association of Scientific, Technical, and Medical Publishers (STM): STM Code of Conduct for Journal Publishers Elsevier: Elsevier Publishing Ethics Resource Kit Springer Nature: Springer Nature Research Integrity Policies and Guidelines Research Ethics Ethics in research is of utmost importance as it ensures the proper and responsible conduct of studies. With the vast amount of information available online, researchers must take the necessary measures to ensure the integrity of their research. Regardless of the field of study, it is essential to adhere to ethical, legal, and safety requirements to protect the rights and well-being of individuals involved in the research process. Researchers must recognize that their work has an impact on people's lives and must be conducted with sensitivity and respect. The Ngā Tikanga Paihere framework, developed by the Department of Statistics, provides guidelines for ethical and culturally appropriate data use in Aotearoa New Zealand. Familiarizing oneself with this framework and following its guidelines is crucial to protect the privacy rights of the people and ensure ethical research practices. By adhering to ethical guidelines, researchers demonstrate their commitment to upholding the rights and well-being of participants, ensuring informed consent, safeguarding privacy, and addressing any potential risks or harms that may arise from the research process. Ethical considerations also extend to the responsible use of data, ensuring its accuracy, reliability, and appropriate dissemination. Researchers must continuously engage in reflective practices, evaluating the ethical implications of their research at every stage. This involves critically examining research protocols, data collection methods, analysis techniques, and the ways in which research findings are shared with the public and relevant stakeholders. By following ethical guidelines and frameworks like Ngā Tikanga Paihere, researchers can contribute to the advancement of knowledge in a manner that is respectful, responsible, and aligned with the principles of ethical conduct. Ethical research practices not only protect the rights and well-being of individuals but also enhance the credibility and integrity of the research field as a whole. Source:

<https://www.roehampton.ac.uk/general-information/ethics-principles-ethos/> Research Ethics Process Research ethics involves the application of moral principles to academic research. As a researcher, it is essential to recognise the ethical responsibility that comes with conducting research and consider the potential impact on various stakeholders, including yourself, your research team, your educational institution, colleagues, and broader communities. Regardless of your field of study, whether it be information technology, liberal arts, natural sciences, or fine arts, ethical considerations may arise at different stages of your research, such as data collection, writing, or publishing. While the fundamental principles of research ethics apply universally, it is important to note that specific ethical standards may vary across disciplines. As a postgraduate researcher, it is crucial to proactively address the ethical implications of your research. During the planning phase of your project, it is advisable to consult with a faculty advisor or mentor who can provide guidance on the ethical and legal considerations specific to your discipline. Engaging in these discussions early on will help ensure that your research is conducted in an ethically sound manner, upholding the integrity of your work and safeguarding the rights and well-being of those involved. History of Research Ethics Please watch the following video from Yale University, which is very helpful in understanding the outcry to the abuses that have occurred in the name of research and how the drive to stop these happening formed the basis of research ethics: The story of Research Ethics is an interesting one. Continue by watching this short documentary on the history of research ethics and

consider the impact that ethics has had on research as a whole: Why is Research Ethics so Important? From watching the videos above, you will have gained an understanding of why research ethics matters. Paying attention to ethical practices in research is very important. Please watch the video below which shares some more ideas on this subject: Professional Codes for Researchers The Royal New Zealand Society of Researchers has a Research Charter for New Zealand , with professional codes that members of the society are bound to obey. It is also used on a voluntary basis by others. The Research Charter for Aotearoa New Zealand is a resource suitable to be adopted in full or in part by any organisation developing its own policy or code. The Charter sets out the principles underpinning sound research practice in Aotearoa New Zealand. In practice, ethics and integrity in the research community has several components. Working with human subjects In New Zealand, research involving the collection of personal information must meet established standards of practice. Many research organisations use ethics committees, including independent lay members, to review and approve proposals for their adherence to good practice. Working with animal subjects In New Zealand, research involving the use of animals must meet established standards of practice. New Zealand is a member of the Australia New Zealand Council for Care of Animals in Research and Teaching ( ANZCCART Links to an external site. ). Publication ethics Publication ethics include matters such as plagiarism, self-plagiarism and falsification of published results. The Committee on Publication Ethics Links to an external site. is an international body that has established a model of best practice. Public engagement by researchers, scholars and scientists Modern society expects to be able to openly discuss and debate research findings, and sometimes to participate in the design and execution of research and its dissemination. As Royal Society Te Apārangī, we have published Public Engagement Guidelines for Researchers, Scholars and Scientists for such engagement. Research integrity Research integrity is a term used to describe the adherence to good ethical standards during the conduct of research, excluding the aspects of publication and working with human and animal subjects. The Royal Society of New Zealand's Code of Professional Standards and Ethics includes research integrity. The code of Professional Standards and Ethics in Science, Technology, and the Humanities sets out the standards for exemplary ethical behaviour and world class research and scholarly practices in Aotearoa New Zealand. You can download a poster of the code here. The code has 12 values and principles shown in the figure below. The code also sets out the research and scholarly practices consistent with the values and principles and the responsibilities and standards for Members of the Royal Society Te Apārangī to meet. References 1. Royal Society Te Apārangī. (2021). What is research practice? Retrieved from <https://www.royalsociety.org.nz/what-we-do/research-practice/what-is-research-practice/> Activity: Professional Standards and Ethical Codes for Researchers Read the following and then proceed to the task below: Why does a Code of Ethics matter so much? It provides a common set of rules or standards for all in the profession to adhere to It defines best practices for the profession It provides a basis to meet compliance requirements for the profession It provides a legal standard for the profession It provides a “seal of approval” that those outside the profession can look for when using/selecting our services It helps as a watchdog for the profession Task 1. Download and review the interpretation of the Royal Society Aotearoa NZ Code of Professional Standards and Ethics and the Poster giving an overview of the Code . 2. Consider the following hypothetical scenario: As part of your research, you have to interview people who live in social housing to find out what smart utilities would be useful in their homes. Think about how the code could guide your research behaviour Consider the research tools (surveys, questionnaires,

interviews), the methodology (experimental, qualitative, quantitative etc), and the types of groups you would work with Would you conduct your research alone or with a group? How would you check and publish your results? Write a page of notes about the things that you would need to consider. This could be a couple of paragraphs, bullet points, or a Mindmap. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- [https://www.youtube.com/watch?v=jD-YCDE\\_5yw](https://www.youtube.com/watch?v=jD-YCDE_5yw)
- <https://www.youtube.com/watch?v=Alv90bASoRI>
- [https://www.youtube.com/watch?v=1Aqt64VN4\\_4](https://www.youtube.com/watch?v=1Aqt64VN4_4)

### Links & References

- [Link] [Code of Professional Standards and Ethics](#)
- [Link] [IEEE Compliance and Ethics](#).
- [Link] [ACM Code of Ethics and Professional Conduct](#)
- [Link] [STM Code of Conduct for Journal Publishers](#)
- [Link] [Elsevier Publishing Ethics Resource Kit](#)
- [Link] <https://www.roehampton.ac.uk/general-information/ethics-principles-ethos/>
- [Link] [Research Charter for New Zealand](#)
- [Link] [Research Charter for Aotearoa New Zealand](#)
- [Link] [ANZCCART](#)
- [Link] [Committee on Publication Ethics](#)
- [File] [poster of the code](#)
- [Link] <https://www.royalsociety.org.nz/what-we-do/research-practice/what-is-research-practice/>
- [File] [Royal Society Aotearoa NZ Code of Professional Standards and Ethics](#)
- [File] [overview of the Code](#)

### Responsible Research

Module 9 Responsible Research Icon Progress Bar (browser only) Focus This page introduces: Research Integrity Kaupapa Māori Research Other ethical guidelines There are readings and short videos to support your learning. Further Reading Ethics in Research in Research Methodology, Chandra and Harindran, 2017 Five Principles for Research Ethics (American Psychological Association) The Research Ethics Guidebook: a Resource for Social Scientists (The Economic and Social Research Council, UK) Ethical Guidelines for Good Research Practice (Association of Social Anthropologists, UK) Australian Code for the Responsible Conduct of Research, 2018 (Australian Government) ESRC Framework for Research Ethics 2015 (The Economic and Social Research Council, UK) Research Integrity If all research is ultimately about creating knowledge, then research integrity is about ensuring that this knowledge can be trusted . Trustworthy research maintains its integrity when researchers keep accurate records and data and follow uniform procedures throughout the research process. Ethics can be defined as norms of conduct that distinguish between acceptable and unacceptable behaviour. This

short video below provides an introduction to research integrity and ethics: Massey University has its Code of Ethical Conduct for teaching and research involving human subjects. Download Code of Ethical Conduct for teaching and research involving human subjects. Review this document to see how it compares to the documents of The Royal Society of Aotearoa New Zealand: Research Practice Code-of-professional-standards-and-ethics/ Research practice charter Kaupapa Māori Research Many of the considerations you've encountered on this page have universal applicability. Keep these in mind as we introduce the concept of Kaupapa Māori research, as you will see many relevant crossovers. Kaupapa Māori research is done by Māori within a Kaupapa Māori Research paradigm. This is Māori Researchers working with Māori, as they understand the Māori way or Te Ao Māori, the Māori world view. Kaupapa Māori is literally 'a Māori way'. Graham Smith describes Kaupapa Māori as: Related to 'being Māori' Connected to Māori philosophy and principles Taking for granted the validity and legitimacy of Māori Taking for granted the importance of Māori language and culture, and Concerned with the 'struggle for autonomy over our own cultural well-being' Kaupapa Māori is about the centering of the Māori world and the legitimation of Māori realities. Kaupapa Māori research must take into account the history of Māori experiences with non-Māori researchers and how Māori have been denied sovereignty within these processes. Our practice must also take into account Māori protocols and customs. In this we are primarily accountable to Māori. Seven Māori cultural values can guide Kaupapa Māori research (Cram, 2009; Smith, 1999). These were first mentioned by Linda Smith in her 1999 book, *Decolonizing Methodologies*, and then picked up by Fiona Cram in her writing about Kaupapa Māori and Indigenous research ethics: Aroha ki te tangata - a respect for people that within research is about allowing people to define the research context (e.g., where and when to meet). It is also about maintaining this respect when dealing with research data (e.g., quantitative research), and extends to the physical sciences when research involves, for example, the examination of human tissue samples. He kanohi kitea - being a face that is seen and known to those who are participating in research. For example, researchers should be engaged with and familiar to communities so that trust and communication is developed. Titiro, whakarongo...kōrero - Look, listen and then, later, speak. Researchers need to take time to understand people's day-to-day realities, priorities and aspirations. In this way the questions asked by a researcher will be relevant. Manaaki ki te tangata - looking after people. This is about sharing, hosting and being generous with time, expertise, relationships, etc. Kia tupato - be cautious. Researchers need to be politically astute, culturally safe, and reflexive practitioners. Staying safe may mean collaborating with elders and others who can guide research processes, as well as the researchers themselves within communities. Kaua e takahia te mana o te tangata - do not trample on the mana (dignity) of people. People are often the experts on their own lives, including their challenges, needs and aspirations. Look for ways to collaborate on research reports, as well as research agendas. Kia mahaki - be humble. Researchers should find ways of sharing their knowledge while remaining humble. The sharing of expertise between researchers and participants leads to shared understanding that will make research more trustworthy. Source:

<https://www.katoa.net.nz/kaupapa-maori/kaupapa-maori-research-ethics> Links to an external site. These values have since become known as Community-Up Research Practices in acknowledgement that they are good practices for all researchers to follow. They can also spark communities to come up with own guidelines for how they want to be treated within research projects. Need Help? Looking for help? Seeking an answer to your question? Want to share

some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- [https://www.youtube.com/watch?v=9pCfq5QH\\_v4](https://www.youtube.com/watch?v=9pCfq5QH_v4)

### Links & References

- [Link] [Ethics in Research in Research Methodology, Chandra and Harindran, 2017](#)
- [Link] [Five Principles for Research Ethics \(American Psychological Association\)](#)
- [Link] [The Research Ethics Guidebook: a Resource for Social Scientists \(The Economic and Social Research Council, UK\)](#)
- [Link] [Ethical Guidelines for Good Research Practice \(Association of Social Anthropologists, UK\)](#)
- [Link] [Australian Code for the Responsible Conduct of Research, 2018 \(Australian Government\)](#)
- [File] [ESRC Framework for Research Ethics 2015 \(The Economic and Social Research Council, UK\)](#)
- [File] [Code of Ethical Conduct for teaching and research involving human subjects.](#)
- [File] [https://learn.mywhitecliffe.com/courses/2671/files/427122/download?download\\_frd=1](https://learn.mywhitecliffe.com/courses/2671/files/427122/download?download_frd=1)
- [Link] [Research Practice](#)
- [Link] [Code-of-professional-standards-and-ethics/](#)
- [Link] [Research practice charter](#)
- [Link] <https://www.katoa.net.nz/kaupapa-maori/kaupapa-maori-research-ethics>

### Research Ethics Process

Module 9 Research Ethics Process Icon Progress Bar (browser only) Focus This page introduces: When to consider Ethics in Research Ethics Process Research Misconduct Completing a Research Proposal Ethics Application There are readings and short videos to support your learning. Further Reading Research Ethics Guidance When to consider Ethics in Research? Any research that involves human or animal participants must consider ethics. The ramifications of the research must be discussed with your tutor and evaluated by reference to the institution's codes and policies on research. For many research studies, this process of peer review will provide a sufficient ethical evaluation. Where required by the Tutor, modifications to proposed research design and procedures should be made. Your research proposal must be approved by an ethics committee or, for very low risk research, be formally notified to the Research Ethics Office. All ethics notifications and applications must be read and approved by your tutor or supervisor before being submitted. As you go through your research, there will be many ethical issues to deal with. Watch the video below for a discussion of the ethical issues in research (30.54 mins) There are ethical considerations to take into account throughout the various phases of research. Below are some of the considerations that will need to be taken into account at certain stages in the research process. Research Design & Data Analysis Read this article What are the ethical considerations in research design and list down the key ethical considerations when deciding on your research design. In the Massey University code of ethics

for teaching and research with human subjects, they provide the following table with the considerations that the researcher needs to take into account when working with data. When you are setting up your research, you may need to apply for a full ethical consideration or just give notification if the data or the process of collecting the data is judged low risk. Also, in the Massey code of ethics for teachers and researchers, there is the below flow chart giving advice on the publishing and sharing of Data which may be sensitive. Research Misconduct Scholarly misconduct is the intentional falsification, fabrication, distortion, or misrepresentation of data or another part of the research process. This offence can have severe repercussions within an academic institution and the larger research community and can also have legal ramifications. Scholarly misconduct is especially harmful because it undermines the integrity of not only the offending work but of the entire field of research. For example, take the case of a renowned biologist who falsified data about intelligence in monkeys: The researcher lost his job and was forced to retract a scholarly paper, which had been cited over 100 times in the literature. Even worse, this instance of scholarly misconduct cast doubt on the researcher's other publications, especially those that cited the retracted paper — and some of these other papers had been cited thousands of times in the literature. Ultimately, even when just a small piece of the scientific literature lacks integrity, the science underlying a whole research area can become tainted. Research misconduct can be described as: Fabrication, falsification, plagiarism or deception in proposing, carrying out or reporting the results of research. Failure to declare or manage serious conflicts of interest. Conducting research without ethics approval as required by legislation. Wilful concealment or facilitation of research misconduct by others. Research Ethics Application The need to obtain research ethical approval is common to all research involving human and animal participants. This approval must be obtained before research participants can be approached and before data collection can begin. The process of ethical review is one way that research participants can be confident that possible risks have been considered, minimised and deemed acceptable. This lab explores some of the main issues researchers should consider when planning an application for research ethical approval by answering the following six questions: Do I need research ethical approval? How many applications will I need to make? Where should I apply for research ethical approval? What do I need to include in my application? What do research ethics committees look for?' and What other approvals might I need? Answering these questions will require you to navigate the Whitecliffe ethical review process. Proceed to the Lab activity on the next page, to undertake this process. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

#### Videos

- <https://www.youtube.com/watch?v=S4WaifPcsI0>

#### Links & References

- [Link] [Research Ethics Guidance](#)
- [Link] [What are the ethical considerations in research design](#)

## Lab 4: Ethics - Completing a Research Proposal Ethics Application

IT8101 Lab 4: Ethics - Completing a Research Proposal Ethics Application Summary In the course of this module, we have looked at different organisations' codes of ethics and policies. In this lab you will complete an Ethics Application for your Research Proposal. You will send it to your lecturer for feedback. Ethical Approval The need to obtain research ethical approval is common to all research involving human and animal participants. This approval must be obtained before research participants can be approached and before data collection can begin. The process of ethical review is one way that research participants can be confident that possible risks have been considered, minimised and deemed acceptable. This lab explores some of the main issues researchers should consider when planning an application for research ethical approval by answering the following six questions: Do I need research ethical approval? How many applications will I need to make? Where should I apply for research ethical approval? What do I need to include in my application? What do research ethics committees look for?' and What other approvals might I need? Answering these questions will require you to navigate the Whitecliffe ethical review process. Access the following documents: Research and Ethics Policy Research Policy guidelines Student Research Proposal Ethics Application Form. Further Resources Read through the following resources: Applying for Ethical approval: Basic Principles Ethics for research with Māori or Pasifika in Health FAQ about Ethics Approval AUT nz Information about the Ethics process SIT WHO Ethics approval processes Now work through the following questions in relation to the Whitecliffe Ethics Approval process: How often does the Ethics Committee meet? How long in advance of a meeting do you need to submit your application? What are the Ethics Committee Terms of Reference? Where can you get support in filling out your Student Research Proposal Ethics Application Form? What are the key principles in considering applications for ethical approval that the Ethics Committee will be guided by? Which principles relate to your research proposal? For each principle what information will you be required to supply? Will you need any approval from your research community? How will you get this approval? Complete the Student Research Proposal Ethics Application Form for your research proposal and send it to your lecturer for feedback.

### Links & References

- [File] [Research and Ethics Policy](#)
- [File] <https://learn.mywhitecliffe.com/courses/2671/files/426295/download>
- [File] [Research Policy guidelines](#)
- [File] <https://learn.mywhitecliffe.com/courses/2671/files/428190/download>
- [File] [Student Research Proposal Ethics Application Form.](#)
- [File] <https://learn.mywhitecliffe.com/courses/2671/files/428181/download>
- [Link] [Applying for Ethical approval: Basic Principles](#)
- [Link] [Ethics for research with Māori or Pasifika in Health](#)
- [Link] [FAQ about Ethics Approval AUT nz](#)
- [File] [Information about the Ethics process SIT](#)
- [File] <https://learn.mywhitecliffe.com/courses/2671/files/427918/download>
- [File] [WHO Ethics approval processes](#)
- [File] <https://learn.mywhitecliffe.com/courses/2671/files/426641/download>

## Module 10: Bibliographic Tools

### Introduction to Module 10

Module 10 Bibliographic Tools: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Ten. This module focuses on referencing in research. You will look closely at the problem of plagiarism and explore citation methods and tools to ensure correct and diligent referencing is achieved, investigating different styles of citation - APA, Harvard, Chicago, etc. This module guides you further on how to evaluate the reliability of a source and how to identify the perspective or argument of an author. Lastly, you will be introduced to useful bibliographic tools and reference management systems, which will be invaluable for all your future research activities. Objectives At the completion of this module, you will be able to: Define plagiarism and how to avoid it Identify different citation styles Evaluate the reliability of sources Gain confidence using reference management systems and bibliographic tools

### Plagiarism

Module 10 Plagiarism Icon Progress Bar (browser only) Focus This page introduces the subject of plagiarism and explains the fundamentals of referencing. There are readings and short videos to support your learning. Further Reading 5 Things Students Might Not Know about Plagiarism Avoiding Plagiarism by Purdue Online Writing Lab (OWL) Plagiarism: What It is and How to Recognize and Avoid It by Indiana University Bloomington Plagiarism and Academic Integrity by Harvard College Writing Program Understanding and Preventing Plagiarism by Council of Writing Program Administrators What is Plagiarism? Plagiarism (pronounced: play -juh-riz-um) is the act of taking someone else's words, ideas, or information and passing them off as your own. If you do not give credit to the author of these ideas in footnotes or endnotes and a bibliography, you are committing plagiarism, which is a serious academic offence. Everything you find that is written, whether in print in books and journals, or on the web, should be considered copyrighted. That means that you should think of it as belonging to someone else. Information that you find on the web is not free to take or use – it is someone else's intellectual property. Any material lifted from an original source, including web resources, without proper acknowledgment or credit is considered plagiarised. Inadvertent or accidental plagiarism is still plagiarism. Plagiarism can inadvertently happen if you are not careful about taking notes while you research; it is sometimes difficult to remember exactly where your ideas came from when you are doing research, so remember to cite your sources while you work. This infographic below from academicplagiarism.com summarises the different types of plagiarism. The academicplagiarism.com website has some good tutorials and resources on academic writing and avoiding plagiarism. source:

<https://academicplagiarism.com/wp-content/uploads/2015/05/PLAGIARISM.jpg> It is your responsibility to know what constitutes plagiarism. Not knowing citation standards is not an excuse. When in doubt, err on the side of over-documentation and cite the source. You can also ask your professor, teaching assistant, or a librarian for help in determining what is and is not plagiarism. Watch the video below on Plagiarism Links to an external which gives a great overview of the challenge of plagiarism and some tips you can use to avoid it. Why do I have to Cite Sources? All scholarly or academic work requires that you cite your sources, whether you are writing a long paper or a quick report. Researching and writing a paper ideally involves a

process of exploring and learning. By citing your sources, you are showing your reader how you came to your conclusions and acknowledging the other people's work that brought you to your conclusions. Documenting your research and scholarship Acknowledging the work of others whose scholarship contributed to your work Helping your reader understand the context of your argument Providing information for your reader to use to locate additional information on your topic Establishing the credibility of your scholarship Providing you with an opportunity to demonstrate your own integrity and understanding of academic ethics. When Do I Need to Cite Sources? Sometimes your discipline will have specific requirements about when and how to cite your sources. However, these are some guidelines that generally apply. Always cite your sources when: You quote verbatim from someone else's work. You summarise or paraphrase someone else's work. You are conveying information which may be unfamiliar to your reader (e.g. statistics or factual data). You are not sure whether or not you need to cite. As mentioned above, when in doubt, err on the side of caution. It is better to give your reader more information rather than less. Remember, if you have a question about whether or not you need to cite your source, you can ask your lecturer or a librarian for help. Sometimes citing your sources takes a little bit of time and effort, but it is worth it to avoid plagiarism. Further Resources on LinkedIn Learning We recommend watching the three videos below to help put your learning into practice: What is citation and why cite (2:36 mins) How to cite (4:30 mins) Citation for visual images and media (2:20 mins) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [5 Things Students Might Not Know about Plagiarism](#)
- [Link] [Avoiding Plagiarism by Purdue Online Writing Lab \(OWL\)](#)
- [Link] [Plagiarism: What It is and How to Recognize and Avoid It by Indiana University Bloomington](#)
- [Link] [Plagiarism and Academic Integrity by Harvard College Writing Program](#)
- [Link] [Understanding and Preventing Plagiarism by Council of Writing Program Administrators](#)
- [Link] [academicplagiarism.com](https://academicplagiarism.com)
- [Link] <https://academicplagiarism.com/wp-content/uploads/2015/05/PLAGIARISM.jpg>
- [Link] [Plagiarism](#)
- [Link] [What is citation and why cite](#)
- [Link] [How to cite](#)
- [Link] [Citation for visual images and media](#)

### Citation Styles

Module 10 Citation Styles Icon Progress Bar (browser only) Focus This page introduces you to a variety of different citation styles used widely by academic researchers. You may be familiar with some, or all, of these styles already. There are readings and short videos to support your learning. Citation Styles In this section, we will cover different citation styles used in research work. Whitecliffe School of IT typically uses the APA referencing style. However, as a

researcher, you will come across publications using different referencing styles. The referencing style is usually dictated by the institution or publication. We discuss some of the key styles below. APA Style APA Style provides a foundation for effective scholarly communication because it helps writers present their ideas in a clear, precise, and inclusive manner. Whitecliffe uses the APA reference style. APA Style originated in 1929, when a group of psychologists, anthropologists, and business managers convened and sought to establish a simple set of procedures, or style guidelines that would codify the many components of scientific writing to increase the ease of reading comprehension. They published their guidelines as a seven-page article in *Psychological Bulletin* describing a “standard of procedure, to which exceptions would doubtless be necessary, but to which reference might be made in cases of doubt”. Since then, the scope and length of the *Publication Manual* have grown in response to the needs of researchers, students, and educators across the social and behavioural sciences, health care, natural sciences, humanities, and more; however, the spirit of the original authors’ intentions remains. The *Publication Manual of the American Psychological Association, Seventh Edition* is the official source for APA Style. With millions of copies sold worldwide in multiple languages, it is the style manual of choice for writers, researchers, editors, students, and educators in the social and behavioural sciences, natural sciences, nursing, communications, education, business, engineering, and other fields. Known for its authoritative, easy-to-use reference and citation system, the *Publication Manual* also offers guidance on choosing the headings, tables, figures, language, and tone that will result in powerful, concise, and elegant scholarly communication. It guides users through the scholarly writing process--from the ethics of authorship to reporting research through publication. The seventh edition is an indispensable resource for students and professionals to achieve excellence in writing and make an impact with their work. The seventh edition has been thoroughly revised and updated to reflect best practices in scholarly writing and publishing. Here is a table with examples of in-text citations in APA 7th edition style for different source types: Source Type In-Text Citation Example Book with a single author (Author's Last Name, Year, p. Page Number) Journal article (Author's Last Name, Year) Webpage with no author ("Title of Webpage," Year, para. Paragraph Number) Source with multiple authors (First Author's Last Name et al., Year) Source with organization author (Abbreviated Name of Organization, Year, p. Page Number) Online report/document (Author's Last Name, Year, section/chapter name, para. Paragraph Number) Source with no date available (Author's Last Name, n.d., p. Page Number) Online video with timestamp (Author's Last Name, Year, timestamp mm:ss) Here is a table with the common types of citations in APA 7th edition style: Type of Source Format Book with a single author Last name, Initial(s). (Year). Title of book. Publisher. Book with two authors Last name, Initial(s), & Last name, Initial(s). (Year). Title of book. Publisher. Book with three to twenty authors List the first 19 authors' names, followed by an ellipsis (...) and the last author's name. Book with an editor Last name, Initial(s) (Ed. or Eds.). (Year). Title of book. Publisher. Chapter in an edited book Last name, Initial(s) of chapter author. (Year). Title of chapter. In Last name, Initial(s) (Ed.), Title of book (pp. page range of chapter). Publisher. Journal article Last name, Initial(s) of author(s). (Year). Title of article. Title of Journal, Volume(Issue), page range. DOI or URL (if available). Magazine article Last name, Initial(s) of author(s). (Year, Month Day). Title of article. Title of Magazine, page range. Newspaper article Last name, Initial(s) of author(s). (Year, Month Day). Title of article. Title of Newspaper, page range. Website article Last name, Initial(s) of author(s) or organization. (Year, Month Day). Title of article. Title of Website. URL. Webpage Name of Organization or

Author. (Year, Month Day). Title of webpage. Title of Website. URL. Online report or document  
 Last name, Initial(s) of author(s) or organization. (Year). Title of report/document (Report No. if applicable). Publisher. URL or DOI. Thesis or dissertation Last name, Initial(s) of author. (Year). Title of thesis/dissertation (Publication No. if applicable) [Doctoral dissertation or Master's thesis, Name of Institution]. Repository or URL.

APA Tutorials The official webpage of APA contains a handful of resources to learn APA style citation. Below are few links to learn this style and also includes few sample papers written by using this citation style. APA Style website Style and Grammar Guidelines Examples of most commonly requested formats Scribblr Sample Papers Chicago Style The Chicago Manual of Style Online is the venerable, time-tested guide to style, usage, and grammar in an accessible online format. It is the indispensable reference for writers, editors, proof-readers, indexers, copywriters, designers, and publishers, informing the editorial canon with sound, definitive advice. The Chicago Manual of Style presents two basic documentation systems: The humanities style (notes and bibliography) The author-date system Choosing between the two often depends on subject matter and nature of sources cited, as each system is favoured by different groups of scholars. If you are unsure what system you should use for your paper, please ask your professor, teaching assistant, or publisher which you should follow. You must choose to use either the notes and bibliography system ([N] and [B]) or the author-date system ([T] and [R]). Do not mix the two systems. Again, ask your professor if you are unsure which is preferred.

Humanities Style The humanities style is preferred by many in literature, history, and the arts. This style presents bibliographic information in notes and, often, a bibliography. It accommodates a variety of sources, including esoteric ones less appropriate to the author-date system. Author-Date System The more concise author-date system has long been used by those in the physical, natural, and social sciences. In this system, sources are briefly cited in the text, usually in parentheses, by author's last name and date of publication. The short citations are amplified in a list of references, where full bibliographic information is provided.

Chicago Tutorials The official website of The Chicago Manual of Style is a good starting point to learn more about this citation style: The Chicago Manual of Style IEEE Style IEEE is a numbered referencing style, where citations are provided as a bracketed number in the text (i.e. "as demonstrated in [1]"), which corresponds to a numbered reference list at the end of the document. The reference list is therefore arranged in the order that works are cited in the text, not in alphabetical order. Only one reference is given per reference number. Repeated references to a publication carry the same reference number; a resource is only listed once in the reference list. In the text, citations should be given simply as "in [1] ...," rather than as "in reference [1] ..." It is not necessary to mention either author names or publication dates in the text unless they are specifically relevant to the discussion. If a specific page or figure needs to be highlighted, this can be done in-text, i.e., [5, pp. 4-6], or [10, Fig. 1]. The reference list should only include the references, with no commentary or introductory phrases such as "For example, see." If necessary, such commentary should be included in the text or in a footnote on the same page as the citation.

Example Text: As demonstrated in [1]... Other studies [2], [3] have found... References: [1] M. M. Chiampi and L. L. Zilberti, "Induction of electric field in human bodies moving near MRI: An efficient BEM computational procedure," IEEE Trans. Biomed. Eng., vol. 58, no. 10, pp. 2787–2793, Oct. 2011, doi: 10.1109/TBME.2011.2158315. [2] G. Veruggio, "The EURON roboethics roadmap," in Proc. Humanoids '06: 6th IEEE-RAS Int. Conf. Humanoid Robots, 2006, pp. 612–617, doi: 10.1109/ICHR.2006.321337. [3] B. Klaus and P. Horn, Robot Vision. Cambridge, MA, USA: MIT Press, 1986 More details on IEEE style citation can be found on below links: IEEE

Author Centre IEEE Editorial Style Manual Harvard Style The Harvard writing style, is also known as parenthetical style or referencing, encompasses two main types of in-text referencing: author-date and author-title. This type of citation is used by many universities and is often a preferred style for arts and humanities. Below are some links to guides to using the Harvard styles from university libraries that use this style. Citation Guide Monash Uni Harvard Referencing Auckland University An example of the Harvard Referencing style: A Chapter in a book: Author, X., Author, Y., and Author, Z. Year. Title of the Chapter: Subtitle. In: Editor M and Editor N (eds) Title of the Book. Place: Publisher, 11–22. ACM Style The Association for Computing Machinery (ACM), the world's largest educational and scientific computing society, delivers resources that advance computing as a science and a profession. For a technology researcher this is a good website to explore. An ACM referencing style guide is presented here - ACM Referencing Guide Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [APA Style website](#)
- [Link] [Style and Grammar Guidelines](#)
- [Link] [Examples of most commonly requested formats](#)
- [Link] [Scribblr](#)
- [Link] [Sample Papers](#)
- [Link] [The Chicago Manual of Style](#)
- [Link] [IEEE Author Centre](#)
- [Link] [IEEE Editorial Style Manual](#)
- [Link] [Citation Guide Monash Uni](#)
- [Link] [Harvard Referencing Auckland University](#)
- [Link] [ACM Referencing Guide](#)

### Evaluating Sources

Module 10 Evaluating Sources Icon Progress Bar (browser only) Focus This page introduces the idea that not all sources are equal and provides tips for choosing which sources are reliable and which are best to avoid. There are readings and short videos to support your learning.

Understanding the Importance of Using Good Sources Understanding the importance of using good sources is vital for students conducting research. Here are some key reasons why using good sources matters: Accuracy and Credibility: Good sources provide accurate and reliable information. When you use reputable sources, you can trust that the information you're citing is valid and supported by evidence. It adds credibility to your own work and helps build trust with your readers. Building a Strong Foundation: Good sources help you build a solid foundation for your research. By referring to high-quality and relevant sources, you show that you have thoroughly explored the topic and considered different perspectives. This strengthens the validity of your arguments and positions your work within the existing body of knowledge. Avoiding Misinformation: Good sources help you avoid misinformation and inaccuracies. By relying on reliable sources, you decrease the chances of including false or misleading information in your research. This ensures that your work is based on facts and contributes to

the overall accuracy of your findings. **Gaining Insights and Understanding :** Good sources provide valuable insights and a deeper understanding of your topic. When you use well-regarded sources, you benefit from the expertise and knowledge of established researchers and scholars. This broadens your own understanding and enhances the quality of your work.

**Ethical Responsibility:** Using good sources demonstrates your ethical responsibility as a researcher. By properly citing your sources, you acknowledge the intellectual property of others and give credit where it is due. This upholds academic integrity, avoids plagiarism, and fosters a respectful academic community.

**Building Trust and Reputability:** Using good sources helps you build trust with your readers and establish your own reputation as a reliable researcher. When you consistently cite high-quality sources, it demonstrates that you have done thorough research and have taken the time to evaluate the reliability of your information. This enhances the credibility of your work and encourages others to view you as a trusted source of information.

**Examples of Good vs Bad Source**

**Bad Source:** A personal blog written by an anonymous author titled "The Top 10 Programming Languages of All Time." The blog lacks citations, references, and any indication of the author's expertise or qualifications in the field of programming languages. It primarily consists of subjective opinions and anecdotal experiences without any supporting evidence or data. **Why it's a bad source:** This personal blog lacks credibility, reliability, and verifiability. The anonymous author's opinions may not be backed by solid evidence or expert knowledge. The absence of citations or references makes it challenging to verify the information presented. Relying on such a source may lead to inaccurate or biased information, which can compromise the quality and validity of your research.

**Good Source:** A peer-reviewed research article titled "A Comparative Study of Programming Languages for Web Development" published in a reputable IT journal. The article is authored by experts in the field of computer science and provides a comprehensive analysis of various programming languages used in web development. It includes empirical data, methodology, and references to support the findings and conclusions presented. **Why it's a good source:** This research article is considered a good source due to several reasons. Firstly, it is published in a reputable journal, indicating that it has undergone rigorous peer review by experts in the field. The authors are recognised professionals with expertise in programming languages and web development. The article provides a systematic study with supporting evidence, making it reliable and credible. The inclusion of references allows readers to explore the sources and validate the information presented.

**Tips for Evaluating Sources**

**Below are tips for evaluating different sources to avoid bias and critically analyse them:**

**Checking for signs of bias**

Does the author or publisher endorse political or religious views that could affect objectivity? Is the author or publisher associated with a special-interest group, such as Greenpeace or the National Rifle Association that might present only one side of an issue? Are alternative views presented and addressed? How fairly does the author treat opposing views? Does the author's language show signs of bias?

**Assessing an argument**

What is the author's central claim or thesis? How does the author support this claim—with relevant and sufficient evidence or with just a few anecdotes or emotional examples? Are statistics consistent with those you encounter in other sources? Have they been used fairly? Does the author explain where the statistics come from? (It is possible to "lie" with statistics by using them selectively or by omitting mathematical details.) Are any of the author's assumptions questionable? Does the author consider opposing arguments and refute them persuasively? Does the author fall prey to any logical fallacies?

**Evaluating Web sources**

Does the Web site or document have an author? You may need to do some clicking and scrolling to find the author's name. If you have

landed directly on an internal page of a site, for example, you may need to navigate to the home page or find an “about this site” link to learn the name of the author. If there is an author, can you tell whether he or she is knowledgeable and credible? When the author’s qualifications aren’t listed on the site itself, look for links to the author’s home page, which may provide evidence of his or her interests and expertise. Who, if anyone, sponsors the site? The sponsor of a site is often named and described on the home page. What does the URL tell you? The domain name extension often indicates the type of group hosting the site: commercial (.com), educational (.edu), nonprofit (.org), governmental (.gov), military (.mil), or network (.net). URLs may also indicate a country of origin: .uk (United Kingdom) or .jp (Japan), for instance. Why was the site created: To argue a position? To sell a product? To inform readers? Who is the site’s intended audience? How current is the site? Check for the date of publication or the latest update, often located at the bottom of the home page or at the beginning or end of an internal page. How current are the site’s links? If many of the links no longer work, the site may be too dated for your purposes. Video Summary Take a look at the video below, which summarises many of the points that you have just read. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=bZ122WakNDY>

## Bibliographies and Tools

Module 10 Bibliographies and Tools Icon Progress Bar (browser only) Focus This page explains the bibliography and how the use of reference management tools can be beneficial to keeping track of your citations ahead of compiling your bibliography. There are readings and short videos to support your learning. Suggestion We suggest you make notes as you work through these topics to help with your learning and preparation for the upcoming activities and assessments. What is a Bibliography? A bibliography is different from a reference list as it lists all the sources used during your research and background reading, not just the ones you refer to in your writing. It is good practice to include all sources you consulted, even if not referenced directly, because they may have contributed to your thinking and your conclusions more than you realise. Bibliographies allow your sources to be found by your reader. It also gives credit to authors you have consulted for their ideas. All references cited in the text must appear, except for personal communications (such as conversations or emails) which cannot be retrieved. Bibliographies can be compiled manually, following one of the citation styles referenced earlier, or they can be generated automatically if you have been using a reference/citation management system to keep track of your sources. You will read more about this below: Citation Management Tools Citation Management tools are handy to manage your references in an organised way. Typically, they enable you to: Import citations from databases, websites, catalogues Organize citations using folders and tags Attach PDFs, images, etc. to your citations Annotate your citations and/or PDFs Output auto-formatted bibliographies and in-text citations (APA, Chicago, & hundreds more styles) There are different citation tools available and these vary in their functionality and attributes. We will explore EndNote, Mendeley, and BibTex for LaTeX. Two other common citation management tools you may like to explore independently are Zotero and RefWorks. If you choose to use LaTeX to write your research, BibTex will manage the bibliography as well. You may need other citation tools as well as LaTeX

- it depends on how you want to do and manage your research. Reference Managers Watch the YouTube video below to get an overview of the tools, from a student who is using them, with some points about LaTeX as well. Then go to the relevant pages to learn more about the tools that interest you.

**EndNote** EndNote is a reference management programme widely used by the academic community. EndNote allows you to transfer groups of references retrieved from online bibliographic databases into your EndNote database and create bibliographies in a variety of formats using a variety of word processors. Store and organise references Import references from article databases and catalogues Instantly generate bibliographies in different referencing styles Sync with Word to insert in-text citations and full references directly into assignments Support collaboration by making it easy to share references Download EndNote The following videos are helpful resources for downloading/installing EndNote. They also offer guidelines on how to add references using EndNote.

**Mendeley** Mendeley is a tool for reference management, research data management and citation solutions. It provides desktop software and an online library. You may use one or both versions. You can sync your desktop library with your online library easily. Unlike EndNote, it costs nothing to set up an account. Features of Mendeley Free to download Versions available for Windows, Mac and Linux operating systems Use the Web Importer to capture references, and full-text documents (PDF) if available Share your reference folders with colleagues or other researchers Generate formatted in-text citations and reference lists within word processor documents (e.g. Microsoft Word) Import and annotate PDFs Download Mendeley The following are a few good resources about downloading/installing Mendeley. They also offer guidelines on how to add references using Mendeley. Mendeley.com guides and videos AUT Library Mendeley Guides

**BibTeX for LaTeX** LaTeX is a powerful typesetting system widely used in academic and scientific writing. It offers a convenient way to manage citations and references through its companion tool, BibTeX. With LaTeX and BibTeX, you can create properly formatted citations and generate a comprehensive bibliography for your research papers, theses, and articles. Using BibTeX, you can store your references in a separate file (usually with a .bib extension) and cite them within your LaTeX document using specific commands. This separation of content and formatting allows for easy updates and consistent citation styles across multiple documents.

**Tutorial on LaTeX/BibTeX:** Here is a step-by-step tutorial to help you get started with LaTeX citations and BibTeX:

**Setting up your LaTeX document:** Begin by creating a new LaTeX document or opening an existing one in your preferred LaTeX editor.

**Creating a BibTeX file:** Create a new plain text file with a .bib extension (e.g., references.bib) to store your references. Each reference in the BibTeX file follows a specific format. For example: `@article {citekey, author = {Author, A.}, title = {Title of Article}, journal = {Journal Name}, year = { Year }, volume = {Volume}, number = {Issue}, pages = {Page Range }, doi = {DOI}, }`

**Adding citations in your LaTeX document:** Use the `\cite{citekey}` command within your LaTeX document to insert citations. Replace "citekey" with the appropriate key from your BibTeX file. For multiple citations, use the `\cite{citekey1,citekey2,...}` command.

**Compiling your LaTeX document:** Compile your LaTeX document using the appropriate compiler (e.g., `pdflatex`, `latexmk`). Run BibTeX to generate the bibliography by executing the command `bibtex filename` (replace "filename" with the name of your main LaTeX file without the extension). Compile your LaTeX document again to incorporate the generated bibliography.

**Formatting citation style:** Choose a citation style by specifying a bibliography style in your LaTeX document using the `\bibliographystyle{stylename}` command. Different citation styles are available (e.g., plain, apa, ieee). Choose the one that matches your requirements or consult your academic institution's guidelines.

**Generating the bibliography:**

Place the `\bibliography{bibfilename}` command in your LaTeX document to generate the bibliography. Replace "bibfilename" with the name of your BibTeX file (without the extension). Repeating compilation: To ensure that the citations and bibliography are correctly updated, compile your LaTeX document multiple times in the order: LaTeX -> BibTeX -> LaTeX -> LaTeX. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Videos

- <https://www.youtube.com/watch?v=nWIZlrX9V7M>
- <https://www.youtube.com/watch?v=wg6l3wuctoM>
- <https://www.youtube.com/watch?v=oUmAWEW6pBc>

### Links & References

- [Link] [EndNote](#)
- [Link] [Mendeley](#)
- [Link] [Mendeley.com guides and videos](#)
- [Link] [AUT Library Mendeley Guides](#)

## Module 11: Writing in LaTeX

### Introduction to Module 11

Module 11 Writing in LaTeX: Overview Icon Progress Bar (browser only) Kia ora koutou, hello everyone, Welcome to Module Eleven. In this module we will examine LaTeX - a powerful typesetting tool. You will explore its function and investigate why it can be useful for researchers to learn how to use it for producing research documents. You will learn some LaTeX code, practice using and creating templates in LaTeX, and learn how to create a bibliography. The module will finish with a lab activity where you will design and create a document using LaTeX. Objectives At the completion of this module, you will be able to: Understand the function of LaTeX Produce some basic LaTeX code Create bibliographies using LaTeX Design typeset documents using LaTeX.

### What is LaTeX?

Module 11 What is LaTeX? Icon Progress Bar (browser only) Focus This page introduces you to the fundamentals of LaTeX. You will see how it can be a useful tool for researchers and will learn how to get started using it. There are readings and short videos to support your learning. Introduction to LaTeX LaTeX (pronounced LAY-tek or LAH-tek) is a tool used to create professional-looking documents. It is based on the WYSIWYM (what you see is what you mean) idea, meaning you only have to focus on the contents of your document and the computer will take care of the formatting. Instead of spacing out text on a page to control formatting, as with Microsoft Word or LibreOffice Writer, users can enter plain text and let LaTeX take care of the rest. Why Learn LaTeX? LaTeX is used all over the world for scientific documents, books, as well as many other forms of publishing. Not only can it create beautifully typeset documents, but it allows users to very quickly tackle the more complicated parts of typesetting, such as inputting mathematics, creating tables of contents, referencing and creating bibliographies, and having a

consistent layout across all sections. Due to the huge number of open-source packages available (more on this later), the possibilities with LaTeX are endless. These packages allow users to do even more with LaTeX, such as add footnotes, draw schematics, create tables etc. One of the most important reasons people use LaTeX is that it separates the content of the document from the style. This means that once you have written the content of your document, you can change its appearance with ease. Similarly, you can create one style of document which can be used to standardise the appearance of many different documents. This allows scientific journals to create templates for submissions. These templates have a pre-made layout meaning that only the content needs to be added. In fact, there are hundreds of templates available for everything from CVs to slideshows. Watch the video below to learn the history and background of LaTeX: Getting Started with LaTeX The core of LaTeX is a backend software package that compiles the LaTeX code (.tex file) and creates the final document (PDF). There are a few different programs available to do this depending on the operating system of the user. The video below introduces and explains the features in LaTeX. Watch the video to get a comprehensive overview of what LaTeX can do: Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=9eLjt5Lrocw>
- <https://www.youtube.com/watch?v=NXW4cbHBthY>

## Using LaTeX

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### Getting Started with LaTeX

Module 11 Getting Started with LaTeX Icon Progress Bar (browser only) Focus This page introduces essential commands and basic components of LaTeX. There are readings and short videos to support your learning. LaTeX Structures Commands are a vital structure for creating any kind of document in LaTeX. Commands have many uses such as changing document/text formatting, creating equations, delineating sections of a document, and inputting symbols. To use a command, a backslash is placed before the command. For example,  $\pi$  is the command to insert the symbol  $\pi$  and `\begin{matrix}` is one of the commands needed to input a matrix. Generally, commands have the format `\command[optional argument]{required argument}`. Multiple arguments can be used with some commands by separating each argument with a comma, for example, `\hypersetup{colorlinks=true, linkcolor=blue, urlcolor=blue}` sets the font colour of certain hyperlinks in a document. Some commands, such as `\pi`, have neither an optional nor a required argument. Others, such as `\begin{}`, will only work with a recognised required argument. Some commands can also accept an optional argument which is used to specify what action the command will take, for example, `\documentclass[11pt, twocolumn]{article}` tells LaTeX that the current project is an article with 11pt font and should be formatted with two columns. Leaving out the optional argument, `\documentclass{article}`, will cause LaTeX to use the defaults associated with this command. Lastly, some commands may have more than one required argument, for example, `\frac{numerator}{denominator}` creates a fraction where the first argument is the numerator and the second argument is the

denominator. This link is to a set of important LaTeX commands for beginners . Download this LaTeX Crib Sheet as a useful guide as you learn LaTeX: [latex\\_cribsheet.pdf](https://tug.org/texniques/tn10/latex_cribsheet.pdf) (source: [https://tug.org/texniques/tn10/latex\\_cribsheet.pdf](https://tug.org/texniques/tn10/latex_cribsheet.pdf)). Packages Packages are used to supplement the standard features of LaTeX. If you are trying to use a command and find that it is generating an error, one of the most common issues is that you have not included the needed package in your document. The command `\usepackage{}` with the package name as the argument will tell LaTeX to include a certain package in your document. Packages should be placed in the document preamble (see Document Structure below). If you would like to include multiple packages in your document, you can either include the `\usepackage{}` command multiple times or use it once and separate each package name with a comma, e.g.: `\usepackage{geometry} \usepackage{amsmath} \usepackage{geometry, amsmath, graphicx} \usepackage{graphicx}` It is better to list the packages separately if you intend to use an optional argument with any package; otherwise, it just comes down to personal preference. Many standard packages come with a MikTeX or MacTeX download, but you can also install new ones if needed. Some instructions for installing new packages can be found online. Text Writing text in a LaTeX document is easy. Once you are inside the body of the document, all you have to do is start typing. When you compile the code, LaTeX will take care of all the text formatting based on any commands and packages used. Math Incorporating mathematical symbols and equations is one of the most popular features of LaTeX. There are a number of ways to include math in a LaTeX document, which we will discuss later on. Comments Anything that follows a % in a line of code will be treated as a comment and ignored by the compiler. If you want to include a percentage symbol in your text, use the command `\%` . For example, if you want to write 100% in LaTeX, you would write `100\%` . Below is a YouTube tutorial on the ins and outs of using LaTeX. You can choose to look at the topics in this tutorial, listed below. The whole tutorial is included at the bottom of the page. 00:12 Why use LaTeX? 01:03 Working online vs. offline 04:23 Your first line of code 06:40 Paired commands 08:45 Typesetting (and pronouncing) the word LaTeX 11:40 Using TeXmaker 24:09 Soft returns vs. hard returns 28:19 Math mode 33:20 Saving your files Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Links & References

- [Link] [important LaTeX commands for beginners](#)
- [File] [latex\\_cribsheet.pdf](#)
- [Link] [installing new packages](#)
- [Link] [00:12](#)
- [Link] [01:03](#)
- [Link] [04:23](#)
- [Link] [06:40](#)
- [Link] [08:45](#)
- [Link] [11:40](#)
- [Link] [24:09](#)
- [Link] [28:19](#)
- [Link] [33:20](#)

## Setting up Documents with LaTeX

Module 11 Setting Up Documents with LaTeX Icon Progress Bar (browser only) Focus This page introduces setting up documents using LaTeX. You will learn how to construct the document using relevant packages and commands. There are readings and short videos to support your learning.

**Document Structure** To create PDFs with LaTeX the user must generate a .tex file. The .tex file will consist of the code that is compiled by the back-end LaTeX software to generate the end result PDF document. With LaTeX installed .tex files can be created in notepad or textedit on Windows or OS X computers respectively, though it is strongly recommended to generate these files directly in a LaTeX text editor. A LaTeX document consists of two main pieces, the preamble and the body, the bibliography can be considered a third piece depending on how you structure it. It is recommended that the user creates a new folder on their computer for each LaTeX document they are working on. When LaTeX compiles a .tex file it generates several files in order to create the PDF. Having a folder for each individual document you are working on will help keep all of the files generated organised. Additionally, it is possible the user will want to insert images, figures, or other objects into their document (BibTeX files for example) - it is much simpler to insert items into the document when they are located in the same folder as the .tex file.

**Preamble** The preamble to a LaTeX document is all the information that occurs before the document begins. The preamble starts with the command `\documentclass{}` and ends with the command `\begin{document}`. Below is an image of a document preamble: Different objects are placed in the preamble between the commands `\documentclass{}` and `\begin{document}`. The most common of these items is the command `\usepackage{}`. A package informs the compiler of certain commands that will be used and certain procedures to take while generating the final document. Any number of packages can be included in the preamble, and one of the most common mistakes made by the new LaTeX user is to include a command in their document without including the proper package in the preamble. Other commands such as `\title`, `\author`, and `\date` are included in the preamble and then may be incorporated within the document. Lastly, new commands can be created in the preamble that will only be used in the current document.

**Body** The body of a LaTeX document is where all of the text, equations, figures, tables, etc. will be placed. The body of the document begins with the command `\begin{document}` and ends with the command `\end{document}`. A very simple sample code and its output can be seen below. The code contains both the preamble and body of the document. One of the most popular features of LaTeX is its ability to easily incorporate and format mathematical equations and symbols. There are two important things to remember to ensure LaTeX handles mathematical formula properly. First, make sure that you have included all necessary packages. The most common package for using mathematical formulas in a document is "amsmath". Although this package allows the use of a multitude of mathematical symbols and commands, it does not include everything. If you try to compile a document and an error occurs related to a mathematical object you have included, you will want to check if there is an additional package you need. For example, if you try to include bold lettering in a mathematical equation without the package "bm", an error will occur. Second, enter math mode. Math mode lets LaTeX know that something should be formatted as math and not regular text, there are also some symbols and commands that cannot be used outside of math mode. The image below shows the same text,  $f(x) = 2x+5$ , firstly not using math mode, and secondly, using it. There is a very clear difference between how the equation is formatted inside and outside of math mode. There are several different ways to enter math mode. If you would like to write mathematical expression inside a

line of text all the mathematical text should be book ended by  $$. For example, writing  $x+3$  would tell LaTeX to write  $x+3$  as math. If you want math to appear on its own line it what is called display mode, you can use either the command  $$$$  on both sides of the math, or  $[$  and  $\backslash]$ . The previous example would look like this:  $$$x+3$$$  or  $\backslash[x+3\backslash]$ . Finally, some commands will automatically place the user in math mode, for example, any text appearing between the commands  $\backslashbegin{equation}$  and  $\backslashend{equation}$  will be written in math mode.$

Common Commands Preamble commands:  $\backslashdocumentclass\{\}$  : The first command in any document. It is used to specify what type of document is being made. Some common arguments are article (common for scientific journals), book, IEEEtran (for IEEE transaction format), slides, beamer (for presentations).  $\backslashusepackage\{\}$  : This command imports a package into the LaTeX document. This is important for including any commands that are not standard in LaTeX.  $\backslashtitle\{\}$  : The intended title of the document should be used as the argument.  $\backslashauthor\{\}$  : The author of the document should be used as the argument, for multiple authors use  $\backslashand$  to separate authors name or use the authblk package.  $\backslashdate\{\}$  : The argument is the date associated with the document. Note that leaving the argument empty will create a blank date in the title, and not including the command will default the title to the date compiled.

Body commands:  $\backslashmaketitle$  : Creates a title based on the  $\backslashtitle$ ,  $\backslashauthor$ , and  $\backslashdate$  provided.  $\backslashtableofcontents$ ,  $\backslashlistoffigures$ ,  $\backslashlistoftables$  : Creates a table of content, list of figures, and/or list of tables.  $\backslashnewpage\{\}$  : Creates a new page  $\backslashbackslash$  : Creates a line break  $\backslashbegin\{\}$  and  $\backslashend\{\}$  : Whenever there is a  $\backslashbegin\{\}$  it must be paired with an  $\backslashend\{\}$  with the same argument at some point later in the document.  $\backslashsection\{\}$  : Denotes a new section. The title section should be used as the argument. Sections will be automatically included in the table of contents. Use the commands  $\backslashsubsection\{\}$  and  $\backslashsubsubsection\{\}$  to create sections within sections. Use the QRcode to access a LaTeX commands cheat sheet to store on your phone. View more Go to Learn LaTeX in 30 minutes for a written online guide of how to create your first document using LaTeX. The video below is a practical guide to creating your first LaTeX document. It provides an overview of what is covered on this page: If you are interested in more information about a specific command, Overleaf.com provides a lot of guides or a simple Google search will often provide great results. Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Videos

- <https://www.youtube.com/watch?v=uVM2FcwPCgk>

## Links & References

- [Link] [LaTeX in 30 minutes](#)
- [Link] [Overleaf.com](#)

## Writing LaTeX code

Module 11 Writing LaTeX Code Icon Progress Bar (browser only) Focus This page introduces using templates and writing and adapting code within this environment to compose your document. There are readings and short videos to support your learning. LaTeX Templates A Sample Document Below is the LaTeX code and the PDF output for a simple sample document. First you will see a readable version of the code followed by the LaTeX PDF output. LaTeX Code

crib sheet LaTeX Commands Templates Before reading this section, you should have a basic understanding of how to create a LaTeX document, as well as the basic structure of a document. This section is about creating templates for LaTeX documents. Templates are meant to speed up the initial creation of a LaTeX document. Often it is the case that the same packages and document structure will be applicable to many of the documents you are working on. In this case, using a template will save you the time of having to input all of this information every time you create a new document.

**Using and Creating Templates** A good front-end LaTeX software package will contain at least some standard templates for different document types such as articles, beamers, and books. A great one will also let you create your own template. Templates can be very useful when there are certain document types you need to create often such as class notes, homework assignments, and lab reports. In these cases, a template will create consistency between documents and greatly simplify the creation of the document.

**Using Templates** How to use a template depends on what programme you are using to write your LaTeX code. If you are not using any front end software, a template would simply be an already written .tex file you used to start your document. It is important to make sure that you do not save over the template as you create the new document, as this would destroy the template file. Different front-end software have different methods for using templates. In TexStudio for example under "File" there is a "new from template" option. TexStudio comes with a number of pre-programmed templates but also allows you to add your own. TexMaker on the other hand, under file, has the option "new by copying an existing file". In this case you would need to create your own template file, but you would not have to worry that you might accidentally save over it. If the template you are looking for is not built into the software you are using, there are online resources such as LaTeX Templates where you can download templates for a variety of purposes. A more advanced LaTeX user may also want to consider creating their own template.

**Creating Templates** When creating a template there are several important questions to ask yourself: What document class will I need and will this ever change? It doesn't make sense to have a template if you will need to change the document class often. It is also important to recognise which document best suits your needs (i.e. if you want sections use article but if you want chapters use book). What packages will I need for the documents I am making? One of the best reasons to have a template is that you won't have to write the preamble every time you start a new document, therefore it is very important to include all the packages you will need. It's better to have extra packages than not enough. How do I want the document to be structured? You can save time by incorporating certain common elements into your preamble, such as title, author, and date. You may also want to include certain structures in the document, for example, if you were making a lab report you may already include all the sections the report requires (Introduction, Experimental Setup, Results, Conclusions, etc...). Are there any new commands that I will want to create? Perhaps you are using your template for a certain class that often uses matrices or a mathematical symbol with a long name. Creating a new command as part of the template can help simplify the writing process.

**Sample Template Code**

```
% This is a template for doing homework assignments in LaTeX
\documentclass{article}
% This command is used to set the type of document you are working on such as an article,
book, or presentation
\usepackage{geometry} % This package allows the editing of the page
layout
\usepackage{amsmath} % This package allows the use of a large range of mathematical
formula, commands, and symbols
\usepackage{graphicx} % This package allows the importing
of images
\newcommand{\question}[2][\begin{flushleft}]{\textbf{Question #1}: \textit{#2} \end{flushleft}}
\newcommand{\sol}{\textbf{Solution}:} % Use if you want a boldface solution
```

line \newcommand{\maketitletwo}[2][\begin{center} \Large{\textbf{Assignment #1}}  
Course Title} % Name of course here \vspace{5pt} \normalsize{Matthew Frenkel % Your name  
here \today} % Change to due date if preferred \vspace{15pt} \end{center}} \begin{document}  
\maketitletwo[5] % Optional argument is assignment number % Keep a blank space between  
maketitletwo and \question[1] \question[1]{Here is my first question} YOUR SOLUTION  
HERE \question[2]{Here is my second question} YOUR SOLUTION HERE \question[3]{What is  
the \Large{\$\int\_0^2 x^2 \, dx\$} \normalsize{. Show all steps}} \begin{align\*} \int\_0^2 x^2  
&= \left. \frac{x^3}{3} \right|\_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3} \end{align\*} \end{document} Sample Template Output You can check the output of the above  
code by compiling it. Below is the sample template output: Writing and Adapting LaTeX Code A  
Sample Document Below is the LaTeX code and the PDF output for a simple sample document.  
First you will see a readable version of the code followed by the LaTeX PDF output. Below is the  
LaTeX code of example above. Play around with this code and compile a document on your  
computer. Code: % This is a simple sample document. Note that everything that comes after a  
% symbol is treated as comment and ignored when the code is compiled. \documentclass{article} % \documentclass{} is the first command in any LaTeX code. It is used  
to define what kind of document you are creating such as an article or a book, and begins the  
document preamble \usepackage{amsmath} % \usepackage is a command that allows you to  
add functionality to your LaTeX code \title{Simple Sample} % Sets article title \author{My  
Name} % Sets authors name \date{\today} % Sets date for date compiled % The preamble  
ends with the command \begin{document} \begin{document} % All begin commands must be  
paired with an end command somewhere \maketitle % creates title using information in  
preamble (title, author, date) \section{Hello World!} % creates a section \textbf{Hello World!}  
Today I am learning \LaTeX. % notice how the command will end at the first non-alphabet  
character such as the . after \LaTeX \LaTeX{} is a great programme for writing math. I can  
write in line math such as  $a^2+b^2=c^2$  % \$ tells LaTeX to compile as math . I can also give  
equations their own space: \begin{equation} % Creates an equation environment and is  
compiled as math  $\gamma^2+\theta^2=\omega^2$  \end{equation} If I do not leave any blank  
lines \LaTeX{} will continue this text without making it into a new paragraph. Notice how  
there was no indentation in the text after equation (1). Also notice how even though I hit enter  
after that sentence and here  $\downarrow$  \LaTeX{} formats the sentence without any break.  
Also, look how it doesn't matter how many spaces I put between my words. For a new  
paragraph I can leave a blank space in my code. \end{document} % This is the end of the  
document Need Help? Looking for help? Seeking an answer to your question? Want to share  
some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback  
Channel

## Links & References

- [File] [LaTeX Code crib sheet](#)
- [File] [LaTeX Commands](#)
- [Link] [LaTeX Templates](#)

## Creating a Bibliography

Module 11 Creating a Bibliography Focus This page introduces using LaTeX to produce bibliographies. You will learn about storing your citation information and then generating the

bibliography within the LaTeX document, saving you time formatting the data yourself. There are readings and short videos to support your learning. Creating a Bibliography There are two parts to creating a bibliography in a LaTeX document. Part one is the creation of a separate .bib file. This file contains all of the citation information (i.e. title, author, publication date, publisher, etc.), as well as a bibID for each citation. It is strongly recommended to use a citation management tool to create your .bib files. Those tools will greatly speed up the process and help keep the file organised, generate automatic bibIDs (which can be edited if desired), and auto fill all relevant data fields which will prevent typos or bugs within the file. Below is an example of the .bib format for four different common references, if you choose to enter them yourself. Note that not all fields are required, and any unused field should be deleted. The bibID is what is used in the main LaTeX document to create in-line references as well as completed bibliographies. When creating the final bibliography, LaTeX will only include citations that were referenced in the document, so it's okay to have unused citations in your .bib file.

Article  
 Book  
 Online Resource  
 Conference Proceeding

```
@article{ bibID , author = {author}, title = {title},
journaltitle = {journaltitle}, date = {date}, }
@book{ bibID , author = {author}, title = {title},
date = {date}, }
@online{ bibID , ALTAauthor = {author}, ALTeditor = {editor}, title = {title}, date
= {date}, url = {url}, }
@proceedings{ bibID , editor = {editor}, title = {title}, date = {date}, }
```

Integrating Bibliography Information Part two is integrating the information from the .bib file into your main LaTeX document. The three ways for doing so are using BibTex , BibTex with natbib , or BibLaTeX . The hyperlinks will take you to explanation of each method from overleaf.com. Both BibTex with natbib and BibLaTeX have the advantage of optional arguments because they require a \usepackage[]{} command. These optional arguments can be used to fine tune how references appear throughout the document and the formatting of the bibliography. Below is a table that highlights some of the important differences for each method.

| BibTex                                                                                    | BibTex with natbib                                                                 | BibLaTeX                                                                              |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Packages Needed                                                                           | None                                                                               | natbib biblatex                                                                       |
| In document command for citation                                                          | \cite{bibID}                                                                       | \cite{bibID}                                                                          |
| Bibliography styles                                                                       | Use command (place in body): \bibliographystyle{stylename}                         | Common Style names: abbrev, acm, alpha, apalike, ieeetr, plain, siam, unsrt           |
| Use command (place in preamble): \bibliographystyle{stylename}                            | Common Style names: dinat, humannat, plainnat, abbrvnat, unsrnat, rusnat, ksfh_nat | Optional Argument of \usepackage: \usepackage[ style=stylename, ]{biblatex}           |
| Common Style names: numeric, alphabetic, authoryear, authortitle, verbose, reading, draft | Print bibliography command \bibliography{bibfilename}                              | DO NOT INCLUDE .bib \bibliography{bibfilename} DO NOT INCLUDE .bib \printbibliography |

Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

## Links & References

- [Link] [BibTex](#)
- [Link] [BibTex with natbib](#)
- [Link] [BibLaTeX](#)

## Online LaTeX Resources

Module 11 Online LaTeX Resources Icon Progress Bar (browser only) Focus LaTeX is a very powerful program with a huge number of features. Below are a few links to introduce some helpful resources for new users or for more experienced users looking to dive deeper into LaTeX

capabilities. Website LaTeX Guides An introduction to TeX - A website from the Harvard mathematics department that overviews both the TeX and LaTeX environment A Beginners Guide to LaTeX - A short guide from Princeton University The Not so Short Introduction to LaTeX2e - A 160+ page PDF that outlines many of the core functionalities of LaTeX Websites for LaTeX Commands and Symbols The Comprehensive LaTeX Symbol List - A more than 300 page document of LaTeX symbols LaTeX Command Summary - A summary of symbols and commands in LaTeX LaTeX Resource Websites LaTeX Wikibooks - A great resource for LaTeX related questions TeXample - Contains packages for creating graphics in LaTeX also has a forum LaTeX Forums LaTeX Forums to discuss any issues you encounter while writing code: <http://tex.stackexchange.com/> <http://latex-community.org/forum/index.php> Overleaf Tutorials Below are a few very good resources to start working with Overleaf: Tutorials Writing your first piece of LaTeX Introduction to LaTeX (part 1) Introduction to LaTeX (part 2) Introduction to LaTeX (part 3) Need Help? Looking for help? Seeking an answer to your question? Want to share some advice, a resource, or a fun fact with your class peers? IT8101 Support and Feedback Channel

### Links & References

- [Link] [An introduction to TeX](#)
- [Link] [A Beginners Guide to LaTeX](#)
- [Link] [The Not so Short Introduction to LaTeX2e](#)
- [Link] [The Comprehensive LaTeX Symbol List](#)
- [File] [LaTeX Command Summary](#)
- [Link] [LaTeX Wikibooks](#)
- [Link] [TeXample](#)
- [Link] <http://tex.stackexchange.com/>
- [Link] <http://latex-community.org/forum/index.php>
- [Link] [Tutorials](#)
- [Link] [Writing your first piece of LaTeX](#)
- [Link] [Introduction to LaTeX \(part 1\)](#)
- [Link] [Introduction to LaTeX \(part 2\)](#)
- [Link] [Introduction to LaTeX \(part 3\)](#)

### Lab 5: Designing a document using LaTeX

IT8101 Lab 5: Designing a document using LaTeX Summary Writing your own code and working through various situations is the greatest way to learn LaTeX. The puzzles build on one other and become more difficult as you progress. If you get stuck, you can always refer back to the instructions, use the tips, or Google it. Keep in mind that there may be more than one solution. Below are a few problems for you to do, to test your skills. When you have finished working your way through the exercises, send your final document to your lecturer to review. Exercises Exercise 1: I am very excited for about PGDip in Information Technology! Create a document with the title “I am very excited about my PGDip in Information Technology”! Exercise 2: Add some simple math Add to the document you created in exercise 1 the following text. The Research methods and skills course is very exciting. We have to deal with a lot of statistics and we should be familiar with equations. I can write math like  $a^2 + b^2 + c^2 + d^2 = e^2$ . I am also

familiar with mathematical symbols. I can write equations based on it:  $\gamma^2 + \theta^2 = \omega^2 - \lambda^2$ . Exercise 3: More complicated equations In this exercise, you will work with more complicated equations and continue adding to the document that you created in exercises 1 and 2. Write the equation as shown below. Use an align environment and align the equations with the equation names and numbers as shown below. Exercise 4: Get familiar with how to create sections and referencing the equations Labels and references are easy to use in LaTeX, and they can be used with any numbered item, such as figures, equations, and sections, and they are immediately updated when the document is compiled. If you realise you forgot one equation in the middle of your paper, between ten other equations, all the equations after the newly included equation would be immediately corrected. All equations after the newly added equation would be renumbered automatically, and proper references to them would reflect this new numbering! Take the document you created in the last exercise and divide it into two sections: one at the start (Getting Started) and one at the end (What about Matrix Equations?). Add references to each of Maxwell's equations to your paper, and then compose a phrase that cites each equation. You will need to use the commands `\label{}` and `\ref{}`. When labelling in LaTeX, the formats eq:name, fig:name, tab:name, and so on are commonly used, depending on the type of item being labelled. Look at the labels in the solution if you're still unsure. Also, if you want your citations to act as hyperlinks, you must use the package `hyperref`, `\usepackage{hyperref}`. Remember, with this package, you can change the default hyperlink settings with the command `\hypersetup{}` in the preamble i.e. `\hypersetup{colorlinks=true, linkcolor=blue, urlcolor=blue, citecolor=blue}`. Exercise 5: Add a matrix equation In this exercise you will need to add a matrix equation as shown below: Exercise 6: Inserting Tables and Figures Create a new section in your document for figures and tables for this activity. Create a table and a figure in that area. Give each object a caption, with the table caption appearing above the table and the figure caption appearing below. This may be done with any figure you like.  $y = 0.12F(y) + 0.369F(y^2) + 0.93681$  Table 1: A simple table showing different rows, columns and justifications. Figure 1: Whitecliffe Symonds Street Campus Exercise 7: Bibliography To do this exercise, first you need to be familiar with Bib Tex. Refer to the link below to know more about bibtex. <https://www.economics.utoronto.ca/osborne/latex/BIBTEX.HTM> Create a Bib tex file with a few citations in it. Add a sentence that contains these citations and then add a bibliography to your document. Example Bibliography These are very good articles that describe cybersecurity of healthcare IoT-based systems [1,2,3,4] References

## Links & References

- [Link] <https://www.economics.utoronto.ca/osborne/latex/BIBTEX.HTM>

## Assessments / Assignments

### Assessment 1: Literature Review, Research Gap and Questions

**Due: 2026-09-03T11:59:59Z | Points: 100**

IT8101 Assessment 1: Literature Review, Research Gap and Questions This assessment contributes to 50% of the final course grade Assessment Overview In this assessment, your first task is to conduct a comprehensive literature review within the field of Information Technology. This review will form the basis for identifying a research gap or a specific research

problem. After establishing this foundation, you will then formulate research questions that directly arise from the insights gained during your literature review. The next step involves refining these questions by analyzing relevant IT information and data. This process ensures that your questions are not only grounded in the existing body of knowledge but are also sharply focused and tailored to address the identified gap or problem.

**Learning Outcomes** This assessment contributes to: LO 1: Develop an appropriate IT related tikanga-centered research question and associated methodology, emerging from relevant literature.

**Turnitin Canvas Plagiarism Framework** This assignment will be checked for plagiarism and originality using the Turnitin Canvas Plagiarism Framework. You can learn more about this system by reading What is the Turnitin Canvas Plagiarism Framework . Before submitting your final version, you have the chance to upload drafts to Turnitin Draft Coach. To do this, read the sections What is Turnitin Draft Coach and How to use Turnitin Draft Coach and follow the instructions provided there for drafting your assignment. The similarity threshold is 15%.

**Conditions** You can access all course materials and any other resources you wish to use as you work on this assessment. The work you submit must be your own work. You can ask your lecturer to clarify the instructions and for advice, but they cannot assist you in completing the tasks required – you must carry out the tasks yourself!

**Overview** his assessment has two parts: Part 1 : A literature review of a relevant topic chosen from one of the fields of interest below. Applied social cybersecurity Human robot interaction Data privacy and user data control IoT interoperability and Multi-platform integration 5G and Beyond communications for smart cities Machine learning and Artificial Intelligence for 5G and beyond Technologies Another area of interest, agreed with the course lecturer. Make sure it aligns with your area of specialization.

**Part 2:** Focuses on developing two research questions for your chosen topic that emerge from the literature.

**Instructions** Part 1: Literature Review Max Word Limit: 4,000 Conduct a literature review on your chosen topic with the purpose of exploring the interconnection between humans and technology, in relation to digital identity. Aim to consult around 20 scholarly sources, and no fewer than 10. While reviewing the literature, identify the research methodologies in each article and summarise why those particular ones were appropriate.

**Note:** Select only peer-reviewed articles. Read the articles thoroughly and formulate the review in your own words. Do not copy directly from the paper. All sources must be acknowledged using correct APA format for in-text citations and a reference list/bibliography . While conducting your literature review, you may find the following questions helpful in analyzing each paper, article, or book. Feel free to answer the questions that are most pertinent to each specific piece of literature:

**Methodology Insights:** What research methodologies does the paper use, and what might be the rationale behind this choice

**Purpose of the Paper:** What goals or objectives is the paper aiming to achieve?

**Problem Focus:** What are the key problems or challenges that the research is addressing?

**Identification of Gaps:** Are there any gaps or underexplored areas in the field that the paper points out?

**Contributions to the Field:** What new findings, perspectives, or contributions does the paper offer?

**Conclusive Elements:** What conclusions does the paper draw, and what are its key takeaways?

**Results Evaluation:** How does the paper evaluate or interpret its findings?

**Core Research Questions:** What central questions is the paper trying to answer or explore? Remember, not all questions may be relevant for each source. Select the ones that most effectively help you understand and evaluate the contribution of each piece of literature to your research topic.

**Part 2: Research Gap and Questions** that emerge from the literature Max Word Limit: 1,500

**Identifying the Research Gap/Problem:** Briefly describe the research gap or problem you have identified in the existing

literature. Address the following points: What's the problem? : Clearly state the issue or gap in the current literature. Who cares - why is it worth solving? : Explain the significance of this problem in the field of Computer Science. Discuss who is affected by this problem and why resolving it is important. Who's tried to solve it before? : Reference previous attempts or studies that have addressed this problem and summarize their approaches. Why were their solutions/attempts insufficient? : Critically analyze why previous solutions did not fully address the problem. This could involve methodological limitations, scope of research, or new developments in the field. What are you going to do about it? : Briefly propose how your approach or perspective in addressing this problem might differ from previous attempts. This is a precursor to the research questions you will develop. Ngā Tikanga Paihere principles: Explain how your approach/ perspective to address this problem will align with the Ngā Tikanga Paihere principles of Kaitiaki, Mauri and Wairua . Formulating Research Questions: Develop two research questions that directly stem from the identified research gap. When you seek to discover something new about a topic, asking questions is essential. Thus, research questions are crucial in the context of the overall research process, providing necessary guidelines for your investigation. In a small-scale research project, you typically formulate a single research question that addresses one specific problem or issue. In contrast, larger projects may involve multiple research questions. It is fundamental in research that each question is focused on your main research problem. The research questions you develop could eventually lead to your research proposal later in the program. Examples of Well-Formulated Research Questions For a smaller-scale project: What are the effects of [specific algorithm] on the efficiency of [specific process] in [specific context]? For a larger project: How can machine learning techniques improve cybersecurity in cloud computing environments? Justification of Each Question: Provide a justification for each research question (approximately 100-150 words per question). Explain how each question addresses the identified gap and its significance in advancing understanding in the field. Explain how your approach/ perspective to address this problem will align with the Ngā Tikanga Paihere principles of Kaitiaki, Mauri and Wairua . Discuss the potential impact or contribution of research answering these questions. Note : Each research question will be marked separately. Hints and Tips Step 1: Generating Research Questions Read & think, then brainstorm: Challenge Existing Literature : Consider unique angles or unexplored areas within your topic. Originality in your questions can lead to significant contributions in the field. Brainstorming : Write down as many questions as possible about your topic. Selection : Choose the two questions that seem the most robust. Refinement : Clarify and focus these questions. Then ask yourself: Scoping Your Research Questions : Balance : Ensure your question is neither too broad (making it unmanageable) nor too narrow (limiting exploration). Guidance : Use your literature review to define the scope. Assess if your question aligns with the gaps or challenges identified in existing research. Practical Considerations : What data will you need, and how will you gather it? What methods will you use? Estimate the time, cost, and resources required. What skills, experience, or knowledge are necessary? Adjustment : If necessary, modify your questions to make them more practical. Note : Aim for a specific, measurable question, considering the factors identified in your literature review. Step 2: Writing two research questions After narrowing down your topic and gathering background information, focus on refining your research question. Consider the following elements: Construct : What is the key idea or concept you are investigating? Beyond the Data : Don't solely rely on data; your question should extend its implications. Application : Ensure broad relevance to the field. Significance : Address the "so what?" factor. Why is this question

important? For each research question, justify its relevance in relation to the research problem identified in your literature review. Step 3: Placing your questions within a theory/research Context A theory or research context offers a lens through which to view your topic, influencing every aspect of your research, including the question and methodology: Identifying Theories : Determine which theories are applicable to your topic. Question Formulation : Create research questions for each relevant theory. Feasibility Assessment : Evaluate the practicality of your questions. Cultural Consideration : Evaluate your question in terms of Ngā Tikanga Paihere principles, ensuring cultural appropriateness and sensitivity. Use a table, like the one below to work through the steps and refine your Research Questions.

| Idea 1 | Idea 2 | Theory/Research context | Topic | Main Research Question | Secondary questions (sub-questions) | Data needed to answer questions | So what? (what will this contribute to the field/profession) |
|--------|--------|-------------------------|-------|------------------------|-------------------------------------|---------------------------------|--------------------------------------------------------------|
|        |        |                         |       |                        |                                     |                                 |                                                              |

Applying the Ngā Tikanga Paihere concept of Kaitiaki, Applying the Ngā Tikanga Paihere concept of Mauri Applying the Ngā Tikanga Paihere concept of Wairua Submission Instructions Work through each of the tasks above. When they are complete, upload all of the assessment documents to Canvas as PDF files. Policies Success Criteria To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%. Extensions and Late Submissions An extension can only be granted under exceptional conditions normally for a maximum of (5 working days) , unless there is evidence to support a longer timeframe. Complete and return the following form to your Course Lecturer: Assessment Extension Request Form.pdf Assessments received one (1) to five (5) working days past the due date, or an approved extended due date, will be marked but will receive a mark capped at the pass mark i.e. 50% for graded assessments. Assessments received more than five (5) working days past the due date, or approved extended due date, will not be marked and will receive a mark of zero or fail. Academic Integrity and Academic Misconduct Policy Whitecliffe is committed to ensuring the highest level of academic integrity and expect staff and students to conduct themselves as responsible and honest members of the academic community. Where there is evidence that breaches of academic integrity may have occurred, the course of Academic Integrity and Academic Misconduct Policy will be followed. This policy considers academic integrity to mean honesty in all matters relating to academic endeavours. Breaches in academic integrity, or academic misconduct, include all forms of cheating incorporates plagiarism (intentional or unintentional), unauthorised collaboration, impersonation, falsification and collusion. Declaration By submitting your work, you are indicating that you agree to the following declaration: "The work presented in this assessment is to the best of my knowledge original, except as acknowledged in the text, and the material has not already been submitted, either in whole or in part, for any academic award at this or any other tertiary institution. I promise not to share this assessment in part or whole with any other student at Whitecliffe or outside this campus."

## Links & References

- [Link] [Assessment Extension Request Form.pdf](#)
- [Link] [Academic Integrity and Academic Misconduct Policy](#)

## Assessment 2: Research Proposal and Reflection

**Due: 2026-10-29T10:59:59Z | Points: 100**

IT8101 Assessment 2: Research Proposal and Reflection This assessment contributes to 50% of the final course grade Assessment Overview For this assessment you will write a research proposal that includes theoretical and practical details, an extended literature review, theoretical framework and methodology . To deepen your learning, you will reflect on the process of creating the research proposal and related ethical issues, with reference to those that directly relate to Ngā Tikanga Paihere . Learning Outcomes This assessment contributes to: LO 2: Critically reflect on ethical issues and/or potential barriers to undertaking proposed research in the field of IT. Turnitin Canvas Plagiarism Framework This assignment will be checked for plagiarism and originality using the Turnitin Canvas Plagiarism Framework. You can learn more about this system by reading What is the Turnitin Canvas Plagiarism Framework . Before submitting your final version, you have the chance to upload drafts to Turnitin Draft Coach. To do this, read the sections What is Turnitin Draft Coach and How to use Turnitin Draft Coach and follow the instructions provided there for drafting your assignment. The similarity threshold is 15%. Conditions You can access all course materials and any other resources you wish to use as you work on this assessment. The work you submit must be your own work. You can ask your lecturer to clarify the instructions and for advice, but they cannot assist you in completing the tasks required – you must carry out the tasks yourself! Instructions This Assessment has three tasks. Task 1: Research question/s based on feedback from Assessment 1 Task 2: Writing a research proposal. Task 3: Critical reflection on ethical issues and/or potential barriers to undertaking your proposed. In Assessment 1, you formulated research questions based on your chosen topic. In this assignment, Assessment 2, you will write a Research Proposal and critically reflect on ethical issues and/or potential barriers to the research. Research Proposals are a necessary step in a research project allowing you to refine your objectives and explain how you plan to investigate them. Proposals include both theoretical information relevant to the topic (i.e., the literature review) and practical details (e.g., timeline and budget etc.). Task 1: Your Research Question Introduction: Writing a proposal is a process. It takes time to work through all the sections you need to complete. It takes time to think about how you are going to do something, justify why you are doing it, and the details of the process. It is impossible to write a good proposal in a day! Coming to a research proposal, most researchers need to spend time narrowing the topic. Most of us start off with a great big concept in mind. When it comes to figuring out how we are going to tackle it and the resources required, we suddenly realise that it is too big. Sometimes this requires going back to the library, doing more reading, and re-thinking your proposal. Instructions: Take your topic and two research questions from Assessment 1. Review them in the light of what you now know about the topic. You may need to refine both the topic and questions based on the feedback from Assessment 1. When you are happy with your topic and questions you can start Task 2. Note : Task 1 does not need to be included in your research proposal. It is solely an exercise designed to refine and improve your research questions. Task 2: Your Research Proposal Introduction: A research proposal is simply: A structured, formal document that explains what you plan to research i.e., your research topic Why your research topic is worth researching i.e., your justification How you plan to investigate your research topic i.e., your practical approach. The purpose of the research proposal is: To convince your research supervisor, committee or university that your research is suitable. That it meets the

requirements of the postgraduate degree programme. That it is manageable given the time and resource constraints you will face. The most important word here is 'convince'. In other words, your research proposal needs to sell your research idea to whoever is going to approve it. If it doesn't convince them of its suitability and manageability, you'll need to revise and resubmit. This will cost you valuable time, which will either delay the start of your research or eat into its time allowance. At a high level your proposal should include three key elements : What do you intend to do? (your research topic) Why do you intend to do it? (your justification) and How do you intend to do it? (your methodology) Instructions: Using LaTeX, write a Research Proposal based on the topic and research questions from Task 1. Your proposal should be around 5,000 words (including your updated literature review, excluding title pages, toc, references and appendix). All sources must be acknowledged using correct APA format for in-text citations and a reference list/bibliography . Step 1: Focus and extend your Literature Review Bring your novel ideas to the particular research area you have identified as of interest. This novelty can be only determined after a thorough literature search. Based on your Research Questions, review, refocus and extend the literature review you undertook in Assessment 1. You will need to cover at least 35 latest benchmark papers published in IEEE, SpringerLink, ACM Digital Library, Google Scholar, ScienceDirect, MDPI, Hindawi, or other reputable academic repositories. Step 2: Identify and write down the Aims of your Research It is important to take the time to clarify what your research is aiming to do. Take time to decide on your research aims and write these down. Step 3: Decide on your research design and methodology and how the information you collect will be analysed Provide details, especially in your methodology and data analysis section. These are the two areas where most proposals come unstuck. If you are working with different groups of people (according to age, gender, ethnicity), state why you are going to do this. Be explicit in your explanation of how you will analyse the data. It is not enough to say "the data will be collated and analysed". How will you do this? What software or theories will be applied? How will you take Ngā Tikanga Paihere principles into account? Step 4: Write your Research Proposal A sound proposal will answer the following questions: What is the research expected to accomplish, if completed successfully and on time? Why is the research being undertaken? What are the reasons behind it? What is the research underlying rationale? How is the research to be implemented? What work is necessary in order to achieve the objectives? Who is primarily responsible for the research implementation and management? Who are the intended beneficiaries, the target group, who are expected to benefit from the research? Within what period of time is the project to be carried out and are the objectives to be attained? What resources are necessary to achieve the objectives? What external factors are necessary for research success? To address all these questions your proposal should include the following core sections: Title Abstract Introduction and/or background Research question Literature review Research design Methodology Significance of Research Expected outcomes Timeline Budget References Appendices Use the following checklist to ensure you have addressed all the assessment requirements. Does the problem statement/research question follow clearly from the literature review? Are the proposed unit of analysis (subjects, participants, situations, organisations, etc.) clearly described in terms of the following: a) their number (e.g., sample size)? b) their background (e.g., sex, ethnicity, etc. in the case of research participants)? c) how they are to be obtained/recruited? Has it been indicated whether the necessary permission for the cooperation of the participants has been obtained or will be obtained? Has the treatment (e.g., any intervention) of the units of analysis been described or specified clearly? Have the variables to be subjected to analysis been operationalised (e.g., the measurement of data under

investigation been investigated and is available)? Is it clear how the measuring instruments to be used fit in with the theoretical background and/or literature review? Is the proposed procedure for analysing the obtained information (e.g., statistical method) specified? Is the proposal presented in terms of all of the headings set out as required in the Assessment? Are there in-text citations throughout the proposal that reference peer-reviewed, scholarly sources?

**Task 3 - Critical Reflection** The third task in this assessment asks you to critically reflect on ethical issues and/or potential barriers to undertaking the proposed research. Using Ngā Tikanga Paihere as a framework, explore, analyse and critically reflect on the ways in which these principles will be integrated and mitigate emerging ethical issues and/or potential barriers to your research. It is important to understand the significance of Ngā Tikanga Paihere principles and which ones relate to your research and consider their impact to your research.

**You must include this task in your Research Proposal.**

**Submission Instructions** Upload each task as a pdf document into Canvas using acceptable labelling conventions that include your name, student ID and task.

**Policies Success Criteria** To be successful in each assessment you must obtain a minimum of 50% of the total available marks. You will be marked according to the rubrics shown in each assessment. Please take the time to study these as they contain information on how you will be assessed. You are allowed a maximum of two attempts. The maximum percentage to be awarded on a second assessment attempt is 50%. Even if you score higher than 50% on a second attempt, you will only be awarded 50%.

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**Declaration** By submitting your work, you are indicating that you agree to the following declaration: "The work presented in this assessment is to the best of my knowledge original, except as acknowledged in the text, and the material has not already been submitted, either in whole or in part, for any academic award at this or any other tertiary institution. I promise not to share this assessment in part or whole with any other student at Whitecliffe or outside this campus."

## Links & References

- [Link] [Assessment Extension Request Form.pdf](#)
- [Link] [Academic Integrity and Academic Misconduct Policy](#)

## Course File Index

*All files attached in this course. Open the link while logged in to download.*

- [\\_README.txt](#) (text/plain, 1 KB)
- [\\_UeY8aTI6d0.png](#) (image/png, 300 KB)
- [1JX4J\\_kq7sU-1.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-2.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-3.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-3feef37b-93e6-4e16-81f5-2d5329ba14f0.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-4.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-5.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-6.png](#) (image/png, 1.6 MB)
- [1JX4J\\_kq7sU-7.png](#) (image/png, 1.6 MB)
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- [3mt71MKGjQ0.png](#) (image/png, 514 KB)
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- [58Z17lnVS4U.png](#) (image/png, 436 KB)
- [64YrPKiguAE.png](#) (image/png, 636 KB)
- [159751.png](#) (image/png, 844 KB)
- [162553.png](#) (image/png, 972 KB)
- [212286.png](#) (image/png, 473 KB)
- [256262.png](#) (image/png, 489 KB)
- [256381.png](#) (image/png, 711 KB)
- [256455.png](#) (image/png, 852 KB)
- [356040.png](#) (image/png, 467 KB)
- [356043.png](#) (image/png, 595 KB)
- [590022.png](#) (image/png, 394 KB)
- [941555.png](#) (image/png, 474 KB)
- [1122865.png](#) (image/png, 629 KB)
- [1181354.png](#) (image/png, 693 KB)
- [1370298.png](#) (image/png, 665 KB)
- [3790833.png](#) (image/png, 376 KB)
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- [3854130.png](#) (image/png, 343 KB)
- [3862132.png](#) (image/png, 558 KB)
- [5726705.png](#) (image/png, 562 KB)
- [6899120.png](#) (image/png, 254 KB)

- [A9Msi-vUNKg.png](#) (image/png, 789 KB)
- [AC09.22.2 Academic Integrity Policy.pdf](#) (application/pdf, 192 KB)
- [AC20.23.1 Academic Progression and Resubmission Policy FINAL.pdf](#) (application/pdf, 159 KB)
- [Assessment 1 Document Structure.docx](#) (application/vnd.openxmlformats-officedocument.wordprocessingml.document, 12 KB)
- [Assessment 1 Feedback.pdf](#) (application/pdf, 122 KB)
- [Assessment 2 Guidelines.pdf](#) (application/pdf, 251 KB)
- [books20240409\\_110328.jpg](#) (image/jpeg, 98 KB)
- [C7B-ExXpOIE.png](#) (image/png, 350 KB)
- [computer\\_lab-1.png](#) (image/png, 631 KB)
- [computer\\_lab-2.png](#) (image/png, 631 KB)
- [computer\\_lab-3.png](#) (image/png, 631 KB)
- [computer\\_lab-4.png](#) (image/png, 631 KB)
- [computer\\_lab.png](#) (image/png, 631 KB)
- [Discussion.jpg](#) (image/jpeg, 48 KB)
- [Discussion.png](#) (image/png, 266 KB)
- [e\\_prXWsOBV4.png](#) (image/png, 735 KB)
- [ER6\\_i8FhQIw.png](#) (image/png, 516 KB)
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- [FTNGfpYCpGM.png](#) (image/png, 584 KB)
- [gKUC4TMhOiY.png](#) (image/png, 639 KB)
- [hHL08lF7Ikc.png](#) (image/png, 810 KB)
- [hhq1Lxtuwd8.png](#) (image/png, 614 KB)
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- [linkedin-learning-banner-1.png](#) (image/png, 17 KB)
- [MicrosoftTeams-image.png](#) (image/png, 48 KB)
- [Nga-Tikanga-Paihere-Guidelines-December-202020240902\\_143343.pdf](#) (application/pdf, 1.6 MB)
- [OReilly\\_Digital\\_Poster\\_1 \(1\)-1.jpg](#) (image/jpeg, 1.1 MB)
- [pwcKF7L4-no.png](#) (image/png, 680 KB)
- [Research Methods and Skills - 1.pdf](#) (application/pdf, 3.1 MB)
- [Research Methods and Skills - 1.pdf](#) (application/pdf, 3.0 MB)
- [s9CC2SKySJM.png](#) (image/png, 430 KB)
- [sfL\\_QOnmy00.png](#) (image/png, 681 KB)
- [TapuPrinciple-1.png](#) (image/png, 122 KB)
- [tech\\_management\\_banner.png](#) (image/png, 496 KB)
- [UX\\_research.png](#) (image/png, 545 KB)
- [VcTi-OqlVgA.png](#) (image/png, 446 KB)
- [vVHXeuOYNbk.png](#) (image/png, 88 KB)
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- [xj8qrWvuOEs.png](#) (image/png, 513 KB)

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